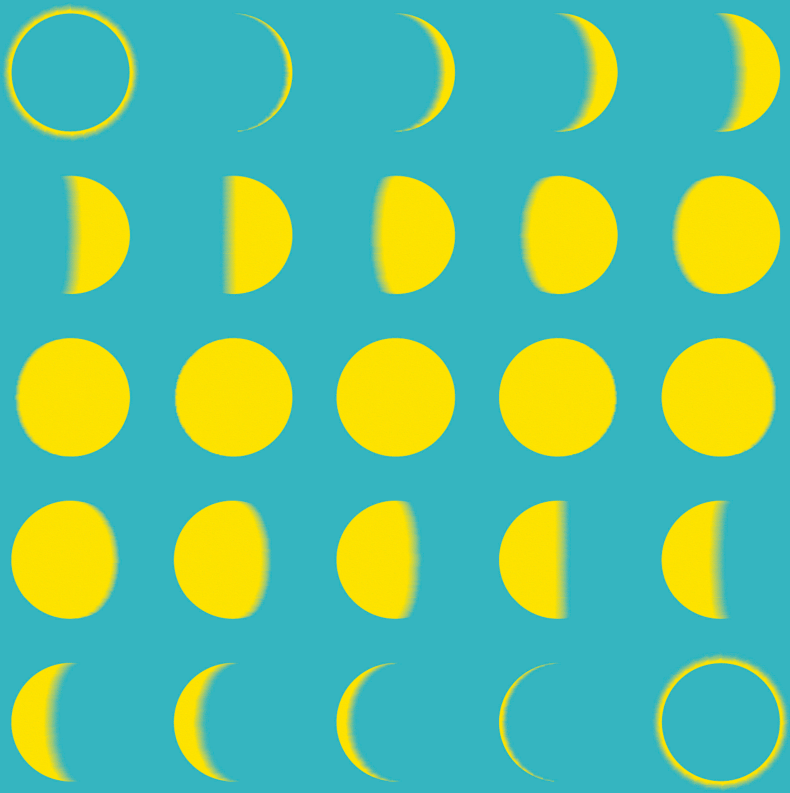
DK

SIMPLY ASTRONOMY



SIMPLY
ASTRONOMY



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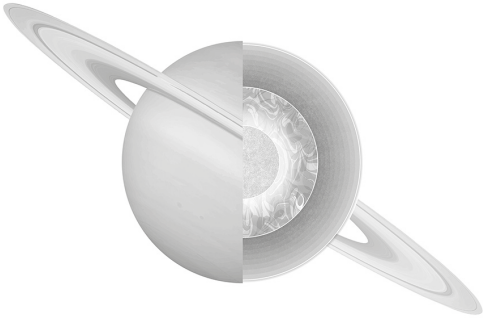
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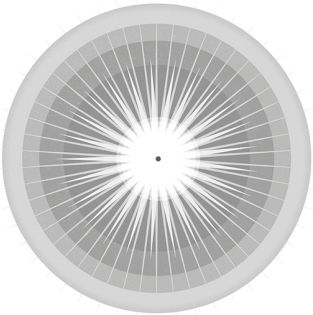
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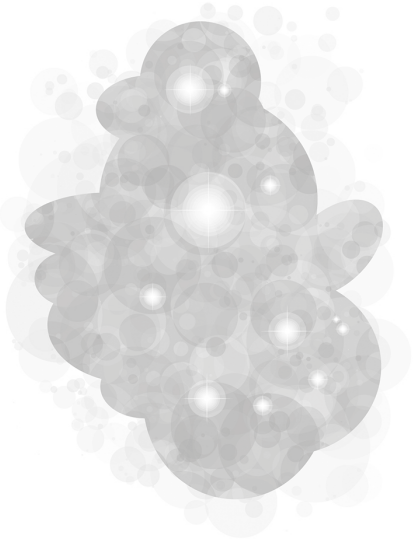
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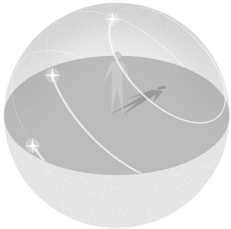
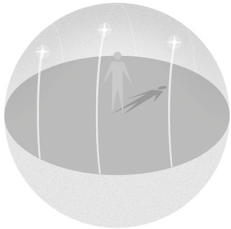
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WHAT IS ASTRONOMY?

THE NIGHT SKY has always inspired mystery and wonder. Astronomy aims to make sense of the Universe, confronting some of our most fundamental questions, such as how did everything begin and is there life beyond Earth? It can explain some of the mysteries but a sense of wonder remains. If anything, modern science reveals a cosmos that is more vast, strange, and complex than might have seemed possible to people even a few centuries ago. For example, to the unaided eye, stars seem like nothing more than points of light. But, with the aid of telescopes and other instruments, we can see that each of these points is a star like our own Sun. We know that a star is a ball of incredibly hot gas that emits energy as a result of nuclear fusion reactions. We have also discovered that stars have complex, often spectacular life histories and that many of them are orbited by systems of planets. From a distance, some of these planets even look similar to Earth.

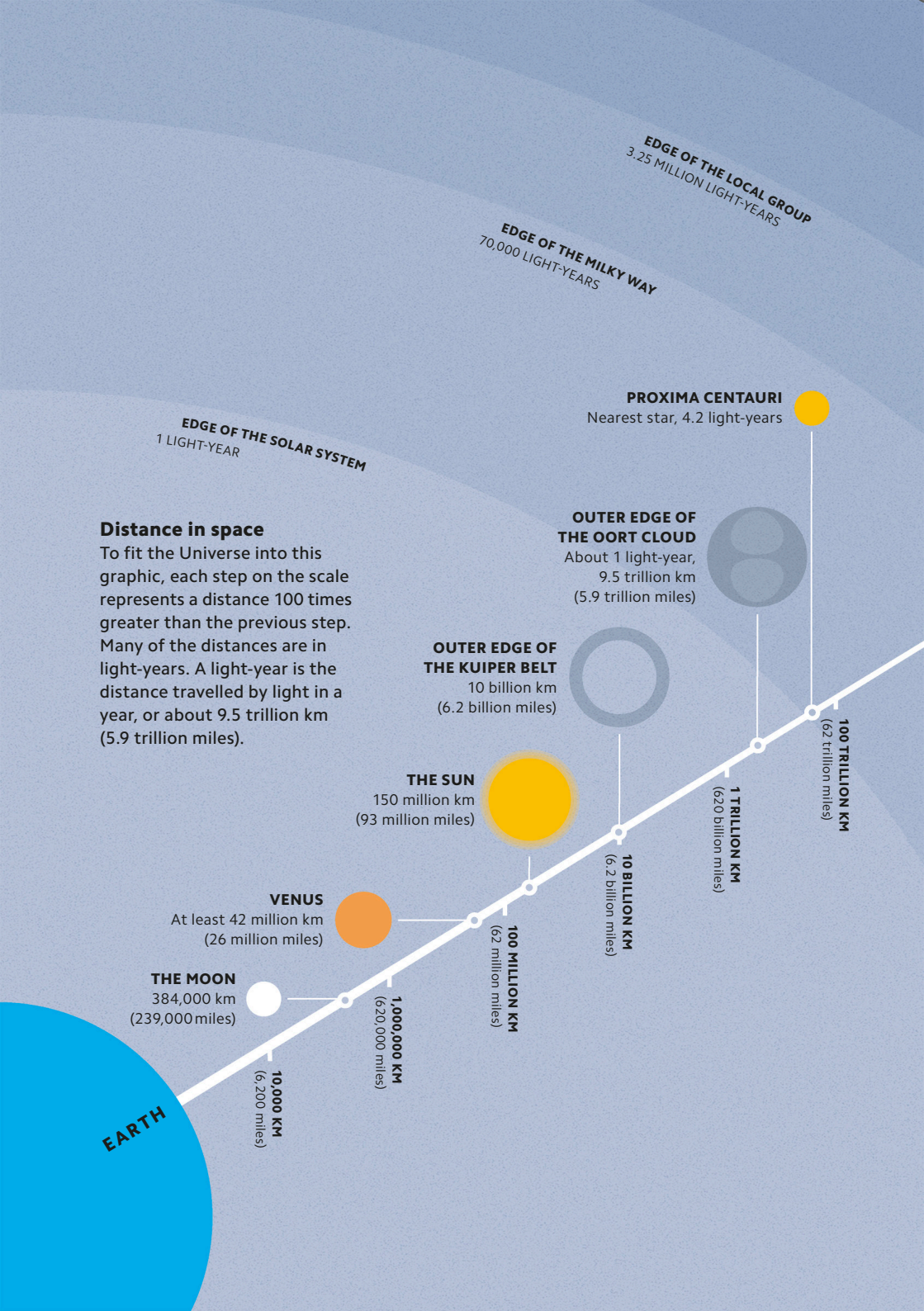
Over the course of its history, astronomy has been propelled forwards by a combination of technological progress and advances in scientific understanding. The last few decades have seen rapid change in both areas. We can now see far out into space (and hence far back in time) by sending telescopes into orbit. We can dispatch robots to explore the surfaces of other planets. But, despite all our achievements, many questions remain. We still cannot describe the Universe's size or shape, and more than 95 per cent of its substance is invisible to us. The Universe has not yet given up all its secrets.

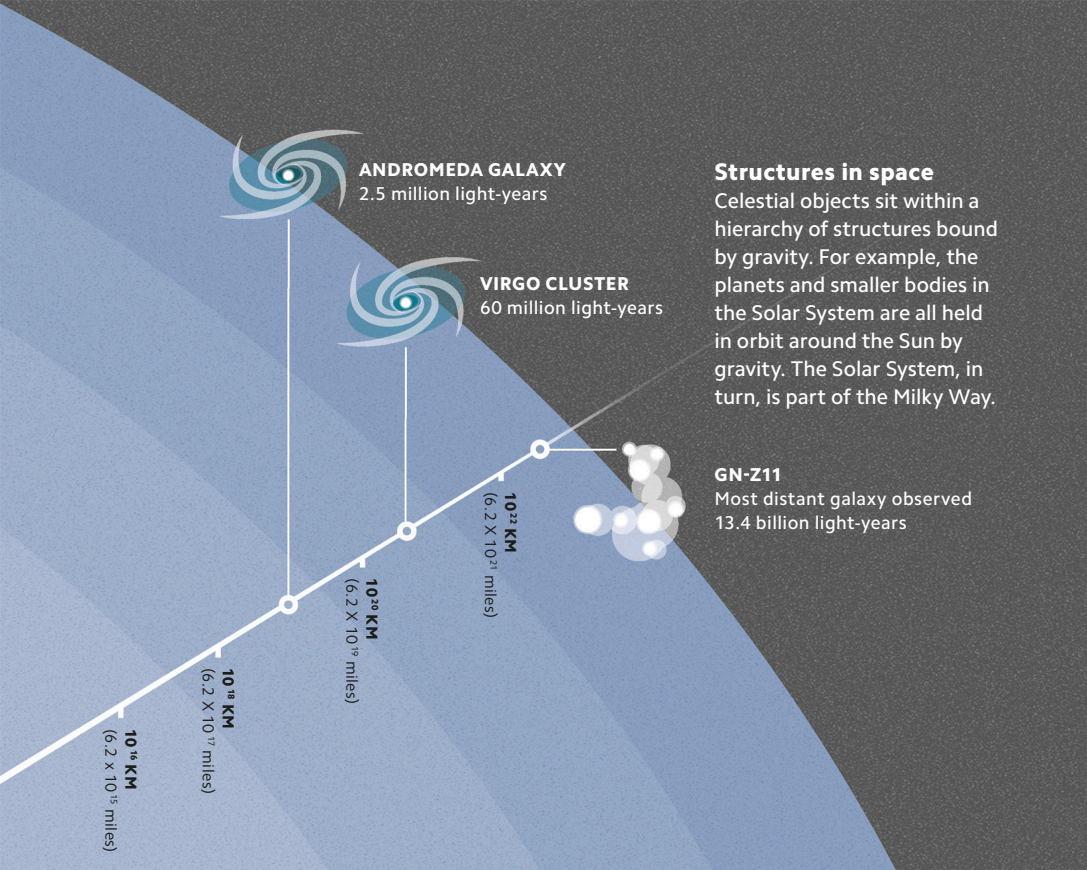
THE

UNIV

E R S E

The Universe is all matter, energy, space, and time. Its size alone can make it seem incomprehensibly vast and baffling. But even the Universe's largest structures are made of the same chemical elements that are around us on Earth – and indeed in our own bodies – and all atoms and subatomic particles are subject to the same fundamental forces. These particles and forces can be traced to the Universe's early history. The story of the Universe started with the Big Bang, when everything expanded from an unimaginably hot and infinitely small, dense point. Ever since, it has been cooling, expanding, and evolving as its original materials have been scattered and recycled across space and time.


EDGE OF THE LOCAL GROUP
3.25 MILLION LIGHT-YEARS
EDGE OF THE MILKY WAY
70,000 LIGHT-YEARS
PROXIMA CENTAURI
Nearest star, 4.2 light-years
EDGE OF THE SOLAR SYSTEM
1 LIGHT-YEAR
Distance in space
To fit the Universe into this graphic, each step on the scale represents a distance 100 times greater than the previous step. Many of the distances are in light-years. A light-year is the distance travelled by light in a year, or about 9.5 trillion km (5.9 trillion miles).
OUTER EDGE OF THE OORT CLOUD
About 1 light-year, 9.5 trillion km (5.9 trillion miles)
OUTER EDGE OF THE KUIPER BELT
10 billion km (6.2 billion miles)
THE SUN
150 million km (93 million miles)
VENUS
At least 42 million km (26 million miles)
THE MOON
384,000 km (239,000 miles)
EARTH
10,000 KM (6,200 miles)
1,000,000 KM (620,000 miles)
100 MILLION KM (62 million miles)
10 BILLION KM (6.2 billion miles)
1 TRILLION KM (620 billion miles)
100 TRILLION KM (62 trillion miles)



Structures in space
Celestial objects sit within a hierarchy of structures bound by gravity. For example, the planets and smaller bodies in the Solar System are all held in orbit around the Sun by gravity. The Solar System, in turn, is part of the Milky Way.

MAKING SENSE OF SPACE

The Universe is everything that exists. We do not know how big it is, partly because we cannot see it all but even the part we can see is vast. Light from its edge, moving at 299,792 km (186,282 miles) per second, has taken 13.8 billion years to reach us. The Universe also includes a huge amount of matter. The Sun is one of up to 400 billion stars in our galaxy – and there could be 100 billion galaxies in the Universe.

EDGE OF THE LANIAKEA SUPERCLUSTER
250 MILLION LIGHT-YEARS

EDGE OF THE OBSERVABLE UNIVERSE
13.8 BILLION LIGHT-YEARS

The Solar System

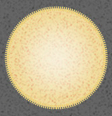
The Solar System is the part of space influenced by the Sun's gravity. The objects orbiting the Sun include rocky planets and larger giant planets, their moons, and countless asteroids and comets.

Gas and dust

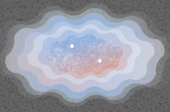
Over 90 per cent of all the Universe's ordinary matter is in the form of low-density gas between the stars and galaxies, most of it hydrogen. In galaxies, stars form in denser clouds of gas and dust.

Stars

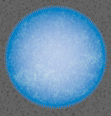
A star is a massive, hot ball of gas that emits radiation as a result of nuclear fusion taking place in its core. The Sun is an average star, but the Universe is also home to much smaller dwarf stars and larger giant stars.



THE SUN



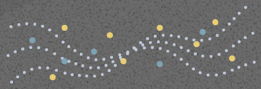
STAR-FORMING NEBULA



GIANT STAR



GIANT PLANET



INTERSTELLAR MEDIUM (GAS AND DUST)



SUNLIKE STAR



ROCKY PLANET



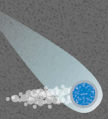
DWARF STAR



MOON



ASTEROID



COMET

CELESTIAL BODIES

The Universe consists of energy, space, and matter. Matter is anything that has mass – that is, anything affected by gravity. Most of the matter drifts through space as gas. But some of it clumps together to form objects such as stars and planets. Not all of the Universe's matter is of the ordinary kind that we can see. In fact, most of it is invisible to us. Astronomers call this dark matter (see p.30).

Star remnants

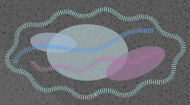
Stars do not shine forever. On using up their fuel, they leave behind an array of remnant objects. These include some of the most extreme objects in the Universe – neutron stars and black holes.

Galaxies

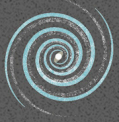
A galaxy is a large collection of stars (some orbited by planets), gas, and dust. Galaxies come in various shapes and sizes, containing from a few million to several trillion stars.

Groupings of galaxies

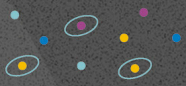
Most galaxies are part of groups (or clusters) held together by gravity. In turn, clusters are part of even larger superclusters. Galaxy superclusters are the largest structures bound by gravity.



SUPERNOVA REMNANT



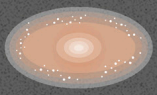
SPIRAL GALAXY



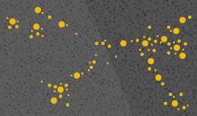
GALAXY CLUSTER



PLANETARY NEBULA



ELLIPTICAL GALAXY



GALAXY SUPERCLUSTER



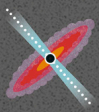
NEUTRON STAR



IRREGULAR GALAXY

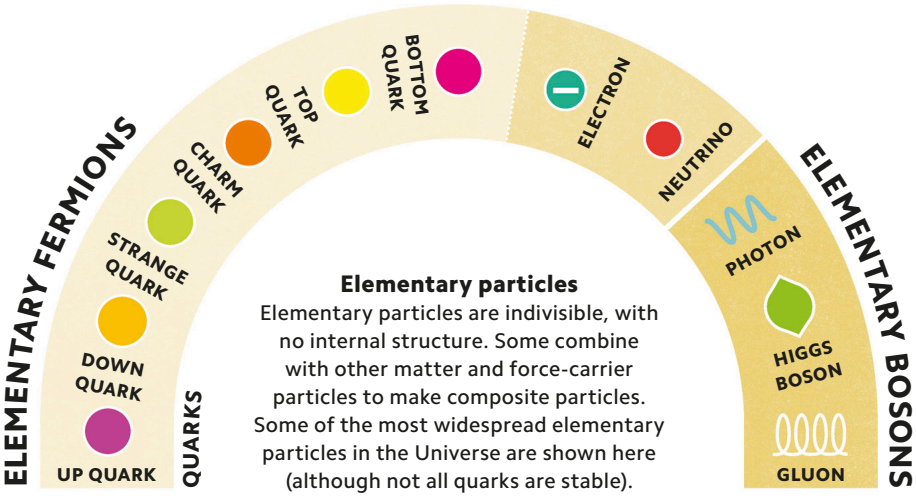


BLACK HOLE



ACTIVE GALAXY

There are about 2 trillion galaxies in the part of the Universe that we can observe.



Elementary particles
Elementary particles are indivisible, with no internal structure. Some combine with other matter and force-carrier particles to make composite particles. Some of the most widespread elementary particles in the Universe are shown here (although not all quarks are stable).

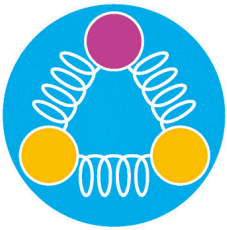
THE SMALLEST UNITS

Objects in the Universe take a vast array of forms – from the familiar surfaces of rocky planets, to intensely hot stars and infinitely dense black holes. Different though they are, these objects are all made up of the same extremely small units, known as particles. As well as particles that form matter (called fermions), there are other particles, known as bosons, that transfer forces between matter particles.



PROTON

Composite particles
Also known as hadrons, composite particles have an internal structure. They include protons and neutrons, which were once thought to be elementary but are now known to be made of quarks held together by gluons.



NEUTRON

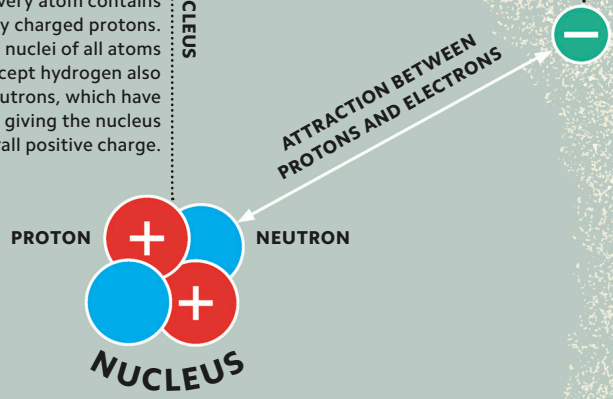
BUILDING BLOCKS

ELECTRONS

Negatively charged electrons move around in the space surrounding the nucleus. The region where they are likely to be found is called an electron cloud.

NUCLEUS

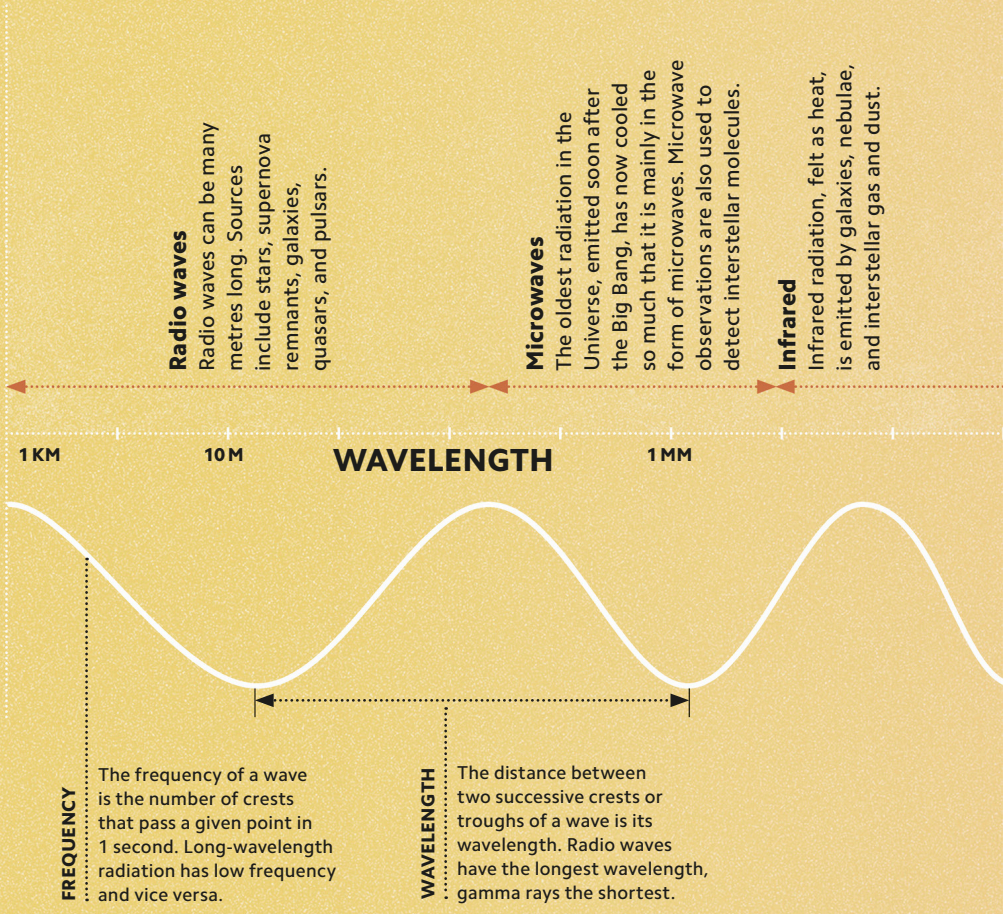
The nucleus at the centre of every atom contains positively charged protons. The nuclei of all atoms except hydrogen also contain neutrons, which have no charge, giving the nucleus an overall positive charge.



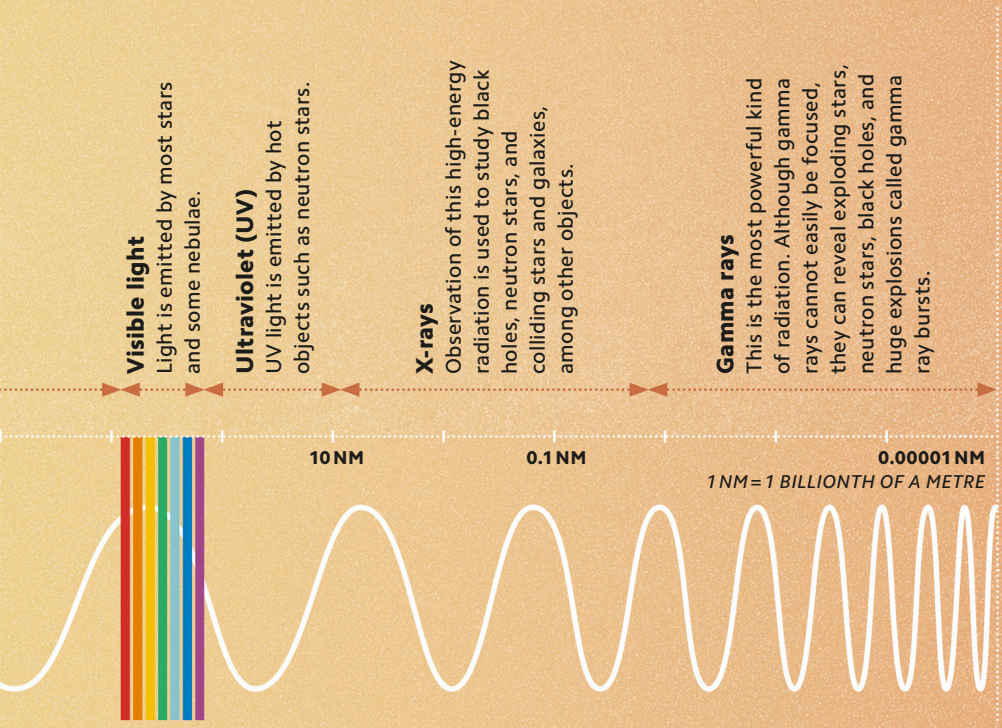
Most large-scale matter is built from atoms, made of protons, neutrons, and electrons. The protons' positive charge is balanced by the electrons' negative charge so an atom has no overall charge. But if it gains or loses electrons, it becomes a charged particle called an ion.

Atomic structure

In each element, the atomic nucleus contains a characteristic number of protons. A neutral helium atom has a nucleus with two protons and two neutrons, orbited by two electrons.



All forms of electromagnetic radiation travel through space at the speed of light.

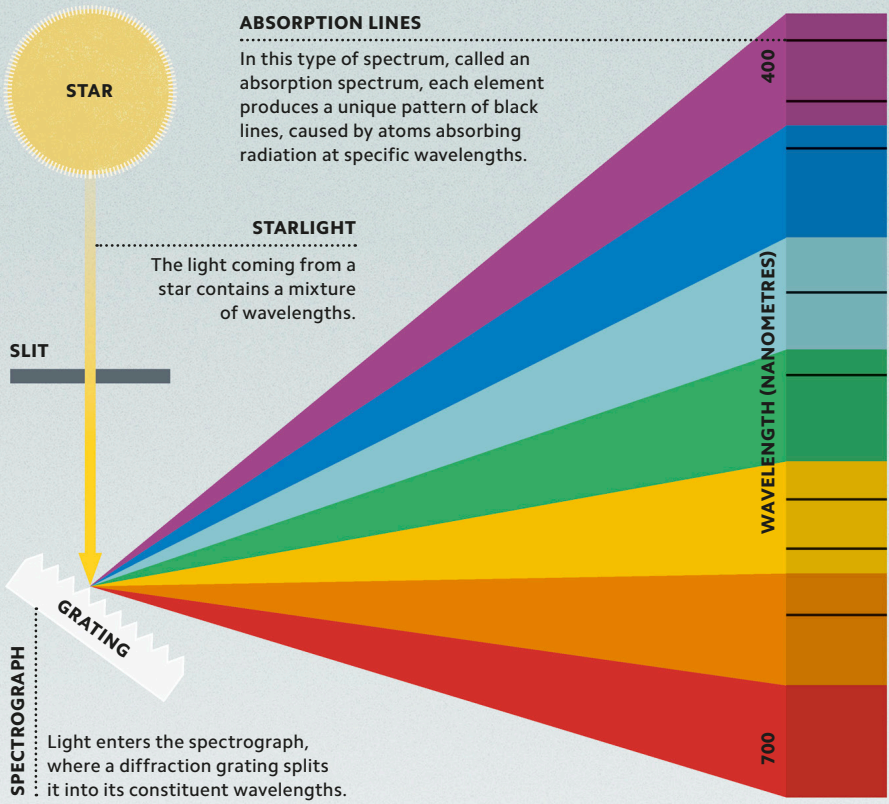


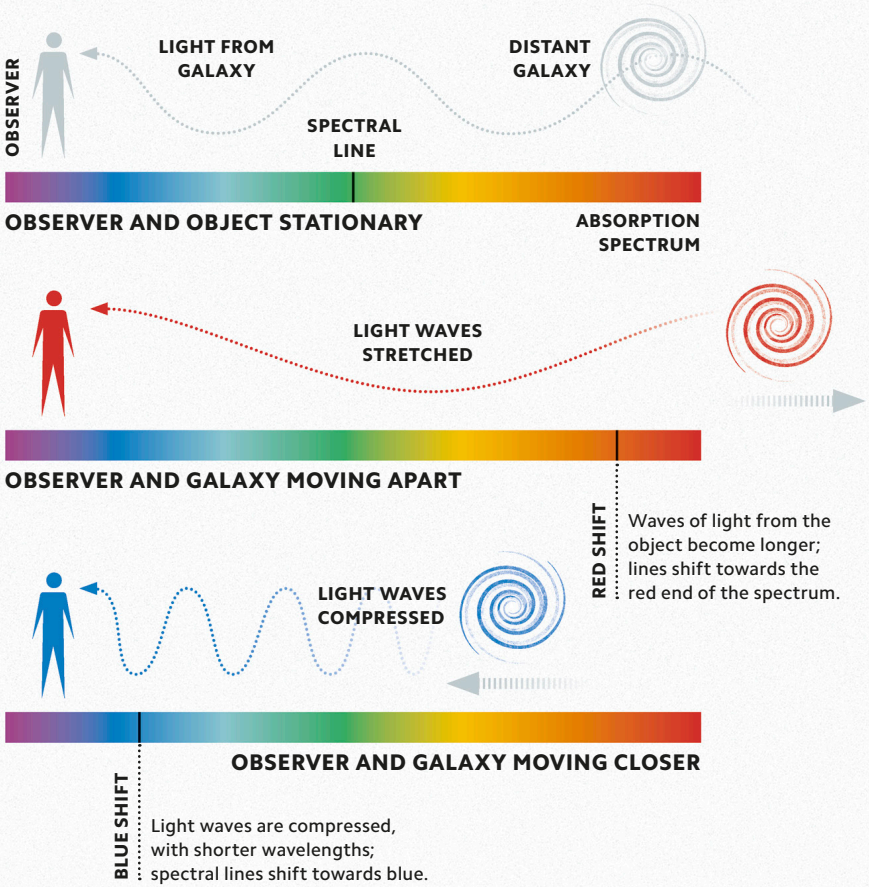
ACROSS THE SPECTRUM

Electromagnetic (EM) radiation is energy that travels in the form of waves. Along with matter, it is one of the main components of the Universe. All forms of ordinary matter emit EM radiation, which can travel through empty space as well as some types of matter. Visible light is the most familiar form of EM radiation. Other forms include infrared, radio waves, and X-rays. These differ in their wavelengths and the way they interact with matter. The full range of wavelengths is called the electromagnetic spectrum.

SPLITTING LIGHT

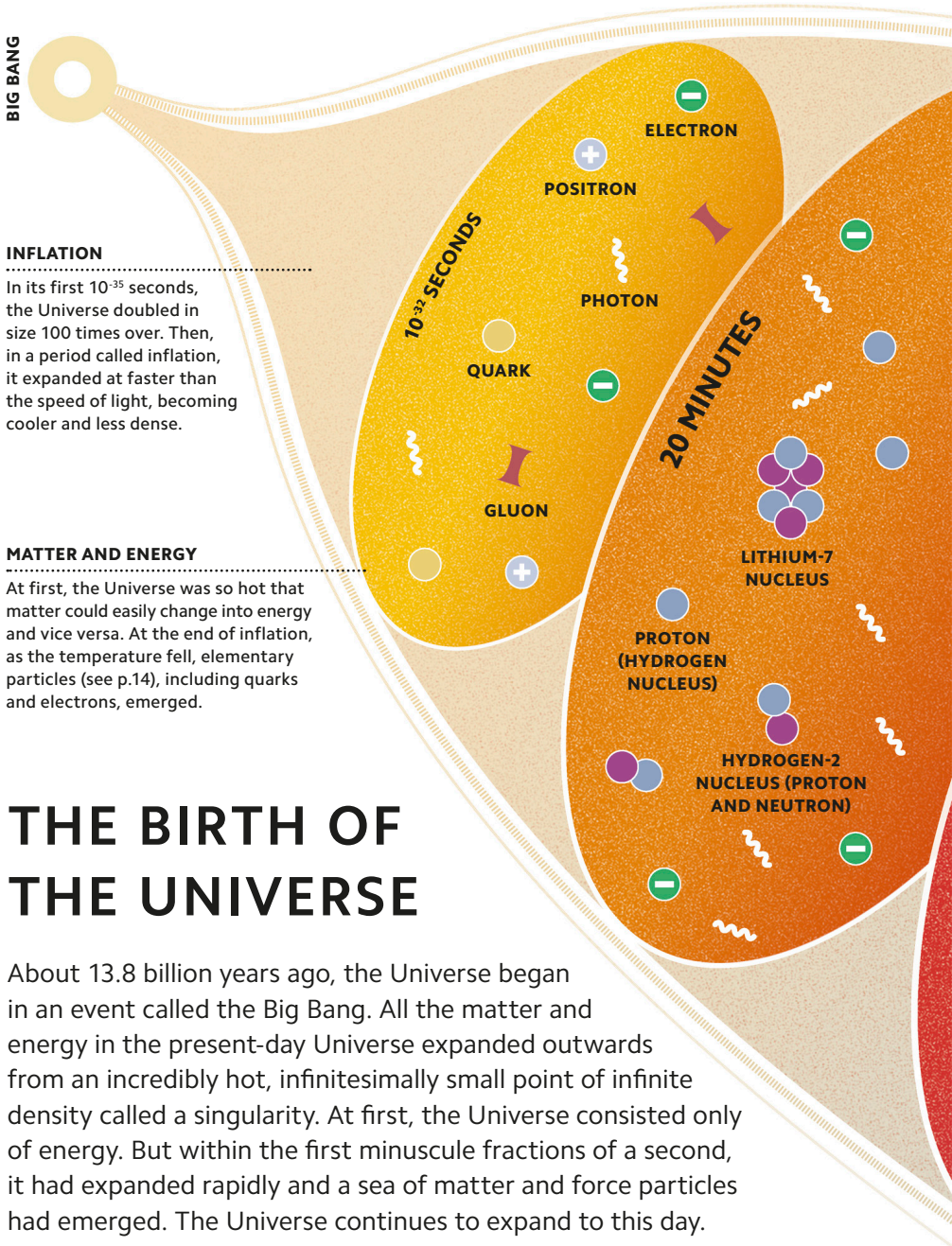
The atoms of each chemical element emit or absorb radiation at characteristic wavelengths (see pp.16–17). This means that by studying the light emitted or absorbed by an object, such as a star or a nebula, astronomers can work out what it is made of. The spectrum of radiation emitted by an object is like a fingerprint. Studying these "fingerprints" is known as spectroscopy, and the instruments used to generate and analyse them are called spectrographs.





SHIFTING LIGHT

The light emitted by an object can reveal not only what that object is made of but also how it is moving. When a source of light, such as a galaxy, and an observer are moving relative to each other, the observer receives light of a different colour compared with when they are not moving. The whole spectrum is shifted towards the red when the object is receding and towards the blue when it is approaching. The faster the relative motion, the greater the shift.



INFLATION

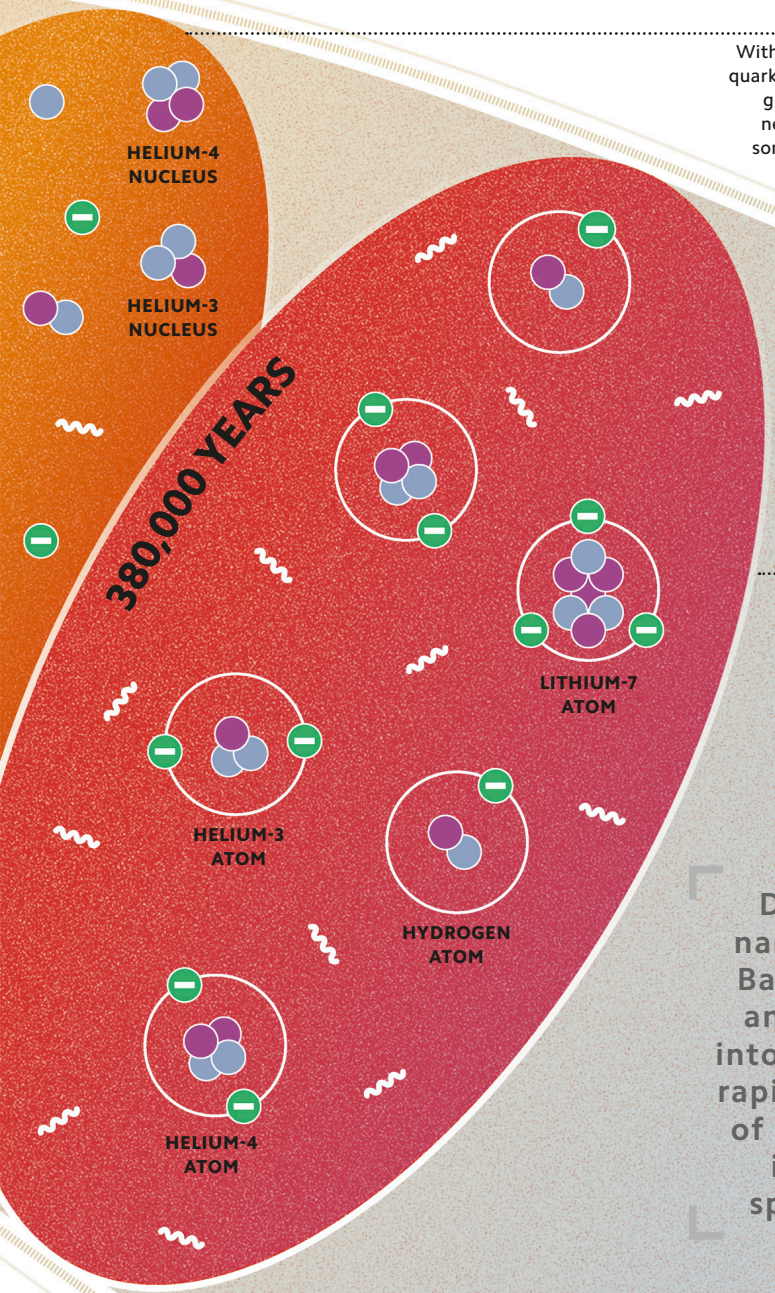
In its first 10^{-35} seconds, the Universe doubled in size 100 times over. Then, in a period called inflation, it expanded at faster than the speed of light, becoming cooler and less dense.

MATTER AND ENERGY

At first, the Universe was so hot that matter could easily change into energy and vice versa. At the end of inflation, as the temperature fell, elementary particles (see p.14), including quarks and electrons, emerged.

THE BIRTH OF THE UNIVERSE

About 13.8 billion years ago, the Universe began in an event called the Big Bang. All the matter and energy in the present-day Universe expanded outwards from an incredibly hot, infinitesimally small point of infinite density called a singularity. At first, the Universe consisted only of energy. But within the first minuscule fractions of a second, it had expanded rapidly and a sea of matter and force particles had emerged. The Universe continues to expand to this day.



ATOMIC NUCLEI FORM

Within a millionth of a second, quarks were bound together by gluons to form protons and neutrons. After 20 minutes, some of these had combined to form atomic nuclei.

THE FIRST ATOMS

For atoms to form, atomic nuclei have to combine with orbiting electrons. But the early Universe was too hot and dense for this to happen. It would be another 380,000 years before conditions were right.

Despite its name, the Big Bang was not an explosion into space but a rapid expansion of everything, including space itself.

FIRST LIGHT

In cosmic terms, the first stars formed just a short time after the Big Bang. These were huge stars that existed for a relatively short time and then exploded as supernovae. Before the Universe was half a billion years old, small galaxies had also emerged. They would go on to merge to form larger galaxies.

The early Universe consisted of atoms of ordinary matter amid another kind of matter known as dark matter (see p.30). This period is called the Cosmic Dark Age because there were no sources of light.

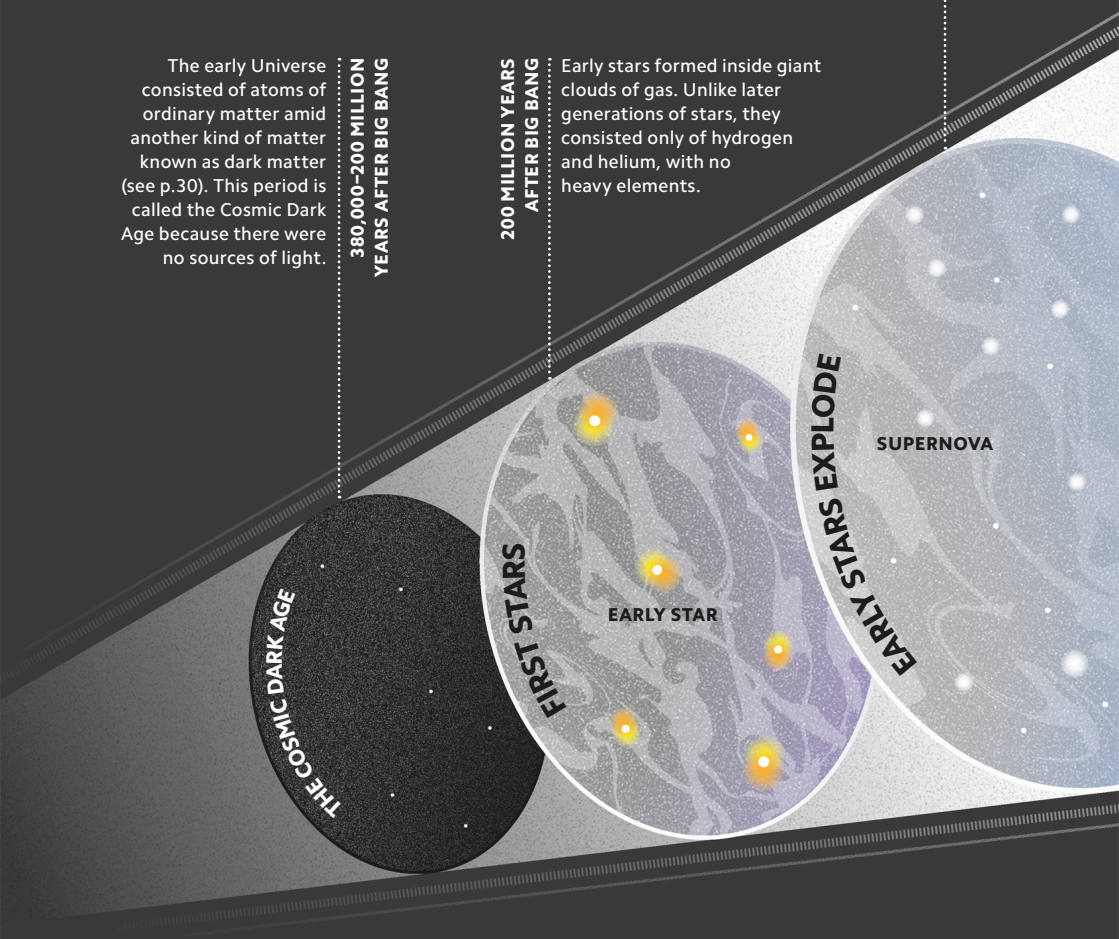
380,000–200 MILLION YEARS AFTER BIG BANG

200 MILLION YEARS AFTER BIG BANG

Early stars formed inside giant clouds of gas. Unlike later generations of stars, they consisted only of hydrogen and helium, with no heavy elements.

Compared with later generations of stars, the early stars consumed their fuel supplies relatively quickly and then exploded as supernovae.

300 MILLION YEARS AFTER BIG BANG



400 MILLION YEARS AFTER BIG BANG

Under the influence of dark matter, stars clumped together to form infant galaxies. These were relatively small structures, known as dwarf galaxies.

GALAXIES FORM

DWARF GALAXY

GALAXIES EVOLVE

MERGING DWARF GALAXIES

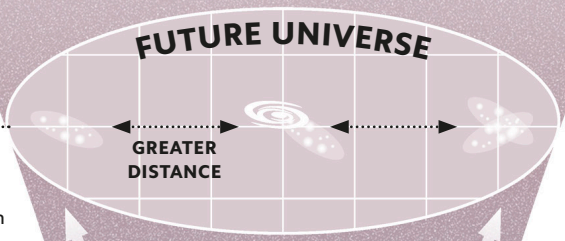
SPIRAL GALAXY

The formation of the first galaxies triggered waves of new star formation. Mergers between the early dwarf galaxies led to the formation of larger galaxies.

800 MILLION YEARS AFTER BIG BANG

SPACE IS GETTING BIGGER

Ever since the Big Bang, the Universe has been getting bigger. This is because space itself is expanding. On large scales, objects in space are getting further apart. This does not apply to objects attracted to each other by gravity, such as stars in a galaxy or even galaxies in a galaxy cluster, but galaxy clusters themselves are getting further apart. Not only are these far-off objects getting further away from us, but the more distant an object is, the faster it is moving away.



SPACE EXPANDS

As space expands, galaxy clusters become further apart. The space in between becomes increasingly empty.



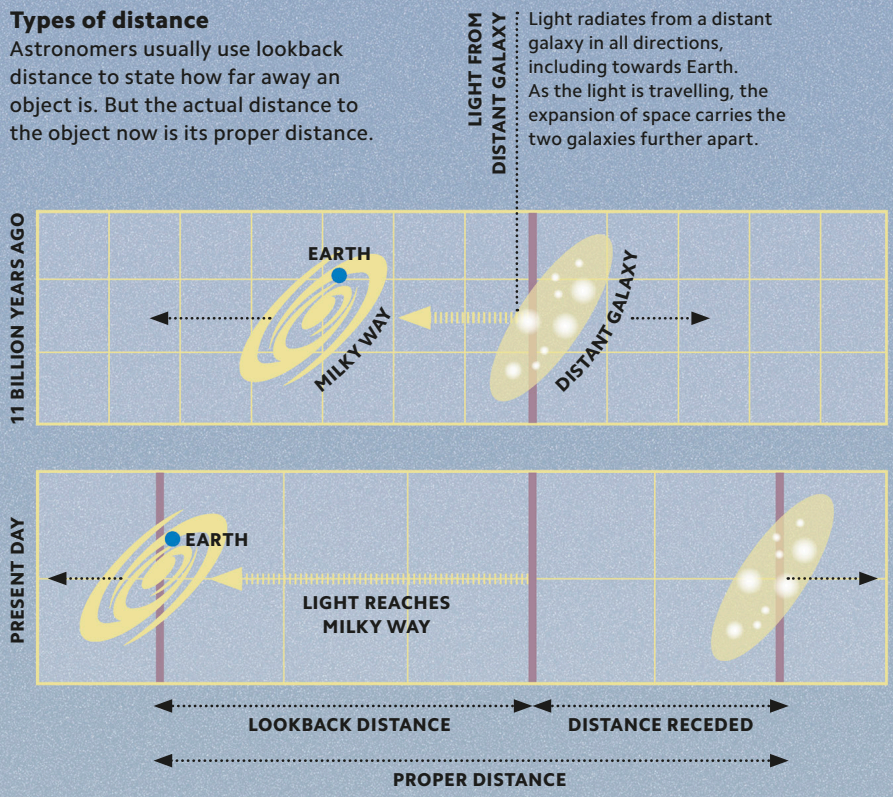
ACCELERATING EXPANSION

For about the last 7.5 billion years, the expansion of the Universe has been speeding up. This might be due to dark energy (see p.31), which acts in opposition to gravity.

How space expands

Galaxies and other objects are not getting larger, nor are they moving through space. Instead, space itself is expanding, carrying objects with it.

Types of distance
Astronomers usually use lookback distance to state how far away an object is. But the actual distance to the object now is its proper distance.

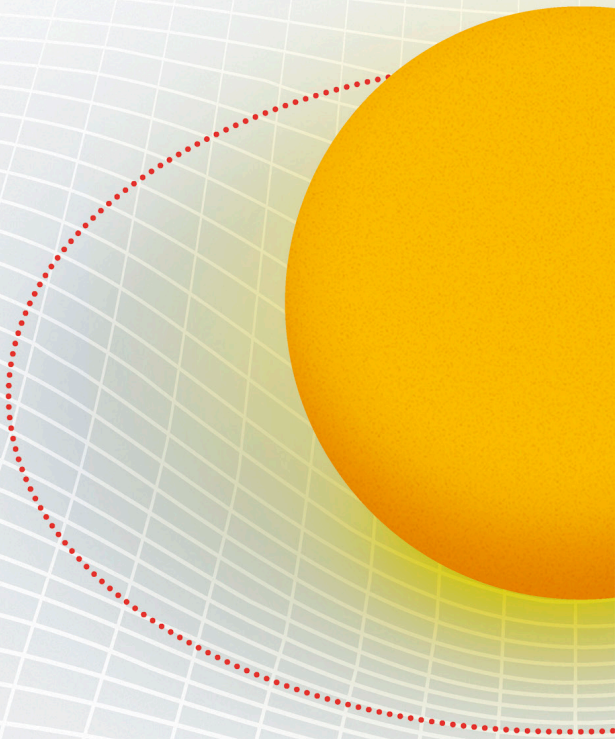


SEEING INTO THE PAST

Light travels incredibly fast, but it still takes time for light from an object in space to reach Earth. So when we look at a celestial object, we see it as it was in the past. The further we look into space, the further we see into the past. When talking about the distance to these remote objects, we have to remember that space is continually expanding. The distance light has travelled from an object to reach us is called the lookback distance. But the current actual distance to the same object is greater and is called the proper distance.

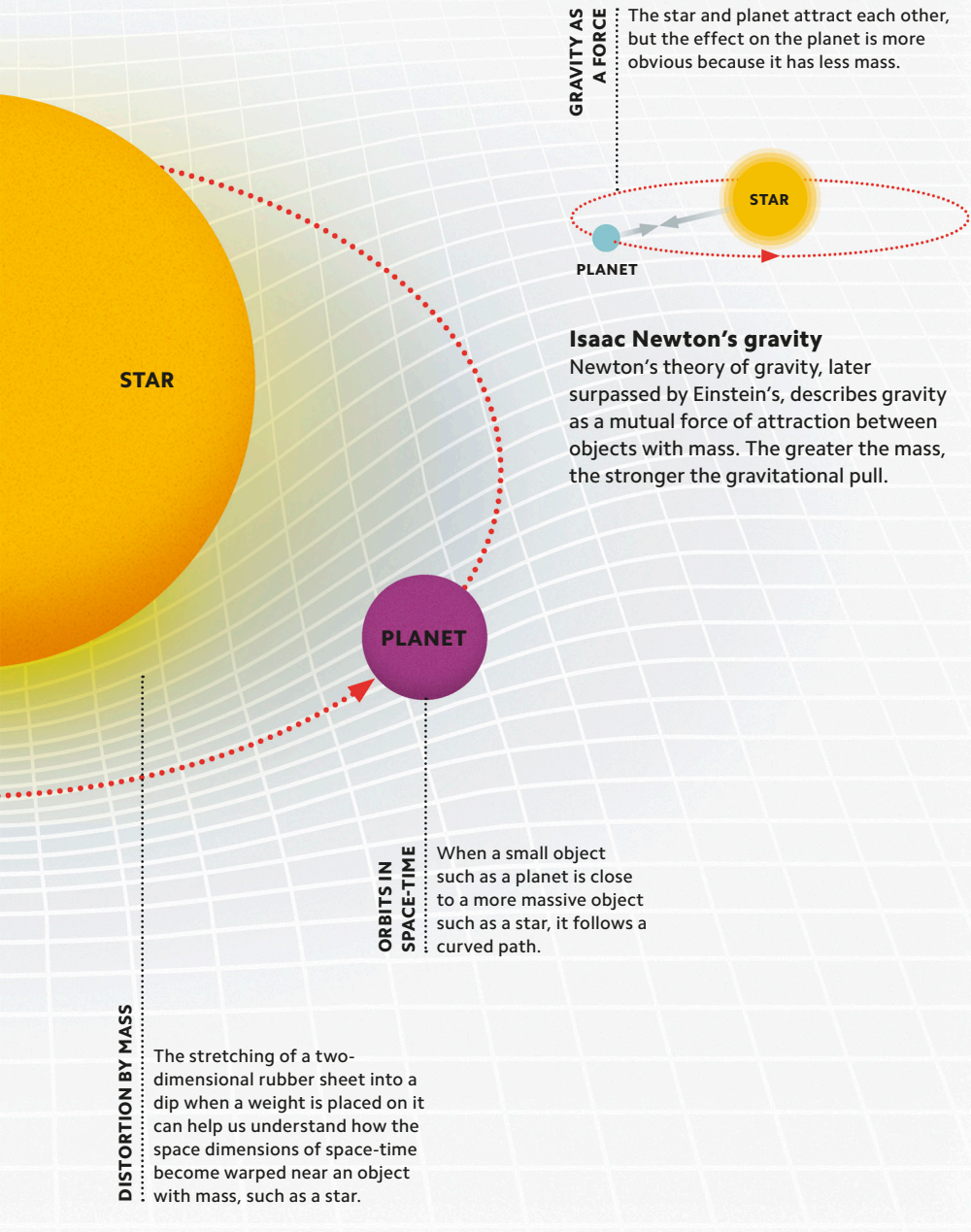
Einstein's gravity
Albert Einstein proposed that gravity is a distortion of space-time. Here, the three dimensions of space are represented by a two-dimensional, stretchable rubber sheet.

SPACE-TIME GRID
An imaginary grid of squares makes it easier to see the distortion of the rubber sheet in this model.

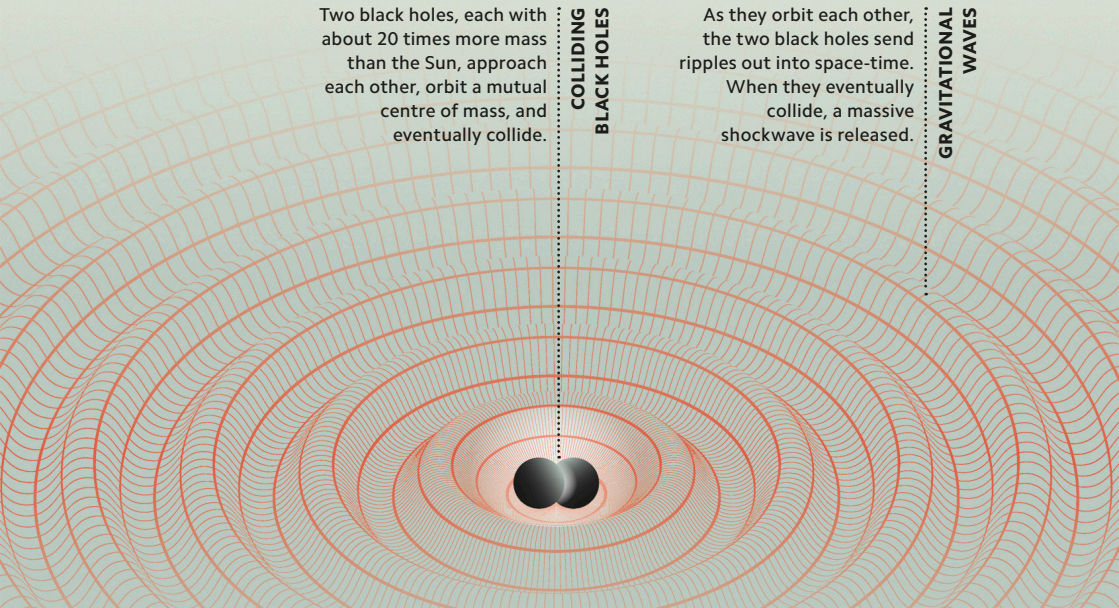


WARPED SPACE AND TIME

In the early 20th century, Albert Einstein and others realized that space and time are not separate entities but are, in fact, closely linked. The three dimensions of space and the single dimension of time can be combined into a four-dimensional realm called space-time. This idea has important implications for how objects move through time and space because mass and energy distort space-time. According to Einstein's general theory of relativity, gravity is the effect that these distortions have on the path that an object takes through space-time.

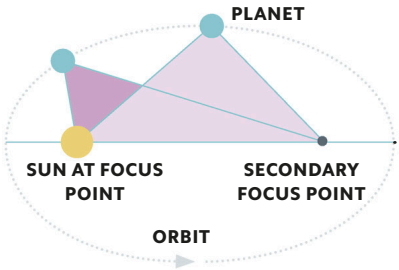

GRAVITY AS A FORCE
The star and planet attract each other, but the effect on the planet is more obvious because it has less mass.
STAR
PLANET
Isaac Newton's gravity
Newton's theory of gravity, later surpassed by Einstein's, describes gravity as a mutual force of attraction between objects with mass. The greater the mass, the stronger the gravitational pull.
STAR
PLANET
ORBITS IN SPACE-TIME
When a small object such as a planet is close to a more massive object such as a star, it follows a curved path.
DISTORTION BY MASS
The stretching of a two-dimensional rubber sheet into a dip when a weight is placed on it can help us understand how the space dimensions of space-time become warped near an object with mass, such as a star.

When an object with mass accelerates (undergoes a change in its velocity), it sends out ripples that travel through space-time (see pp.26–27) at the speed of light. These ripples are called gravitational waves. Astronomers detected a gravitational wave for the first time in 2015, picking up the disturbance caused by a collision between two black holes. Other energetic events, such as supernovae and collisions between neutron stars, might also generate detectable waves.



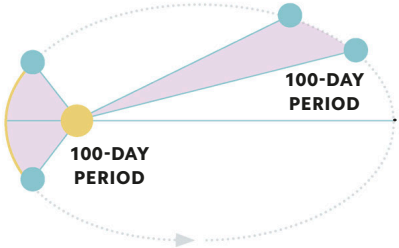
RIPPLES IN SPACE-TIME

The first gravitational waves were detected exactly 100 years after they were predicted by Albert Einstein.



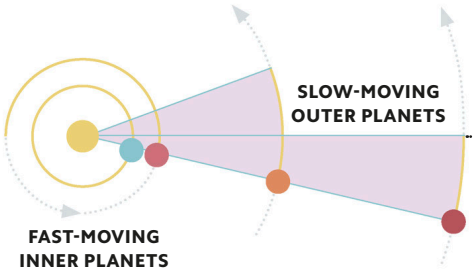
KEPLER'S FIRST LAW

The orbit of a planet is an ellipse with two focus points. The Sun sits at one focus. The planet is always at the same combined distance from these two points.



KEPLER'S SECOND LAW

A planet speeds up as it gets closer to the Sun. An imaginary line from a planet to the Sun sweeps out equal areas in equal periods of time.



KEPLER'S THIRD LAW

Planets take longer to complete an orbit the further they are from the Sun. Kepler's third law links a planet's orbital period to the size of its orbit.

BODIES IN MOTION

The path of a body moving under the gravitational influence of another is called an orbit. The less massive object follows a curved path, constantly changing its direction and speed. The smaller body does not simply orbit the larger. Instead, both move around a point called their joint centre of mass. Where one object orbits another in a closed system (like a planet in the Solar System), the two bodies follow three laws discovered by the astronomer Johannes Kepler.

Dark haloes
The way that most galaxies rotate cannot be explained by the amounts of ordinary matter we can detect. Models of how they form and rotate suggest that galaxies, including the Milky Way, are surrounded by haloes of dark matter.



Hidden matter
The unseen Universe consists partly of dark matter but mostly of dark energy (see opposite).

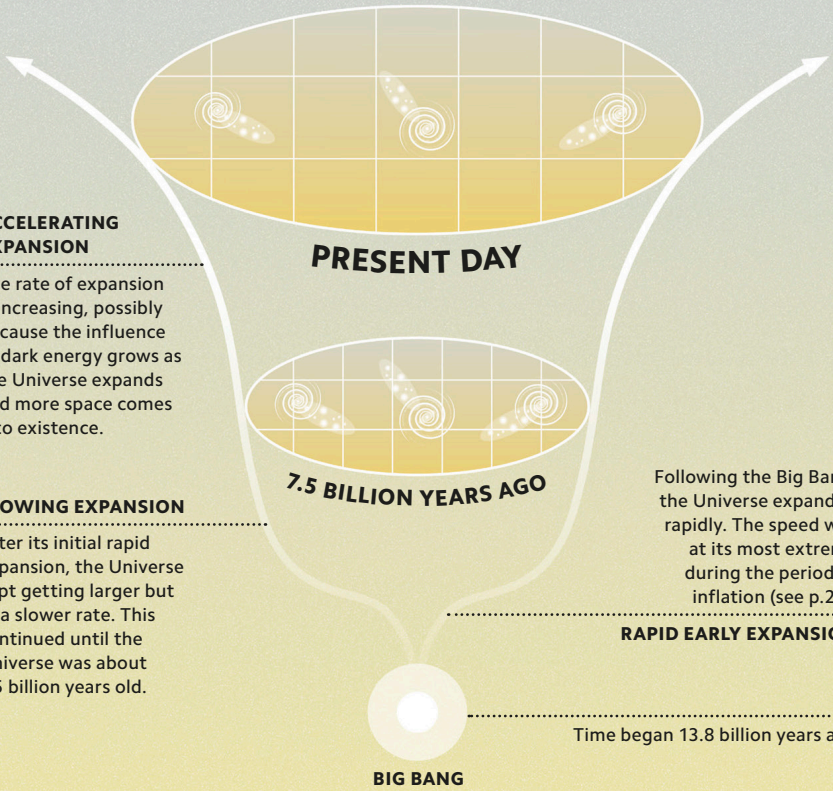
KEY
- **ORDINARY MATTER 4.9%**
- **DARK MATTER 26.8%**
- **DARK ENERGY 68.3%**

THE UNSEEN UNIVERSE

We see objects such as stars and planets because they emit or reflect light or other radiation. But the ordinary matter in these objects accounts for just 5 per cent of all matter. About 25 per cent is matter in an invisible form called dark matter. We know it is there only because its gravity affects ordinary matter. Dark matter could be ordinary matter that does not emit much radiation. But it seems more likely to be made of subatomic particles that do not interact with radiation at all.

THE ENERGY OF SPACE

Most of the mass and energy in the Universe is in the form of a mysterious invisible substance or force known as dark energy that is causing the expansion of the Universe to accelerate. A possible explanation is that space is not simply "nothing" but instead has intrinsic properties that oppose gravity and stop the Universe from contracting. We do not experience the effects of dark energy because, at the scales most familiar to us, gravity binds matter together.



THESSYST

O L A R

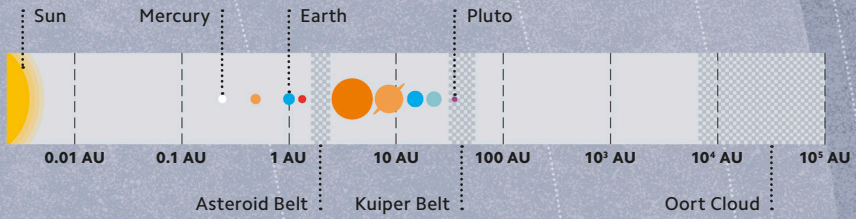
E M

The Sun, our nearest star, dominates the Solar System. Its gravitational influence extends over a vast region of space that includes eight major planets, more than 200 moons, and countless other small bodies, including dwarf planets, asteroids, and comets. As well as holding these objects in their orbits, the Sun is also their primary source of energy. As our neighbourhood, the Solar System is the corner of the Universe that we know best. We can see the Moon's craters and plains and several planets without the help of binoculars or telescopes. All the planets and some of the smaller bodies have also now been visited by spacecraft, which have sent back data, pictures, and even some physical samples.

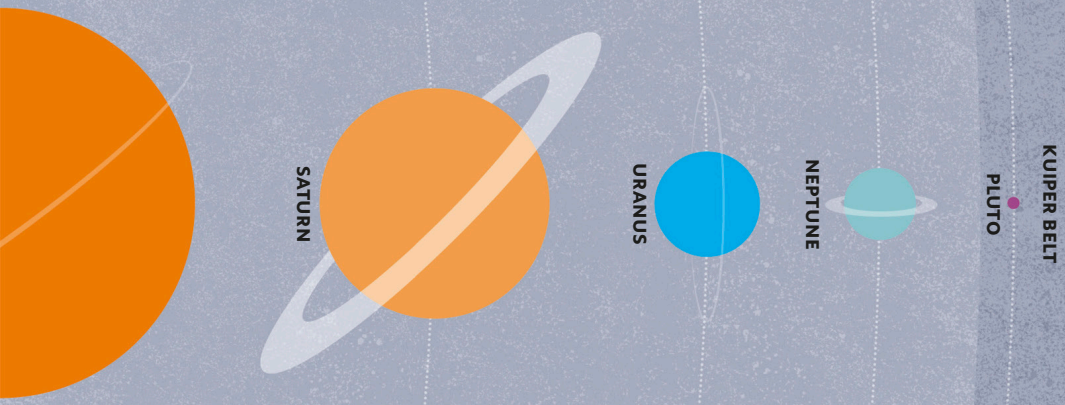
THE FAMILY OF THE SUN

The Sun makes up 99.8 per cent of the Solar System's mass.

Energy source
All planets in our Solar System orbit the Sun. The closer each planet is to the Sun, the more energy and heat it receives.

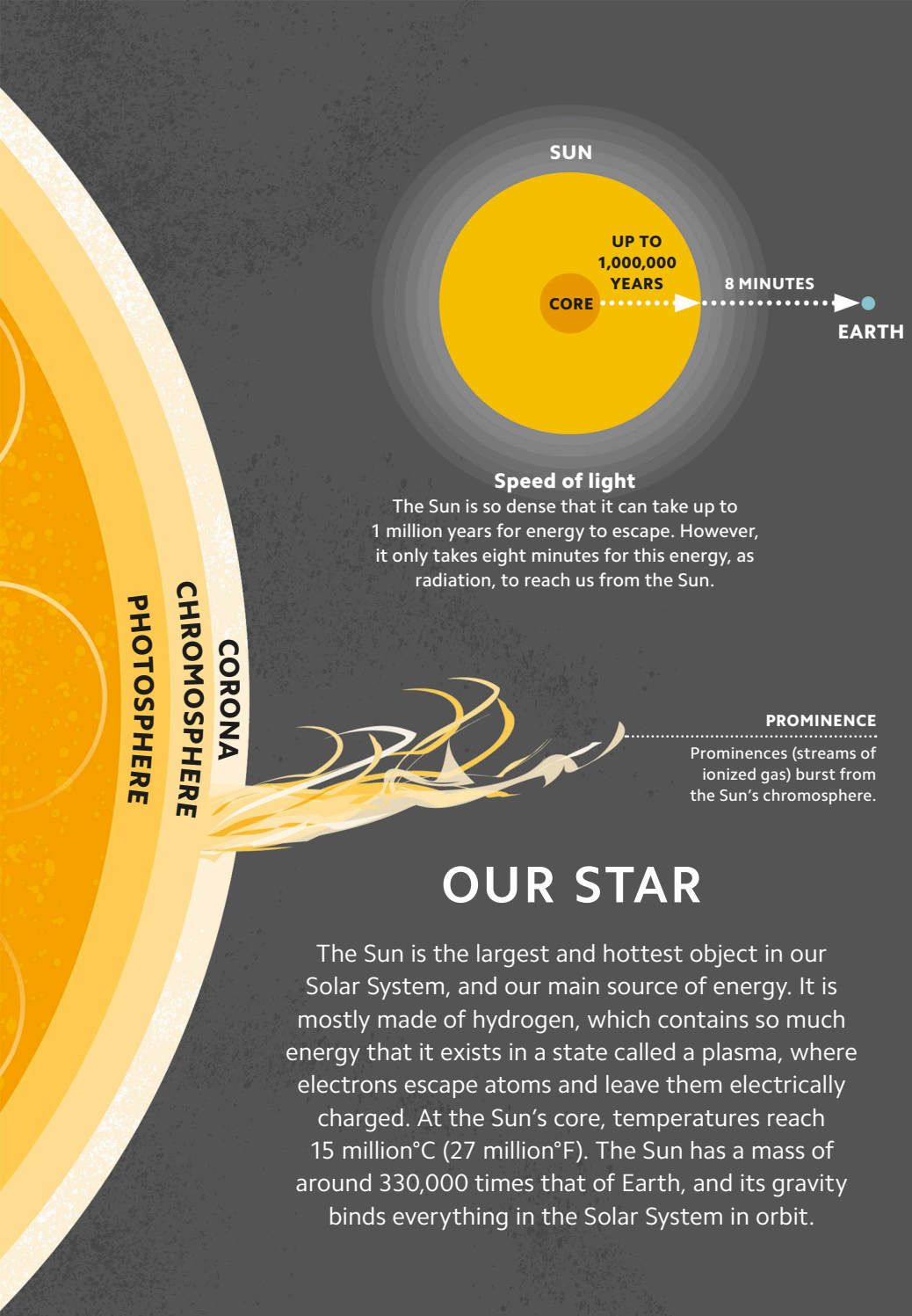


Solar System distances
Earth is 150 million km (93 million miles), or one astronomical unit (AU), from the Sun. On this graphic, the distance markers increase by a factor of 10 with each step away from the Sun.



The Solar System is a family of objects held in position by the gravitational attraction of the Sun. Each planet orbits the Sun at a different speed. Nearest the Sun are the four rocky planets – Mercury, Venus, Earth, and Mars – followed by the Asteroid Belt, an area filled with tens of thousands of rocky objects. The giant planets – Jupiter, Saturn, Uranus, and Neptune – orbit further out from the Sun. Beyond Neptune is the Kuiper Belt, a doughnut-shaped disc of icy objects. Further still, more icy objects are found in the Oort Cloud.


CONVECTIVE CURRENTS
In a repeating cycle, hot gas rises, transfers energy to the surface of the Sun, then cools and sinks down.
Internal structure
Intense heat and pressure in the Sun's core trigger nuclear fusion reactions. This generates energy, which slowly travels through the dense radiative zone before being transferred, via circulating pockets of gas, through the convective zone towards the photosphere.
CORE
RADIATIVE ZONE
CONVECTIVE ZONE
The Sun is 1.4 million km (870,000 miles) wide, meaning that over a million Earths could fit inside it.
The photosphere, the first of three layers comprising the Sun's atmosphere, is transparent to light, creating the illusion of a solid surface.
ATMOSPHERE



Speed of light

The Sun is so dense that it can take up to 1 million years for energy to escape. However, it only takes eight minutes for this energy, as radiation, to reach us from the Sun.

PROMINENCE

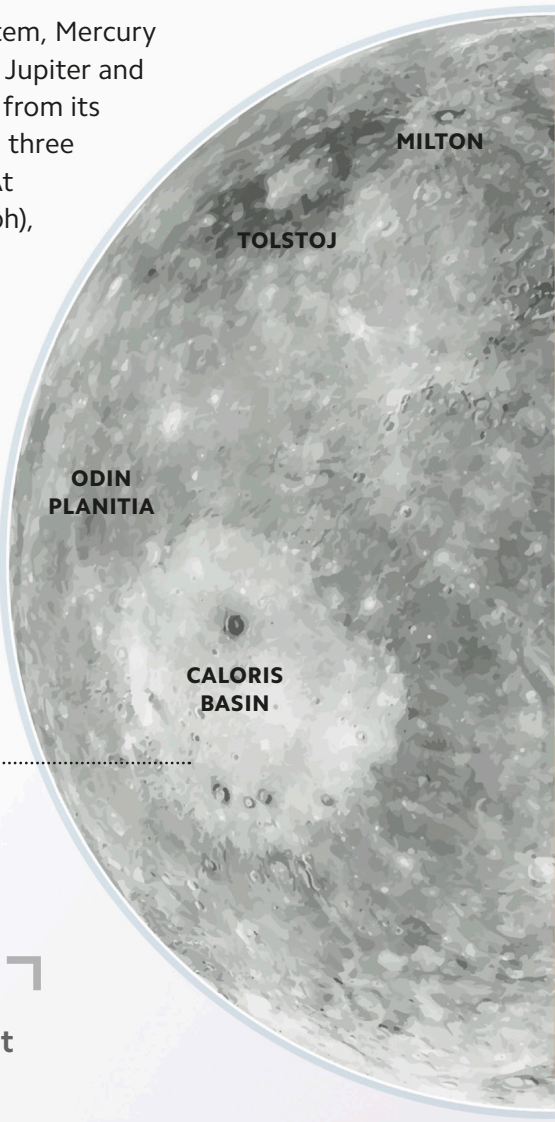
Prominences (streams of ionized gas) burst from the Sun's chromosphere.

OUR STAR

The Sun is the largest and hottest object in our Solar System, and our main source of energy. It is mostly made of hydrogen, which contains so much energy that it exists in a state called a plasma, where electrons escape atoms and leave them electrically charged. At the Sun's core, temperatures reach 15 million°C (27 million°F). The Sun has a mass of around 330,000 times that of Earth, and its gravity binds everything in the Solar System in orbit.

THE SWIFT PLANET

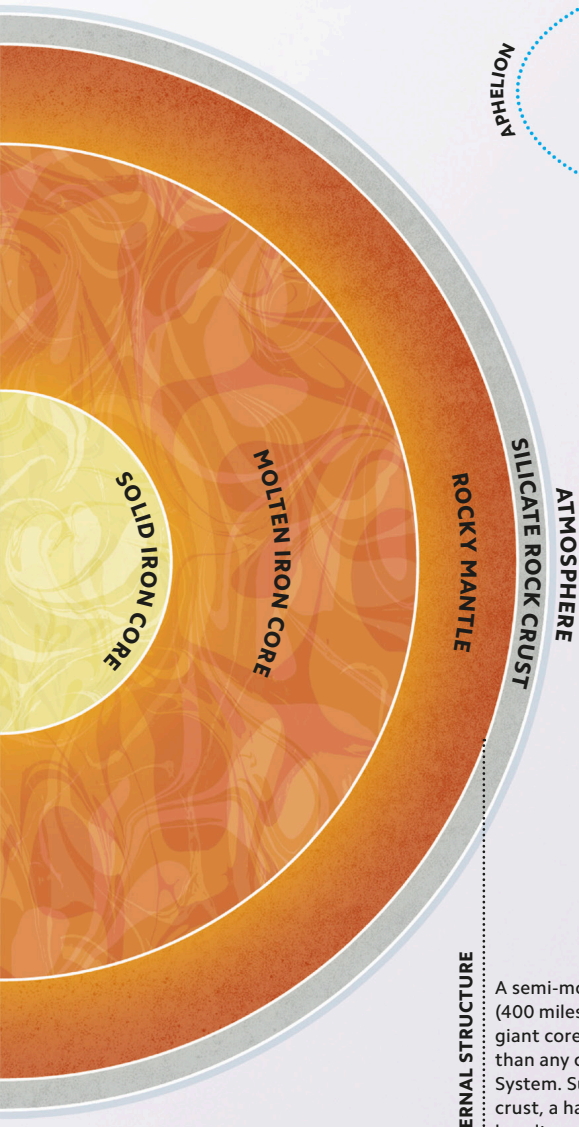
The smallest planet in the Solar System, Mercury is smaller than the largest moons of Jupiter and Saturn. It is so close to the Sun that from its surface, the Sun would seem two to three times larger than it does on Earth. At more than 170,000 kph (106,000 mph), Mercury has the fastest orbit of any planet. However, its slow rate of rotation means that a day on Mercury lasts twice as long as a year. This, along with Mercury's lack of atmosphere, results in the greatest temperature variation of any planet, with highs of up to 430°C (806°F) in the day and lows as cold as -180°C (-292°F) at night.



VAST CRATER

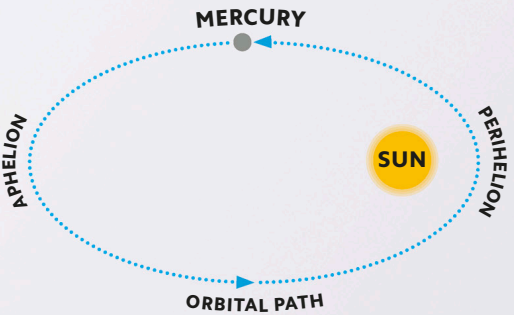
At over 1,500 km (930 miles) across, the Caloris Basin is one of the Solar System's largest impact craters. It is encircled by a ring of mountains 2 km (1.2 miles) high.

A year on Mercury – the amount of time the planet takes to fully orbit the Sun – lasts just 88 days.



INTERNAL STRUCTURE

A semi-molten rock mantle 600 km (400 miles) thick surrounds Mercury's giant core, which boasts more iron than any other planet in the Solar System. Surrounding this is Mercury's crust, a harsh landscape made of basalt and other silicate rocks, and dotted with impact craters.



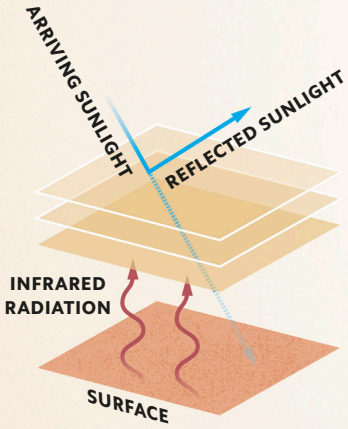
Eccentric orbit

Mercury has the most eccentric (elongated) orbit of all the planets. At perihelion, its closest point to the Sun, Mercury is 47 million km (29 million miles) from the Sun, while at aphelion, its furthest point, it is 70 million km (43 million miles) away.

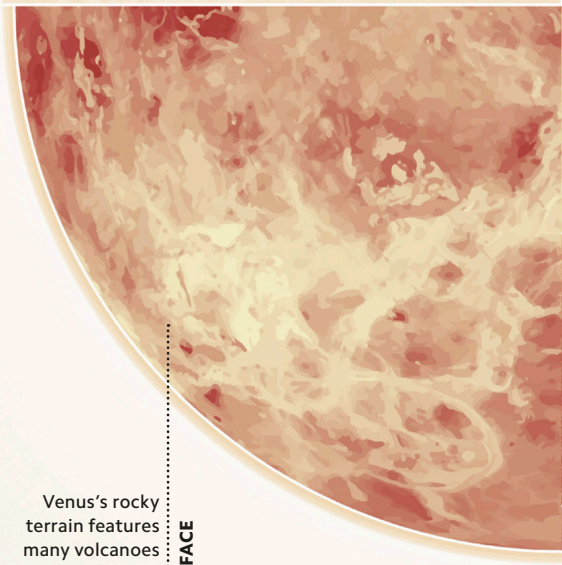
> "There are so many stars shining in the sky ... but when Venus comes out, all the others are waned ..."
>
> Galileo Galilei

Internal structure
With an iron core and a hot, rocky mantle, Venus is similar in structure to Earth and only 650 km (400 miles) smaller in diameter. The two also share a rocky surface.

CLOUD DECK
A thick layer of cloud obscures the surface of Venus.



Greenhouse effect
Only 35 per cent of the sunlight that hits Venus breaks through its clouds. The thick atmosphere prevents any rising radiation from escaping, so heating the planet.

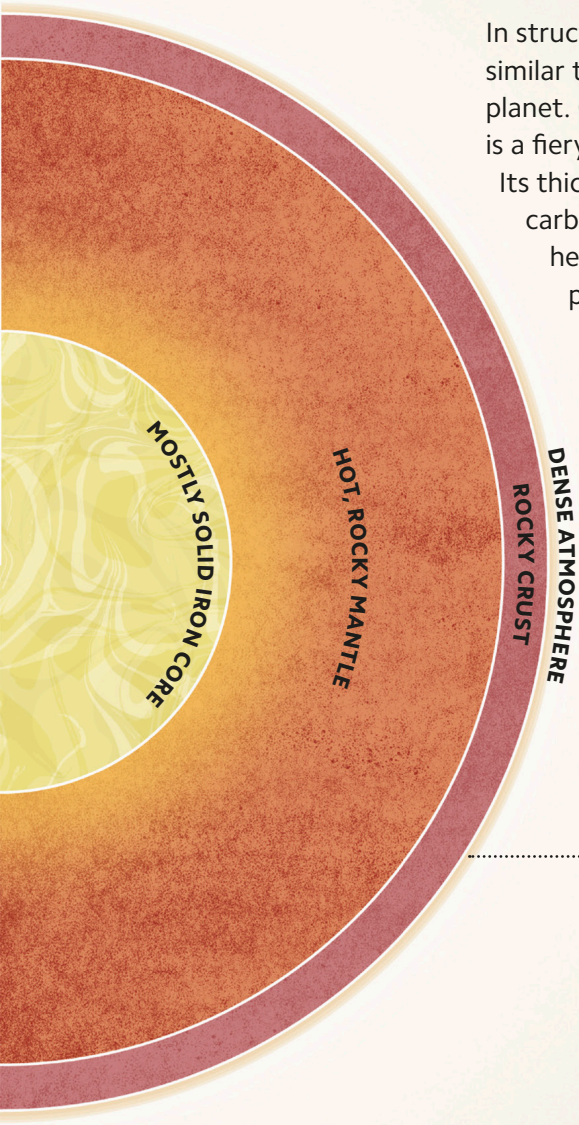


Venus's rocky terrain features many volcanoes and craters, and vast lava plains.

SURFACE

EARTH'S SISTER

In structure and size, Venus is strikingly similar to Earth, its neighbour and "twin" planet. On the ground, however, Venus is a fiery, waterless, and harsh terrain. Its thick atmosphere, at 96.5 per cent carbon dioxide, traps the nearby Sun's heat. This makes Venus the hottest planet in the Solar System, with surface temperatures as high as 471°C (880°F) – hot enough to melt lead. The planet's surface is also littered with volcanoes no more than 500 million years old, a few of which may be active or just dormant.

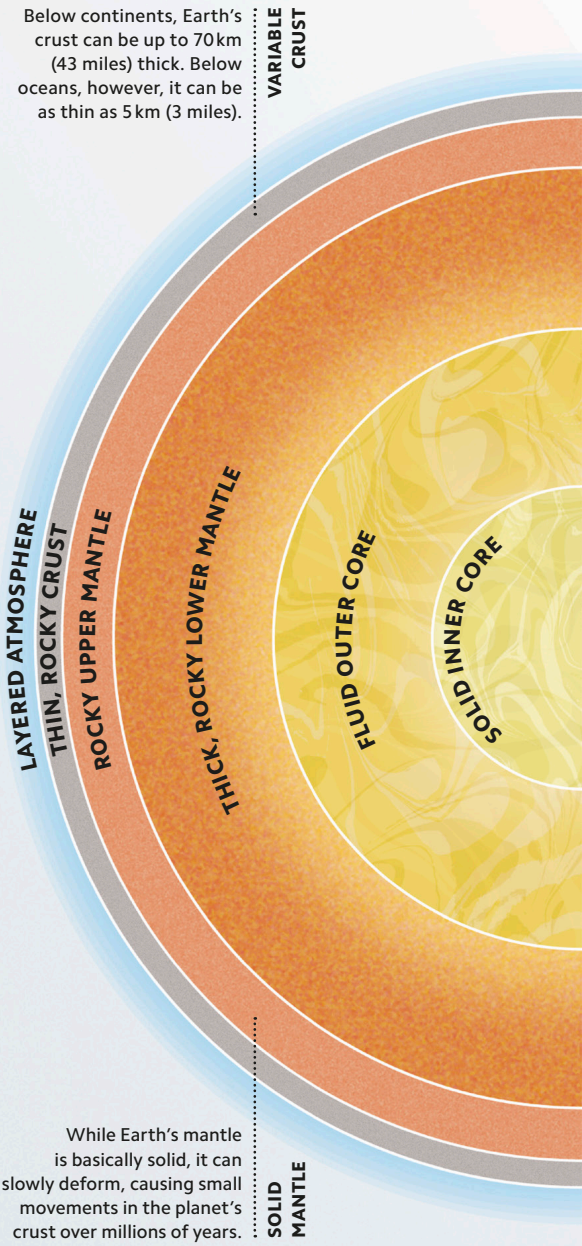


ATMOSPHERE

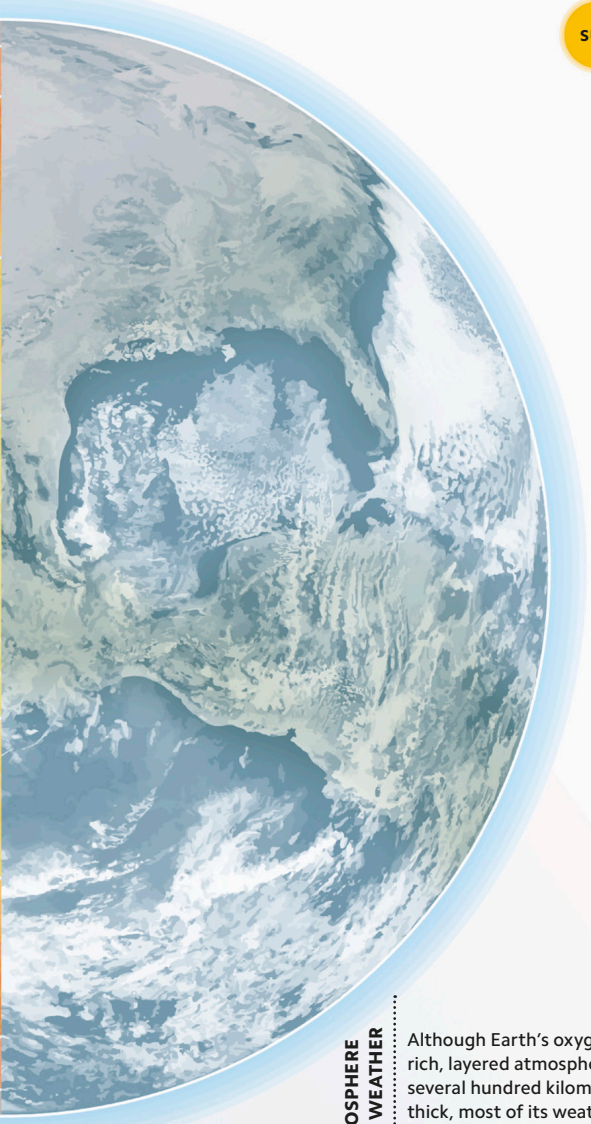
The lower layer of Venus's atmosphere is dense, thick, and extremely hot. Above it, the planet's cloud deck extends 32–90 km (20–55 miles) above its surface. From there, the air gradually thins out into space.

HOME WORLD

Situated between Venus and Mars, Earth is the only planet in the Solar System known to host life. This is largely possible because of the fine balance between Earth's distance from the Sun and the planet's moderately thick atmosphere, along with an abundance of water. Five huge oceans cover 69 per cent of Earth's total surface, and icy polar caps a further 2 per cent, while the rest is land. Earth is a dynamic planet, with slow-moving tectonic plates that, on colliding with each other, give rise to features such as mountain ranges, volcanoes, and deep-sea trenches.

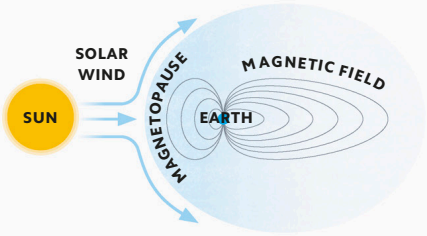


Internal structure
Earth's iron-nickel inner core is encased in a liquid-iron outer core, a mostly solid mantle, and a crust made up of many different types of rock.



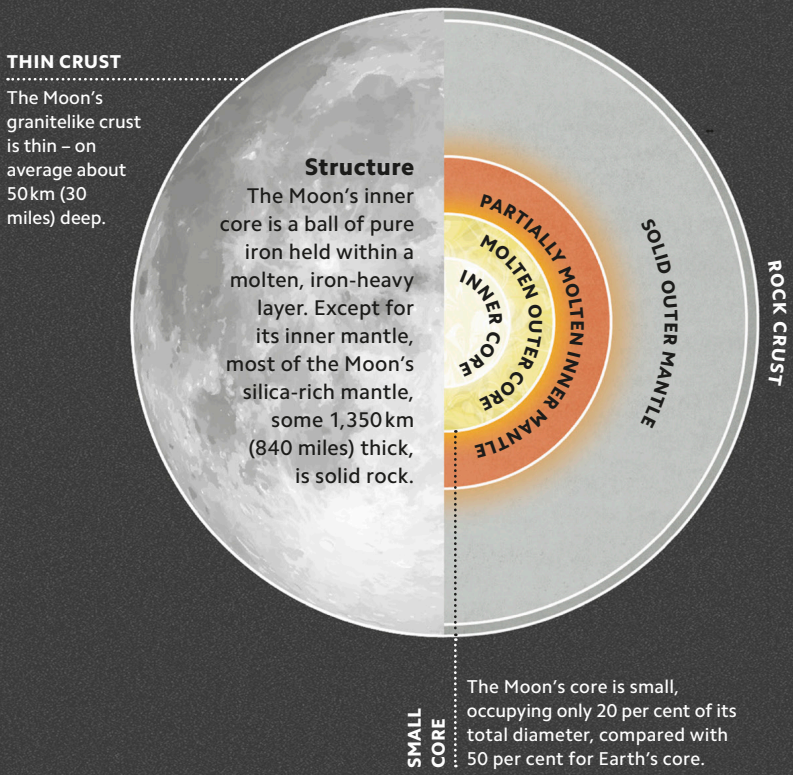
ATMOSPHERE AND WEATHER

Although Earth's oxygen-rich, layered atmosphere is several hundred kilometres thick, most of its weather systems are found in the lowest 10 km (6 miles).



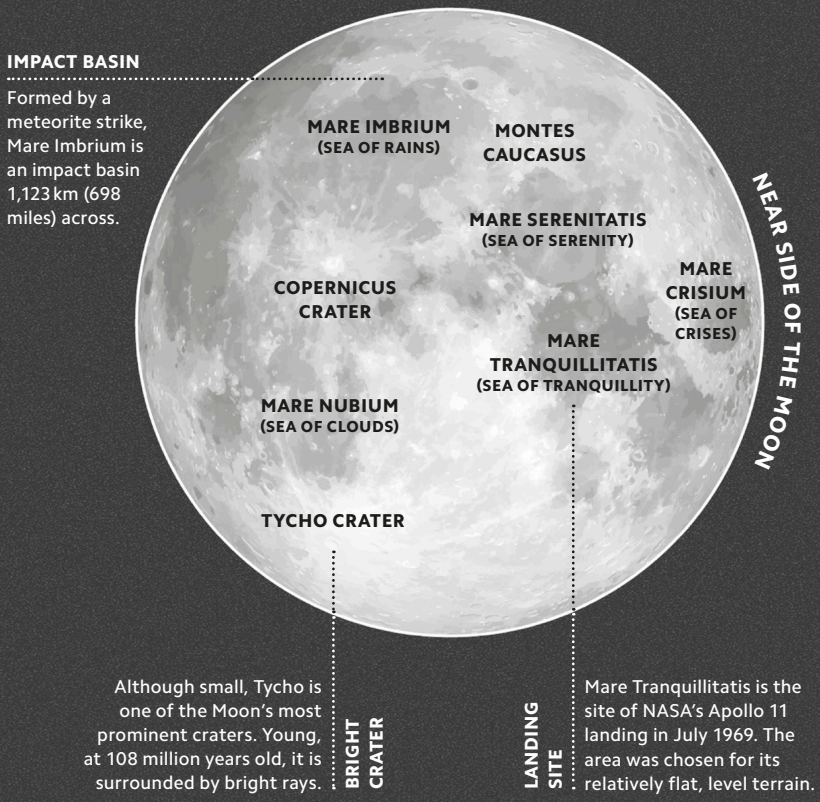
Magnetosphere
Earth is shielded from the solar wind – charged particles ejected from the Sun – by its magnetic field, which creates a magnetic bubble in space called the magnetosphere. The front surface of the magnetosphere, the magnetopause, is where the solar wind is deflected away.

Some parts of Earth's surface experience stronger gravity than others. This is because Earth is not perfectly round but rather slightly squashed, so that it is wider at the equator.



EARTH'S SATELLITE

Earth's only natural satellite, the Moon orbits our planet at an average distance of 384,400 km (238,900 miles). It is the brightest and largest object in the night sky, so much so that some of its surface features are visible to the naked eye. The Moon's gravitational pull is the main cause of Earth's ocean tides. A source of intrigue since prehistoric times, the Moon remains the only extraterrestrial body to have been explored by human astronauts.



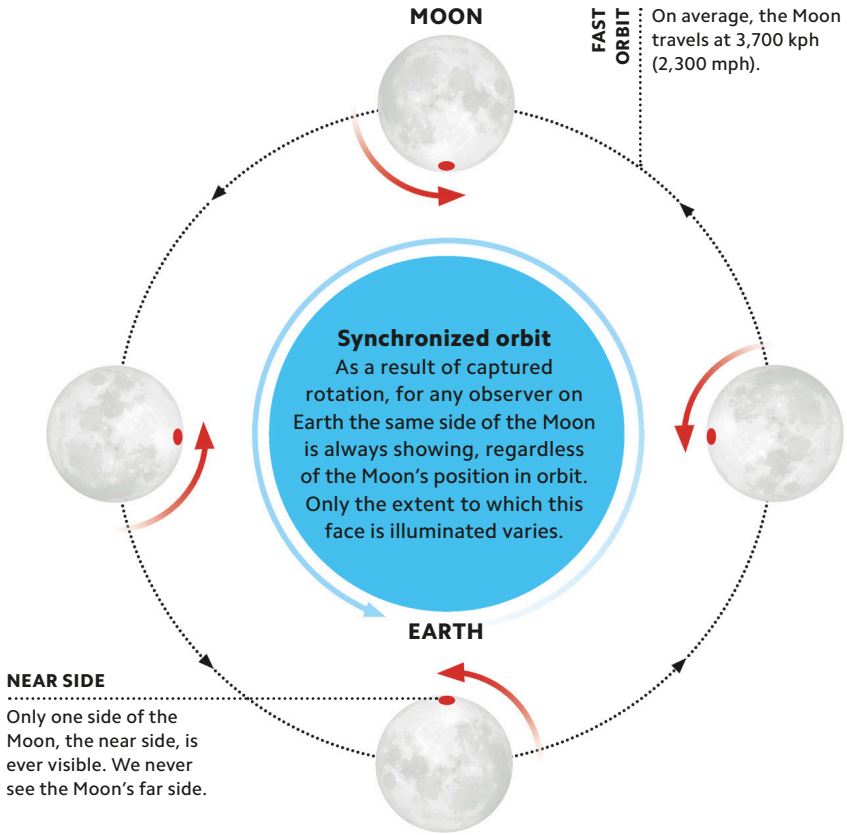
Surface

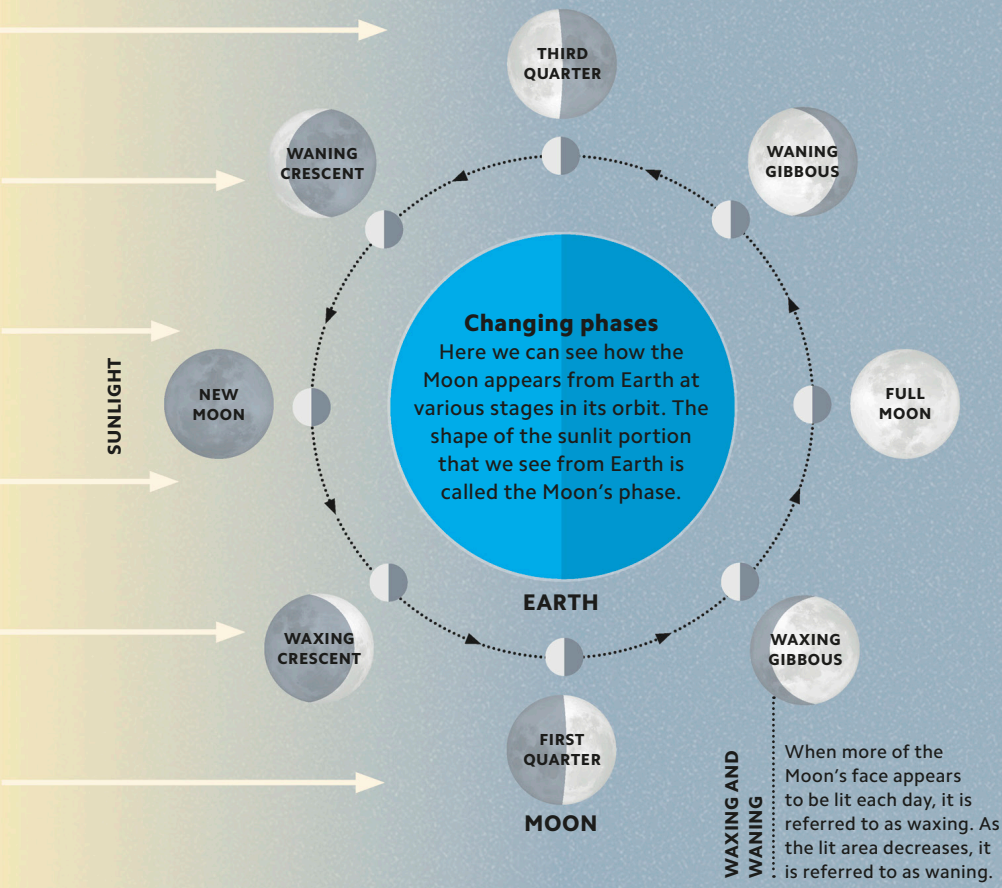
Meteorites have blasted the Moon for billions of years, creating thousands of craters. Dark plains called seas formed when lava from volcanic eruptions flowed into impact basins, then cooled and solidified.

The Outer Space Treaty, created in 1967 and signed by over 130 countries to date, prohibits any nation from claiming sovereignty over the Moon.

TURNING IN STEP

The Moon orbits Earth once every 27.3 days. However, it takes slightly longer (29.5 days) to go through its cycle of phases (see opposite). This is because as the Moon orbits Earth, Earth also orbits the Sun. Millions of years of gravitational interactions between the Moon and Earth have seen the Moon's rate of rotation slow down and become locked at exactly the same rate at which it orbits. This phenomenon is known as captured rotation.





NEW MOON TO FULL MOON

The Moon is visible only because it reflects light from the Sun. As a result, when the Moon sits between the Sun and Earth, at the new Moon stage, there is no light reflecting back to us. This means that we can't see the Moon at all, unless it is silhouetted during a solar eclipse. When at the opposite side of its orbit, the whole of the Moon's Earth-facing side is lit and seen as a full Moon. At different stages between new and full Moon, only certain parts of the Moon's illuminated half are visible from Earth.

IN THE MOON'S SHADOW

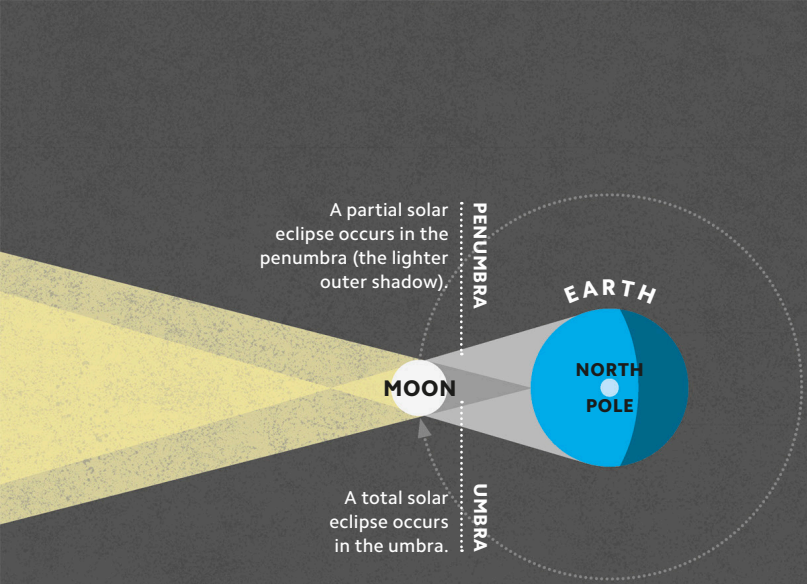
SUNLIGHT

A solar eclipse occurs when the Moon blocks the light from the Sun, casting a shadow on Earth. If the Moon orbited Earth on the same plane as Earth orbits the Sun, a solar eclipse would occur every month. The Moon's orbit, however, is tilted, so an eclipse occurs somewhere on Earth once every 18 months or so. An annular eclipse occurs when the Moon is too far from Earth to fully block out the Sun's light, leaving a narrow ring of light around the Moon.

IN EARTH'S SHADOW

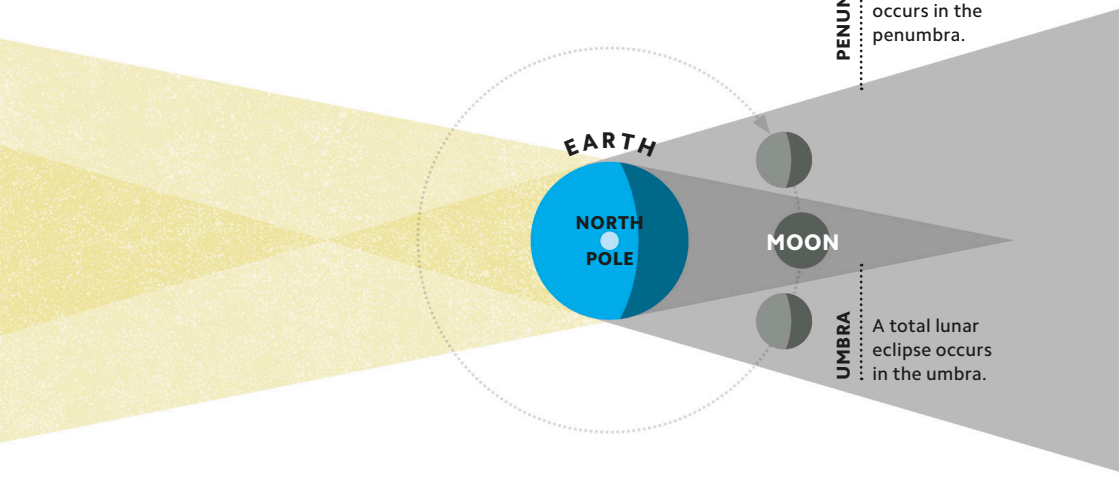
SUNLIGHT

A lunar eclipse occurs when Earth blocks sunlight from reaching the Moon. Normally, the Moon is visible because it reflects sunlight, so when Earth blocks this light out, the Moon darkens. It doesn't totally disappear, however, because Earth's atmosphere still scatters light, which reaches the Moon and reflects back. This smaller amount of light makes the Moon appear much darker, and with a yellow, orange, or red hue.



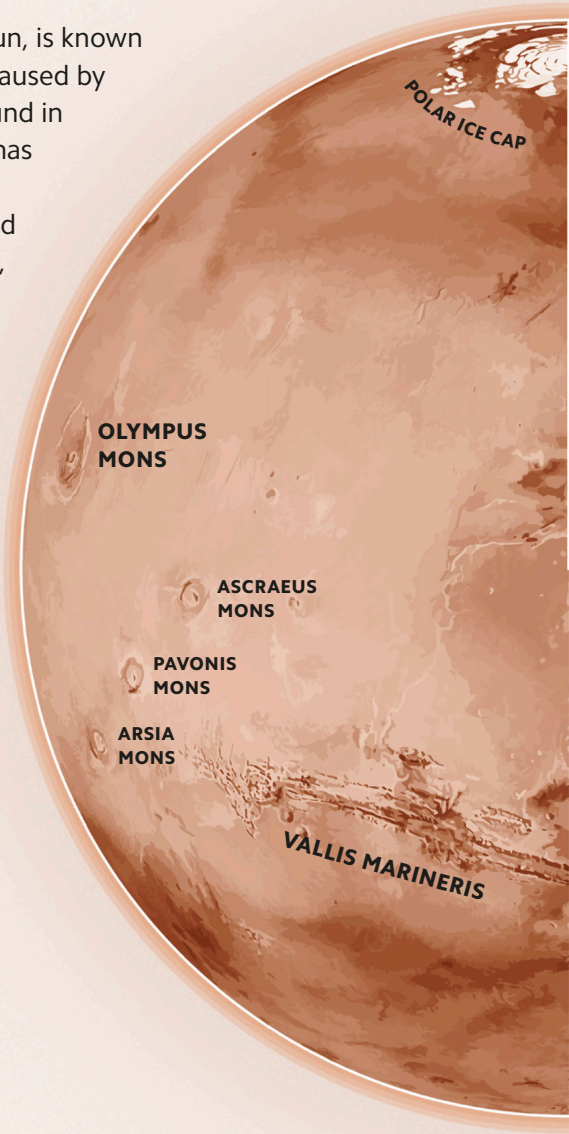
Solar eclipse
Inside the umbra, the darkest part of the shadow cast onto Earth, a total solar eclipse occurs, lasting anywhere from a few seconds to several minutes.

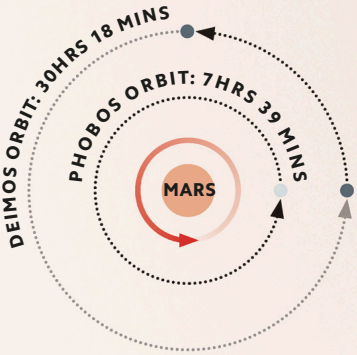
Lunar eclipse
A lunar eclipse lasts much longer than a solar eclipse, because the Moon moves into Earth's shadow, which is much wider than the Moon.



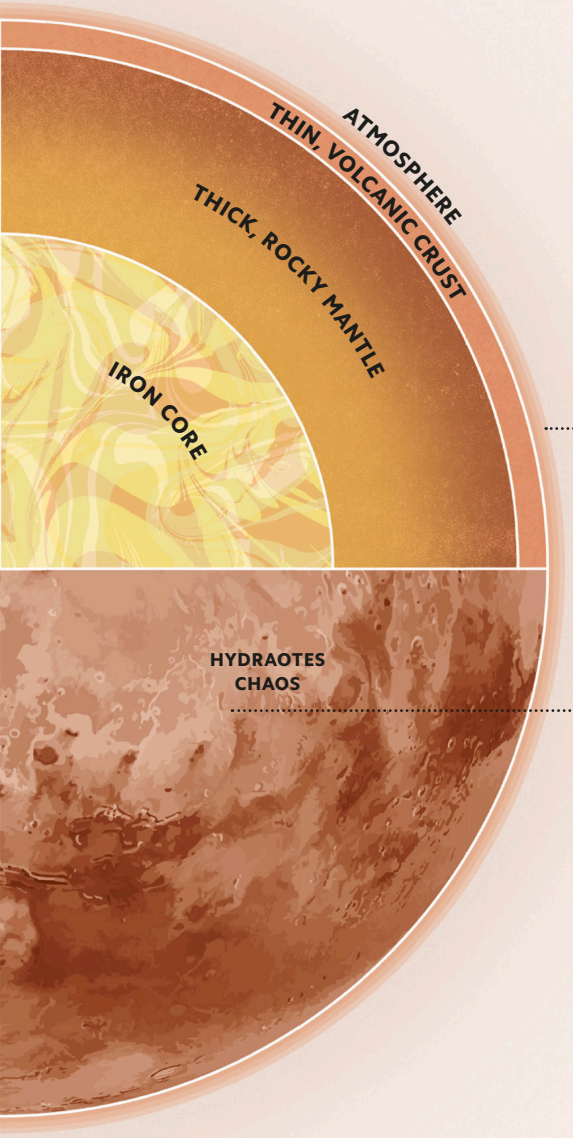
THE RED PLANET

Mars, the fourth planet from the Sun, is known for its striking reddish hue. This is caused by iron oxide, the same compound found in rust, on the planet's surface. Mars has many Earthlike features, including clouds, winds, seasonal weather, and a roughly 24-hour day. Traditionally, these traits have led many to view Mars as the most likely potential host for life, besides Earth, in our Solar System. However, Mars is much colder than Earth, and has no liquid water on its surface and almost no oxygen in its atmosphere.





Martian lunar system
Mars has two moons: Phobos and Deimos. Phobos is closer to Mars and orbits the planet more than four times faster than Deimos.



Structure

Mars has a dense core of iron, nickel, and sulphur, which may be totally or partly liquid, surrounded by a rocky mantle. The planet's crust is mostly volcanic rock. Mars's atmosphere is 95.3 per cent carbon dioxide, a gas that freezes during winter, causing a significant drop in pressure.

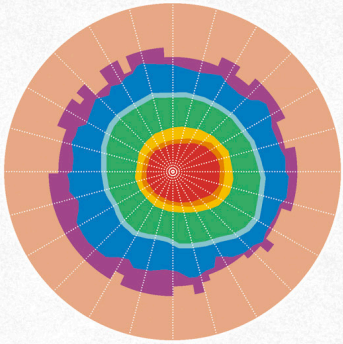
THIN ATMOSPHERE

The Martian atmosphere is about a hundred times thinner than Earth's. Near the surface, winds whip up huge dust storms. Clouds and fog sometimes form.

CHAOS TERRAIN

Hydraotes Chaos is just one example of chaos terrain – areas on a planet with an erratic mix of features such as ridges, hills, valleys, mesas, and troughs.

Vallis Marineris, the largest canyon on Mars, is long enough to stretch from New York to Los Angeles.



Polar ice
In early spring, when temperatures on Mars are cooler, large amounts of carbon dioxide ice build up around its poles. By late spring, these recede.

ICE DEPTH

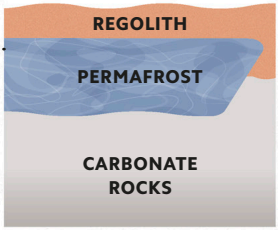
THICKEST THINNEST

MARTIAN ICE

The Mars we know is a sterile and barren wasteland. A variety of surface features visibly shaped by water flow, however, suggest that billions of years ago, it was probably covered in rivers, lakes, and even oceans of water. Today, water is found only on the surface of Mars in ice form. However, in the south polar region, scientists have recently discovered subglacial lakes buried deep underground – the first stable bodies of liquid to be found on the planet.

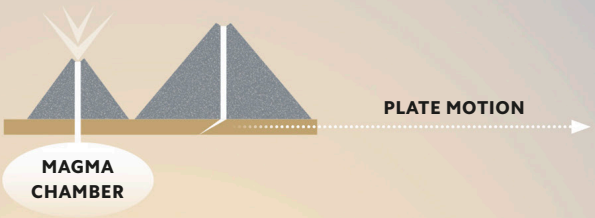
Under the surface, Mars is covered in bands of slow-moving glaciers (permafrost).

FROZEN UNDERGROUND



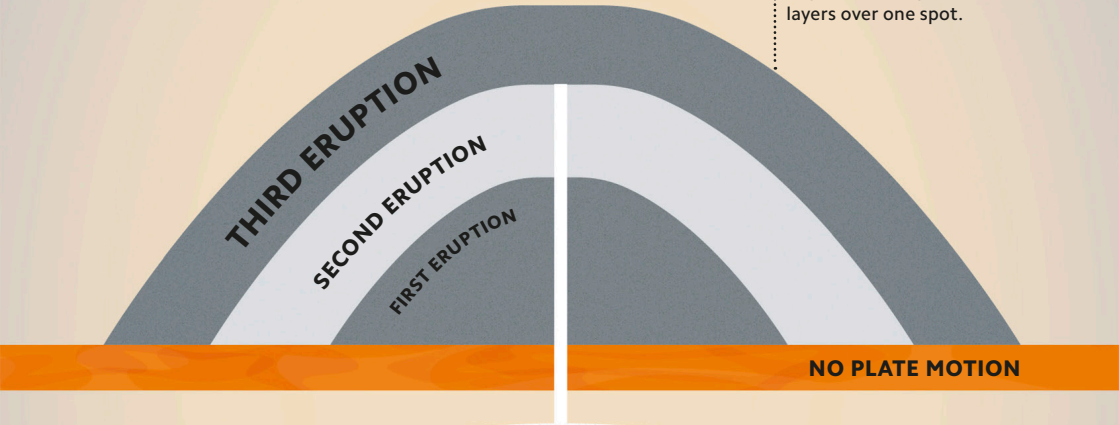
Layered surface
Loose rock and dust (regolith) covers the surface of Mars. Beneath is a layer of permafrost above a solid rock crust.

Volcanoes on Earth
On Earth, shifts in tectonic plates mean that the next eruption from the same magma chamber can often form a new volcano.



SUCCESSIVE LAYERS

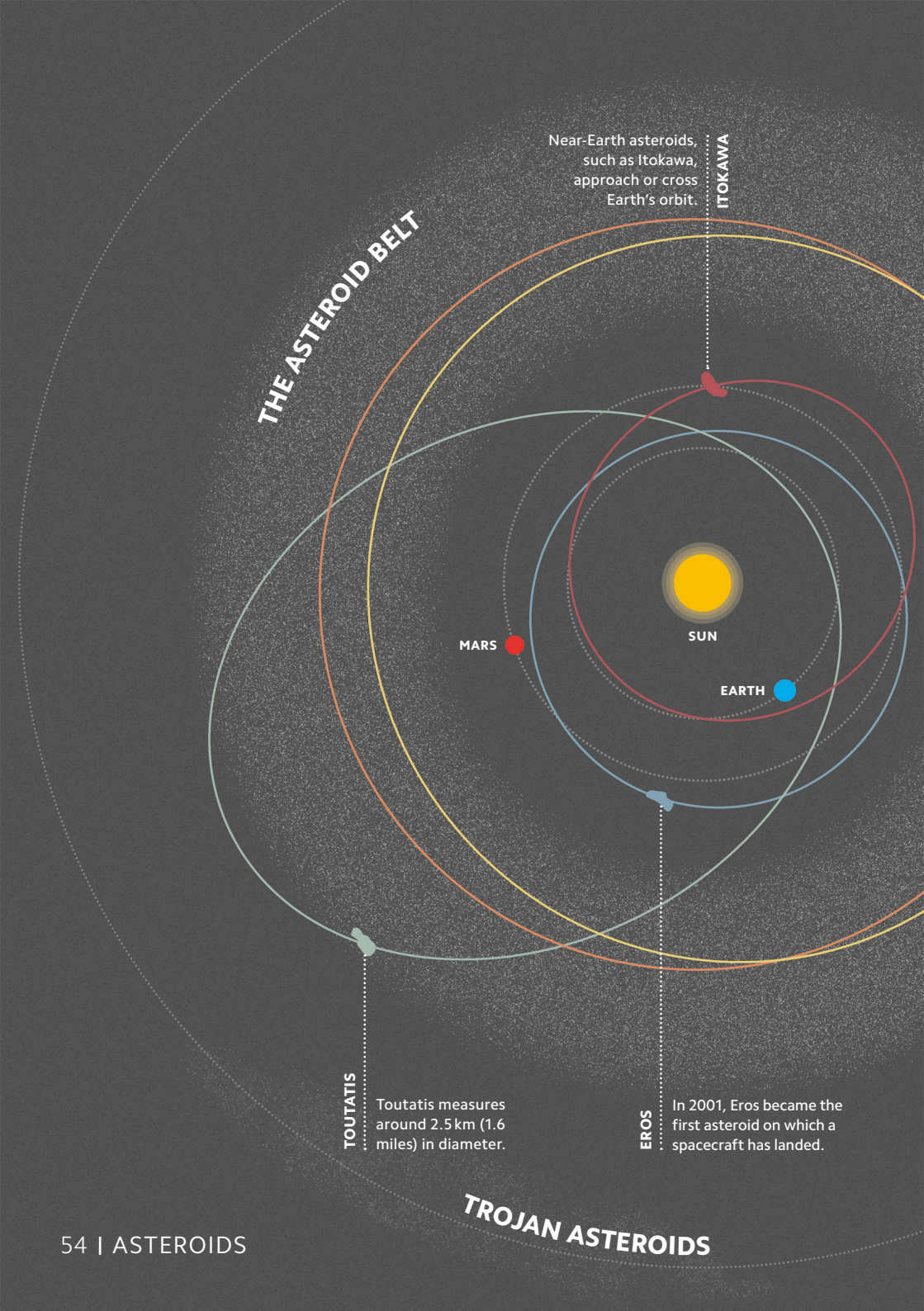
Because Mars's crust, unlike Earth's, is one solid, unmoving piece, eruptions over millions of years build up in layers over one spot.

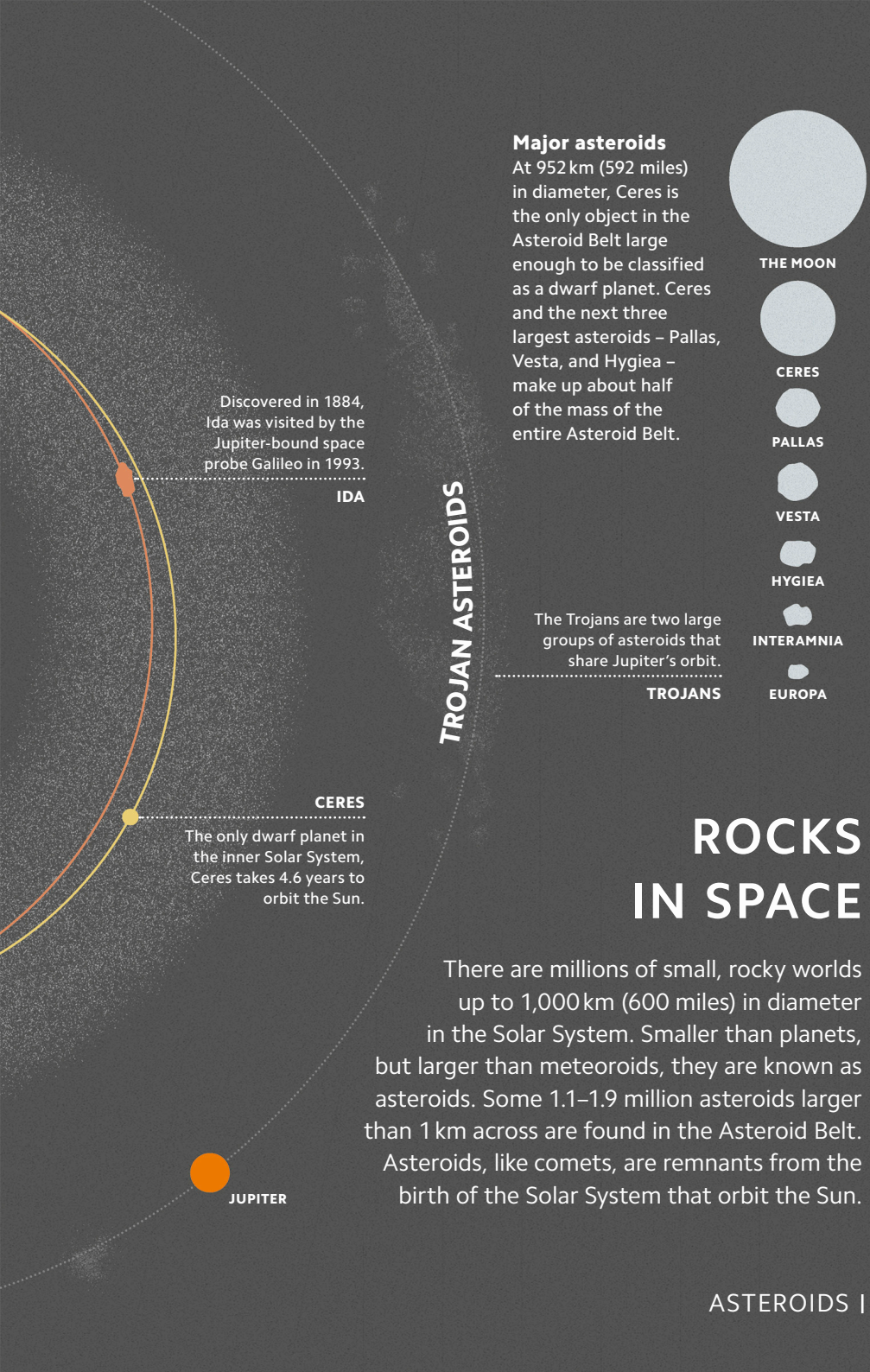


SUPERSIZED VOLCANOES

Mars is home to Olympus Mons, which, at three times taller than Mount Everest, is the largest volcano in the Solar System. However, none of the volcanoes found on Mars are active today, as the crust is too thick to allow molten rock to rise up through it. There is evidence for eruptions as recently as 50,000 years ago, but most Martian volcanoes formed 1–2 billion years ago.

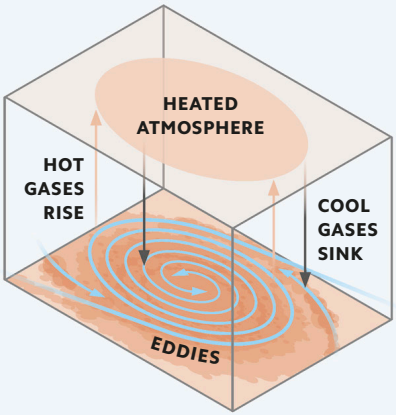
MAGMA CHAMBER


THE ASTEROID BELT
Near-Earth asteroids, such as Itokawa, approach or cross Earth's orbit.
ITOKAWA
SUN
MARS
EARTH
TOUTATIS
Toutatis measures around 2.5 km (1.6 miles) in diameter.
EROS
In 2001, Eros became the first asteroid on which a spacecraft has landed.
TROJAN ASTEROIDS

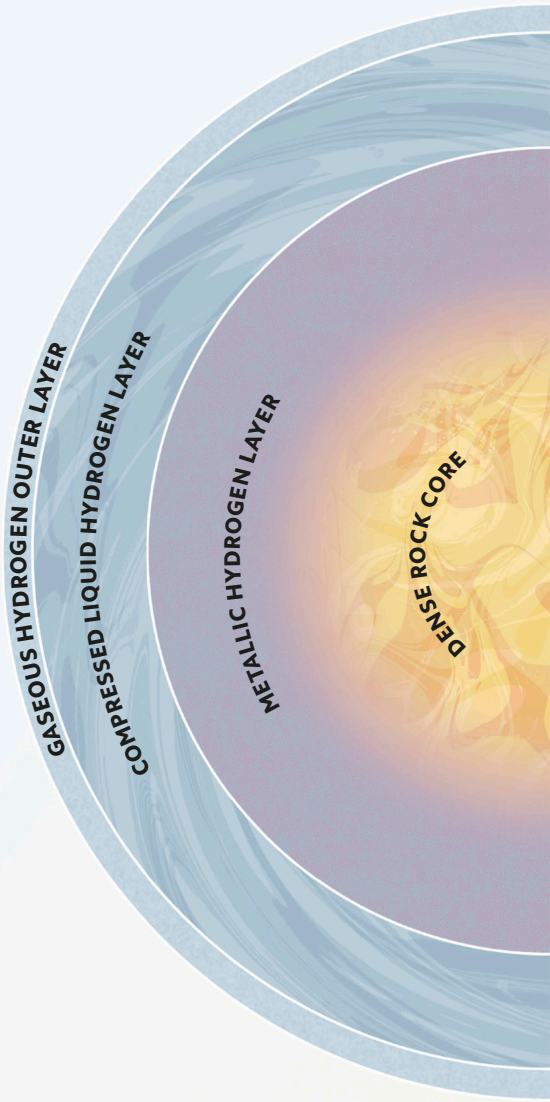


ROCKS IN SPACE

There are millions of small, rocky worlds up to 1,000 km (600 miles) in diameter in the Solar System. Smaller than planets, but larger than meteoroids, they are known as asteroids. Some 1.1–1.9 million asteroids larger than 1 km across are found in the Asteroid Belt. Asteroids, like comets, are remnants from the birth of the Solar System that orbit the Sun.



The Great Red Spot
This huge storm rolls around between two jet streams in the atmosphere, circling in opposite directions. These streams bring swirling eddies of gas, which merge with the Great Red Spot and may help to keep it spinning. Hot gas rising at the centre may also spin upwards through this powerful vortex.



Internal structure
Jupiter's hot core is encased by a high-pressure mantle of metallic hydrogen, which behaves like a liquid metal in that it can conduct electricity and generate magnetic fields. Above this is a vast sea of liquid hydrogen, which gradually thins out into an atmosphere that is 90 per cent hydrogen.

KING OF THE GIANTS

The most massive object in our Solar System apart from the Sun, Jupiter is also large enough for the other seven planets to fit inside it at the same time. It is the fifth planet from the Sun and one of the brightest objects in the night sky due to the amount of sunlight it reflects back at us. This gas giant takes almost 12 Earth years to finish one full orbit of the Sun. However, because it rotates more than twice as fast as Earth, a day on Jupiter lasts only 10 hours.



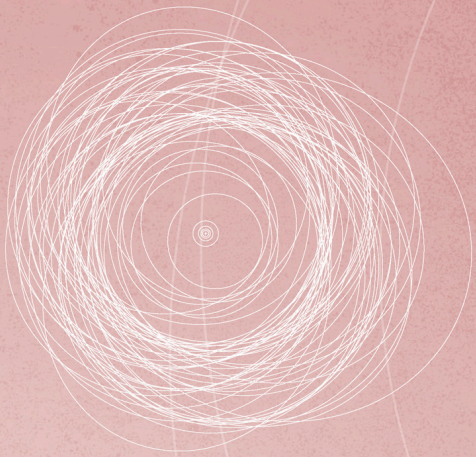
TURBULENT ZONE

A strong jet stream rips through Jupiter's South Tropical Zone, the planet's most active weather region.

OUTER LAYER

Swirling gases form a pattern of dark belts divided by lighter zones in Jupiter's atmosphere, which extends into space for more than 5,000 km (3,100 miles).

The temperature of Jupiter's core is 24,000°C (43,200°F) – more than four times that of the surface of the Sun.



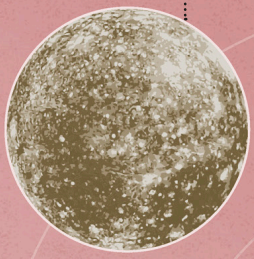
Vast lunar network
Jupiter's larger inner moons orbit in the same direction as the planet rotates. Its smaller outer moons orbit in a messy cloud, and probably used to orbit the Sun before being captured by Jupiter's gravitational pull.

THE JOVIAN SATELLITES

In total, Jupiter has more than 75 confirmed moons. The planet's four Galilean moons are the most massive and inspire the most scientific interest. The Galilean moons and four smaller moons even closer to Jupiter make up the planet's regular satellites, meaning they follow a relatively close and circular orbit. The remaining moons are irregular satellites, meaning their orbits are more eccentric (elongated), distant, or inclined.

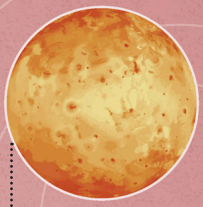
CALLISTO

Callisto is the most distant of the Galilean moons. So many craters fill its battered surface that any new craters tend to erase older ones.



Galilean moons

These four moons are named after their discoverer, Galileo, who declared them the first objects to orbit a planet other than Earth. They are visible with binoculars when far enough away from Jupiter.

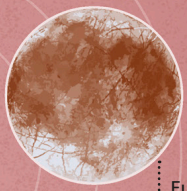


Jupiter's gravity pulls on Io, creating enough heat to power volcanoes on its surface. This makes Io the most volcanically active object in the Solar System.

IO

\+

JUPITER



EUROPA

Europa has a water ocean hidden beneath a smooth, icy surface. This water-ice crust moves in a similar fashion to Earth's tectonic plates.



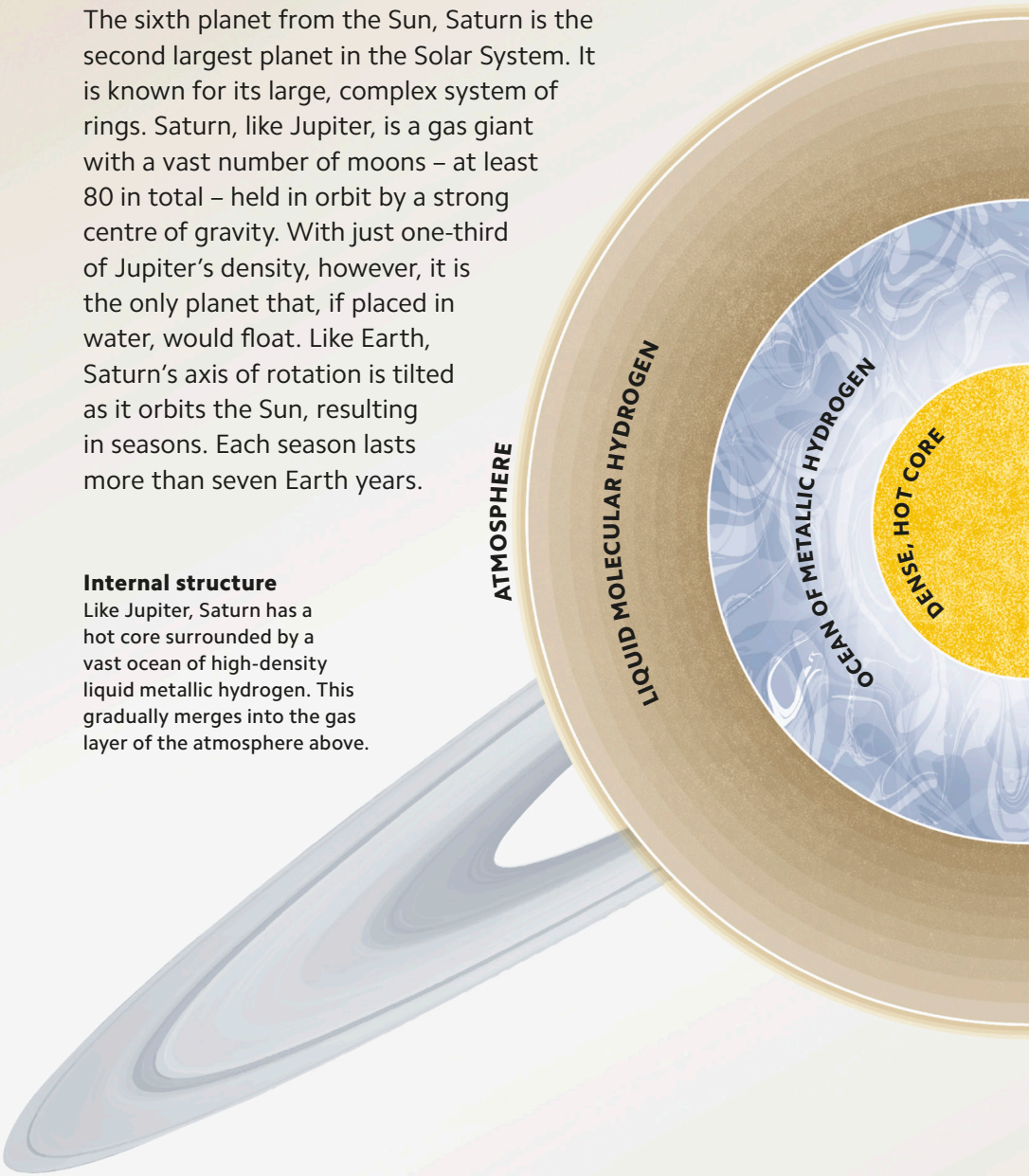
GANYMEDE

Ganymede, at 5,300 km (3,300 miles) wide, is significantly larger than Mercury and the largest moon in the Solar System.

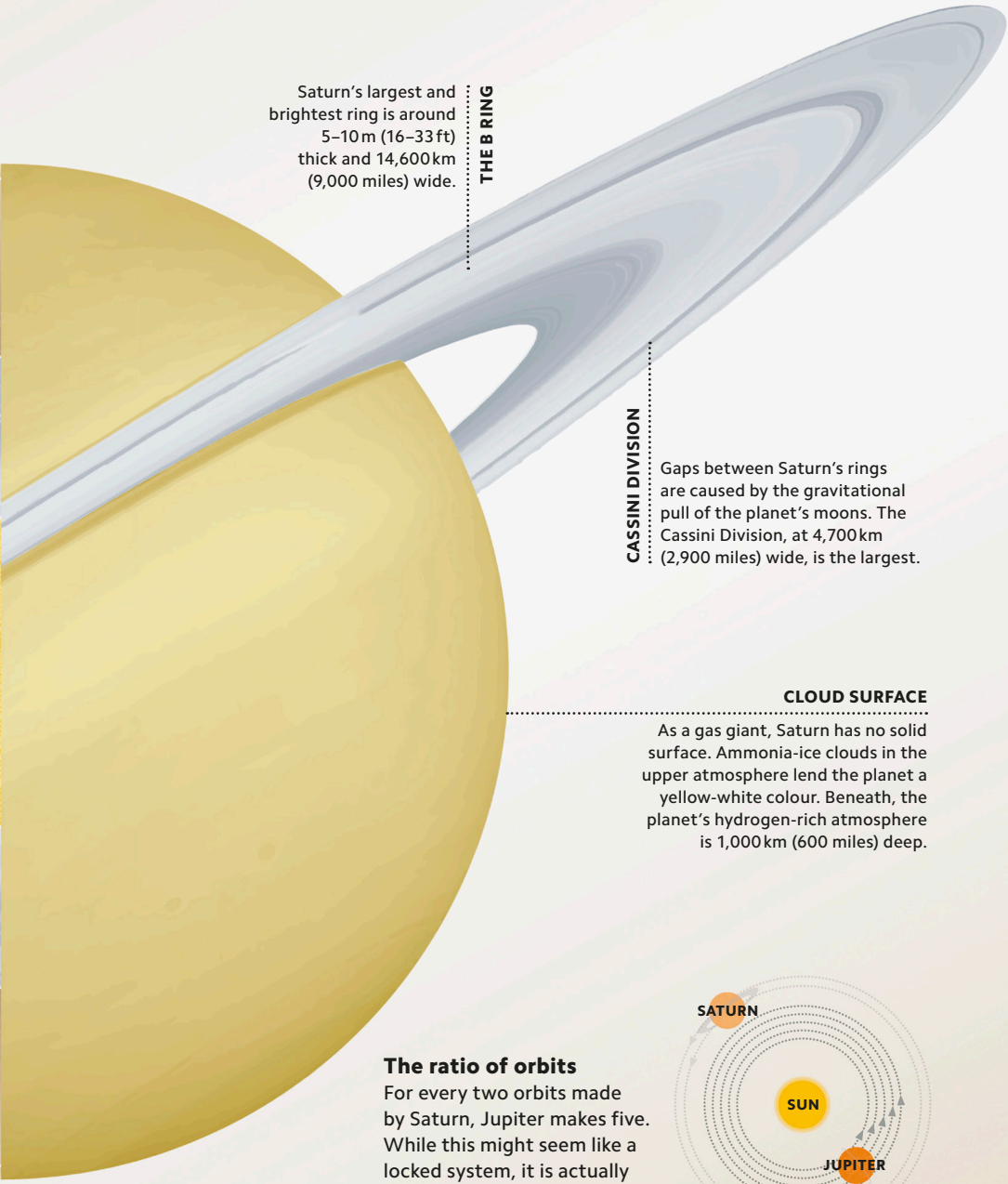
Ganymede is the only known moon in the Solar System with its own magnetic field.

THE RINGED PLANET

The sixth planet from the Sun, Saturn is the second largest planet in the Solar System. It is known for its large, complex system of rings. Saturn, like Jupiter, is a gas giant with a vast number of moons – at least 80 in total – held in orbit by a strong centre of gravity. With just one-third of Jupiter's density, however, it is the only planet that, if placed in water, would float. Like Earth, Saturn's axis of rotation is tilted as it orbits the Sun, resulting in seasons. Each season lasts more than seven Earth years.



Internal structure
Like Jupiter, Saturn has a hot core surrounded by a vast ocean of high-density liquid metallic hydrogen. This gradually merges into the gas layer of the atmosphere above.



THE B RING

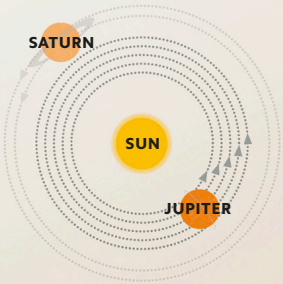
Saturn's largest and brightest ring is around 5–10 m (16–33 ft) thick and 14,600 km (9,000 miles) wide.

CASSINI DIVISION

Gaps between Saturn's rings are caused by the gravitational pull of the planet's moons. The Cassini Division, at 4,700 km (2,900 miles) wide, is the largest.

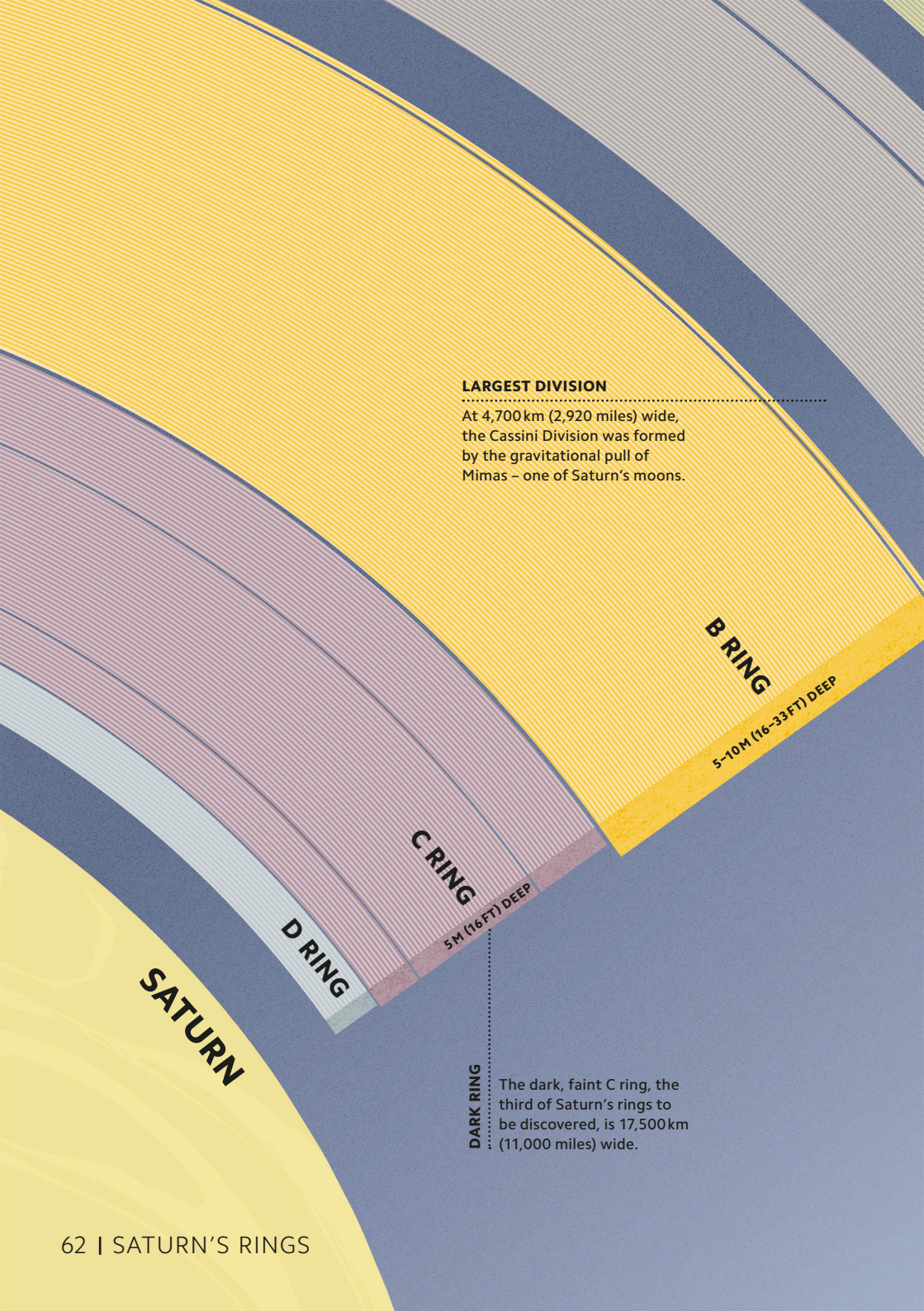
CLOUD SURFACE

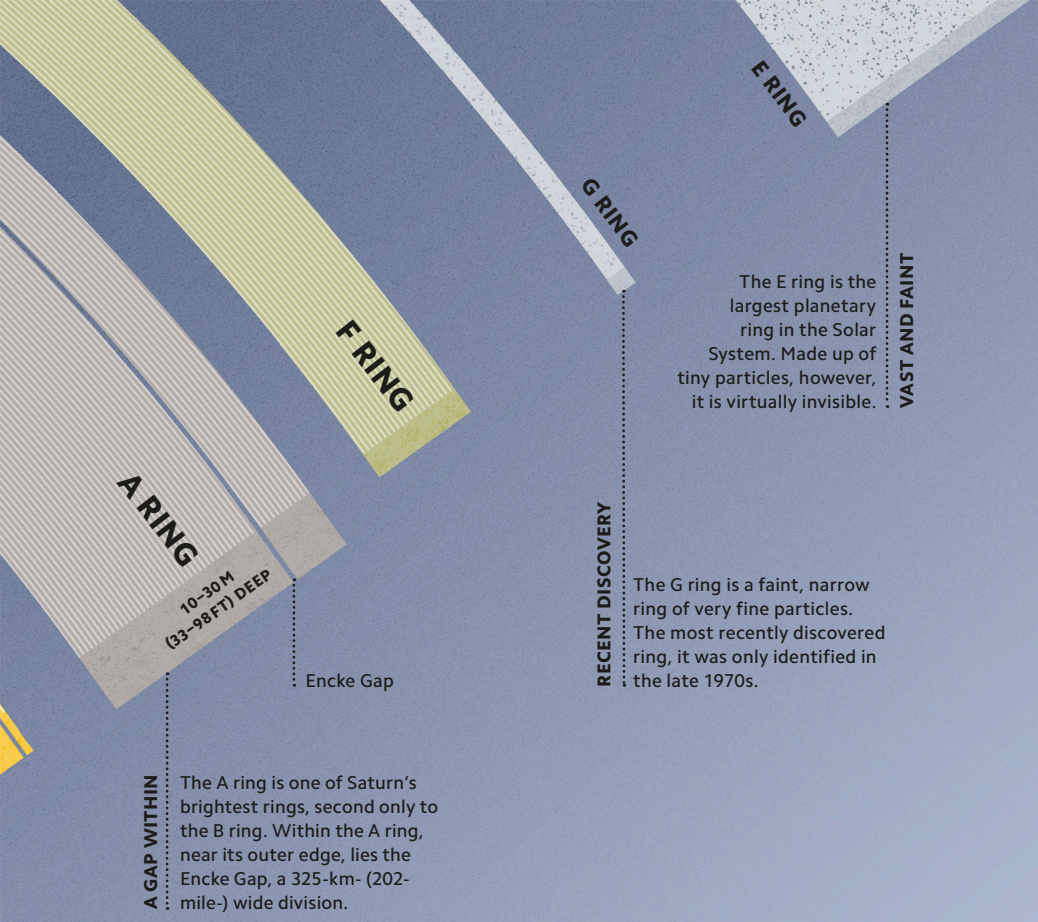
As a gas giant, Saturn has no solid surface. Ammonia-ice clouds in the upper atmosphere lend the planet a yellow-white colour. Beneath, the planet's hydrogen-rich atmosphere is 1,000 km (600 miles) deep.



The ratio of orbits

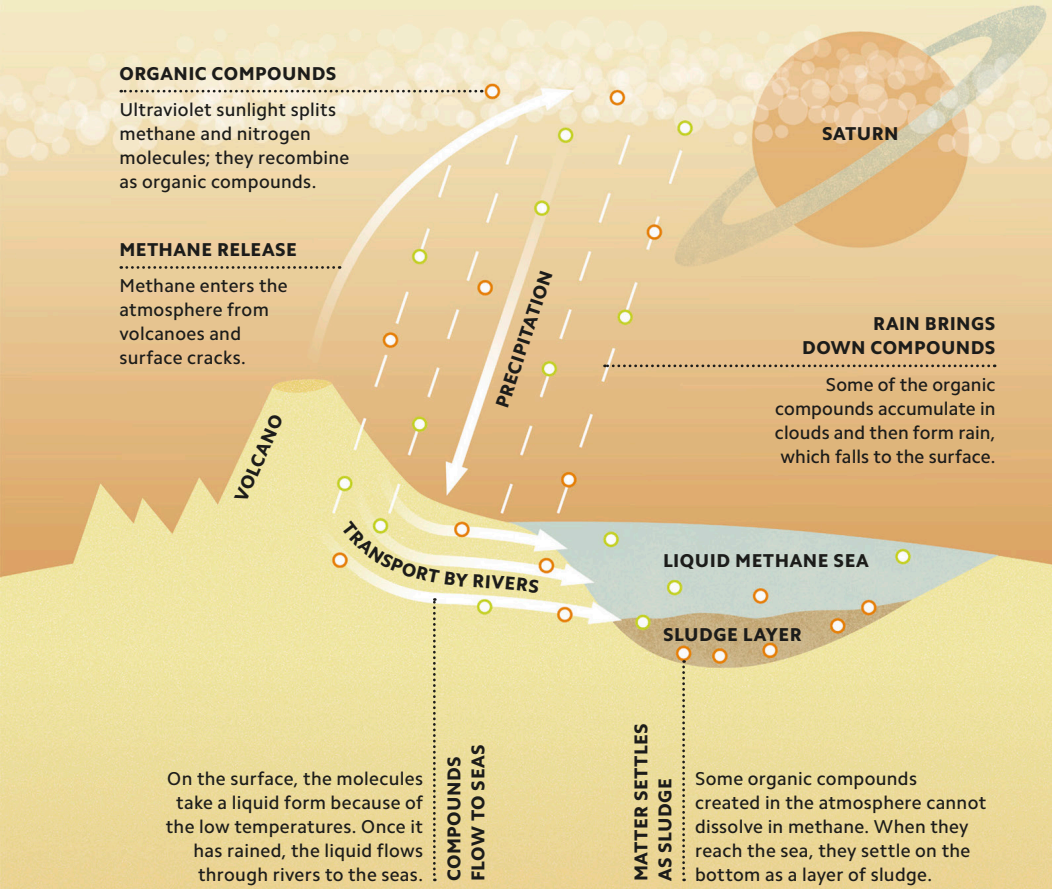
For every two orbits made by Saturn, Jupiter makes five. While this might seem like a locked system, it is actually a coincidence, as the ratio of their orbits is slowly shifting.


LARGEST DIVISION
At 4,700 km (2,920 miles) wide, the Cassini Division was formed by the gravitational pull of Mimas – one of Saturn's moons.
B RING
5–10 M (16–33 FT) DEEP
C RING
5 M (16 FT) DEEP
D RING
SATURN
DARK RING
The dark, faint C ring, the third of Saturn's rings to be discovered, is 17,500 km (11,000 miles) wide.



BANDS OF ICE

Several planets in the Solar System have rings. However, Saturn's are the most spectacular and complex. Its large rings, each made up of many smaller ringlets, extend up to 282,000 km (175,000 miles) from the planet. They consist of billions of chunks of ice (with a tiny amount of rock and dust) ranging in size from a grain of sand to a house. The rings are named alphabetically in the order of their discovery, while gaps between rings are named after astronomers.

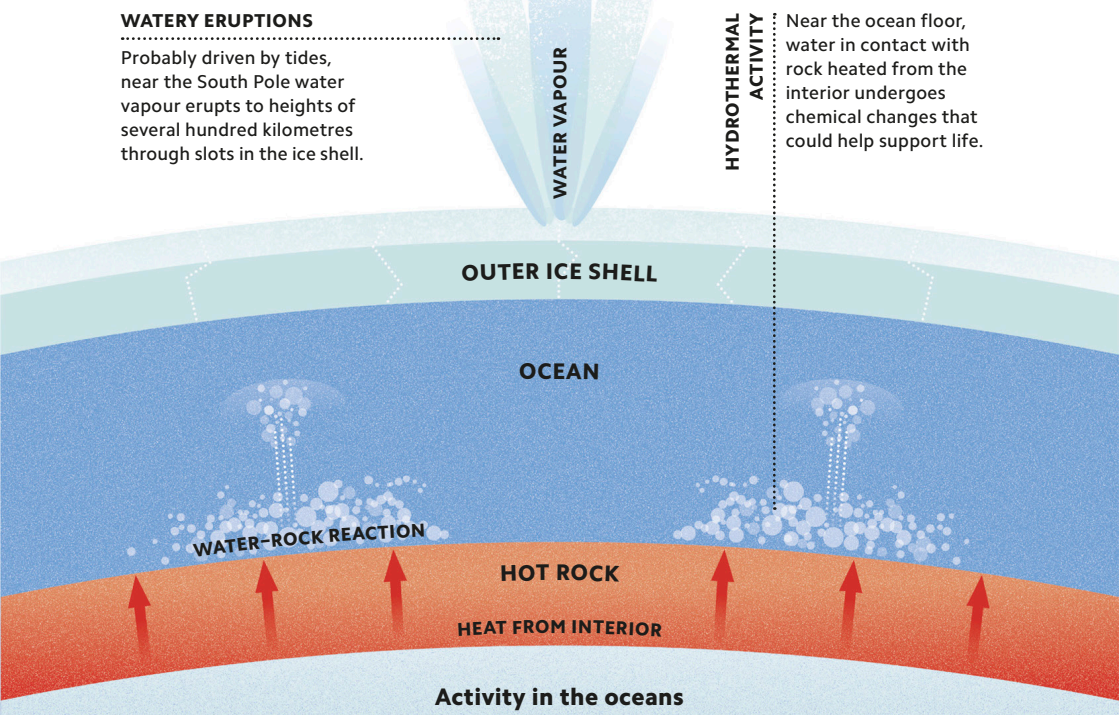


THE METHANE MOON

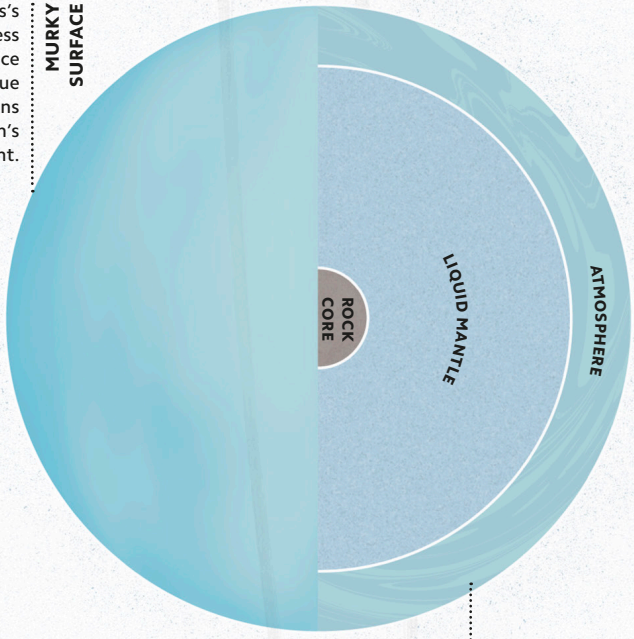
Saturn's biggest moon, Titan is the only body in the Solar System other than Earth to have a weather cycle. But because of the low surface temperatures, of around -180° C (-290° F), Titan's lakes, clouds, and rain are made up of chemical compounds called hydrocarbons, such as methane and ethane, instead of water. Titan's atmosphere is 95 per cent nitrogen and 5 per cent methane, with a layer of ethane that gives the moon a haze.

THE E-RING MOON

Enceladus is a small moon, only about 500 kilometres (300 miles) across, within Saturn's E ring (see pp.62–63). It has all the ingredients needed for life but is extremely cold, with temperatures of about -200° C (-330° F) that keep surface water in solid form. A thick layer of surface snow reflects sunlight, making Enceladus the most reflective body in the Solar System. However, there are some pockets of liquid water, thought to result from heating from the moon's interior.

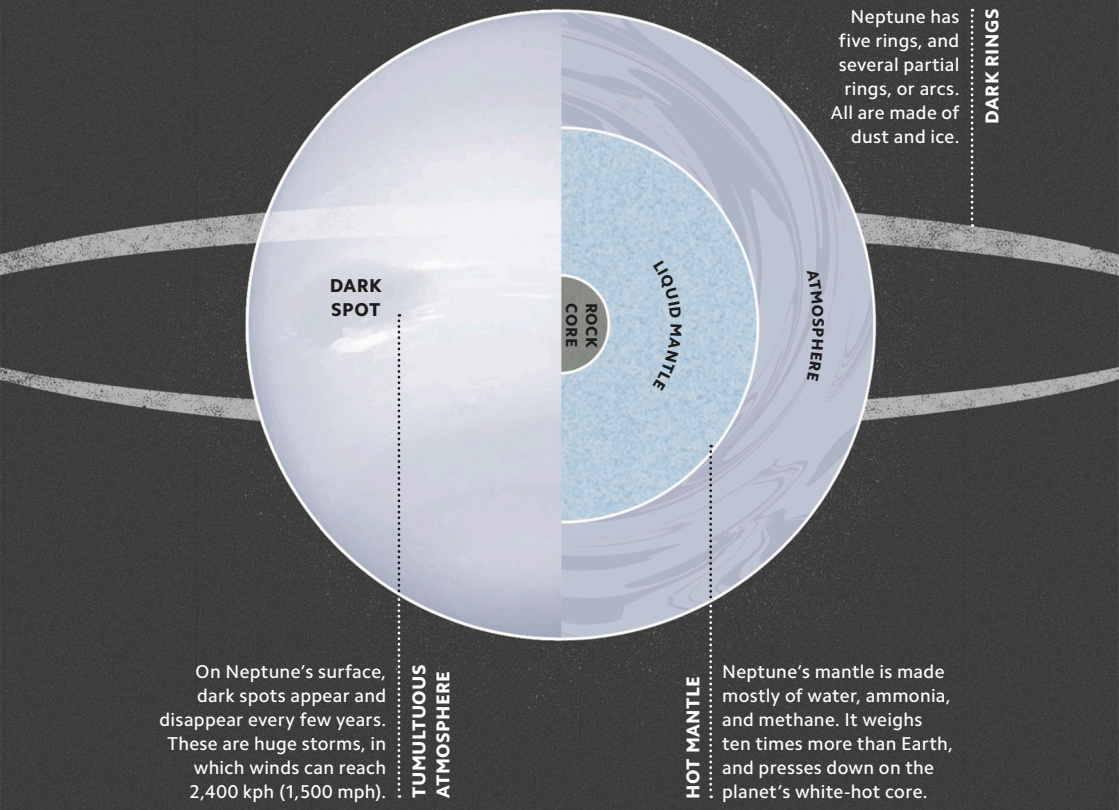


Activity in the oceans
Heat from Enceladus's interior may lead to chemical changes in its ocean water. In places, plumes of ice and water vapour also burst through the ice shell.



THE SIDEWAYS PLANET

Uranus is an ice-giant planet rich in ices of water, ammonia, and methane. The seventh planet from the Sun, it orbits at a distance of around 2.9 billion km (1.8 billion miles). The planet's equator is nearly at a right angle to its orbital plane. This results in the most extreme seasons of any planet as, for a quarter of its 84-year-long orbit, one pole faces the Sun while the other is dark. Uranus also rotates east to west, which is the opposite direction to most of the other planets.

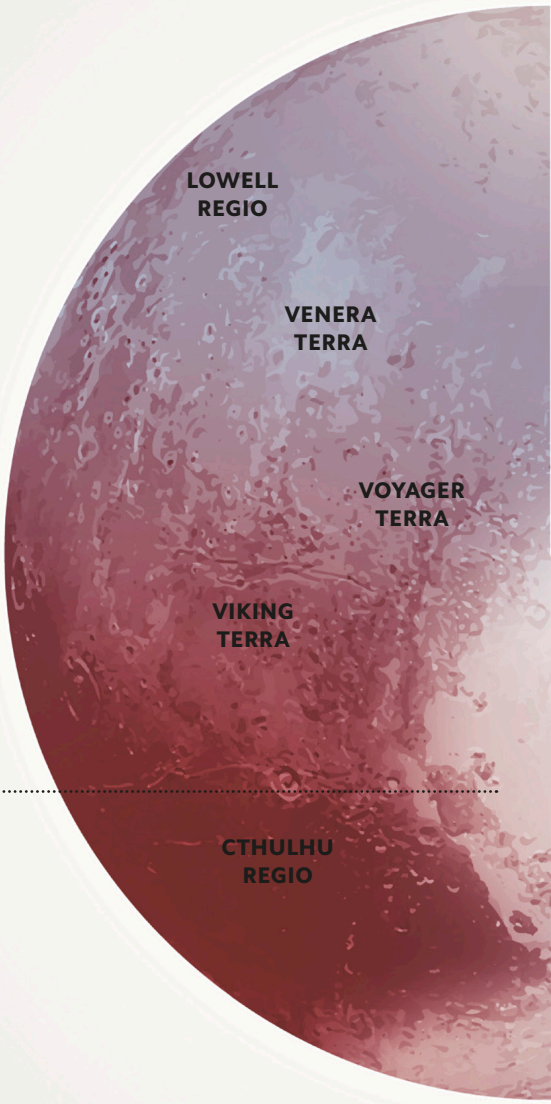


THE OUTERMOST GIANT

Neptune is only slightly smaller than Uranus. However, it orbits at a much greater distance of 4.5 billion km (2.8 billion miles) from the Sun, taking 165 years to complete a single orbit. Both planets share a blue tinge, derived from the methane in their near-identical atmospheres. Neptune's darker hue, however, suggests that it contains another unknown chemical. Within both Uranus and Neptune, intense pressure may have formed an ocean of diamonds between the mantle and the core.

FROZEN WORLD

A tiny world, Pluto has a diameter only two-thirds that of Earth's moon – and when the International Astronomical Union (IAU) formalized its criteria for planet classification in 2006, Pluto was designated as a dwarf planet rather than a planet. Of Pluto's five moons, the largest, Charon, is around half the size of Pluto itself. On Pluto's surface, where the average temperature is -233°C (-387°F), white methane frost sits on water-ice mountain peaks, and there is evidence of active cryovolcanoes – volcanoes that erupt water, ammonia, or methane.



TOMBAUGH REGIO

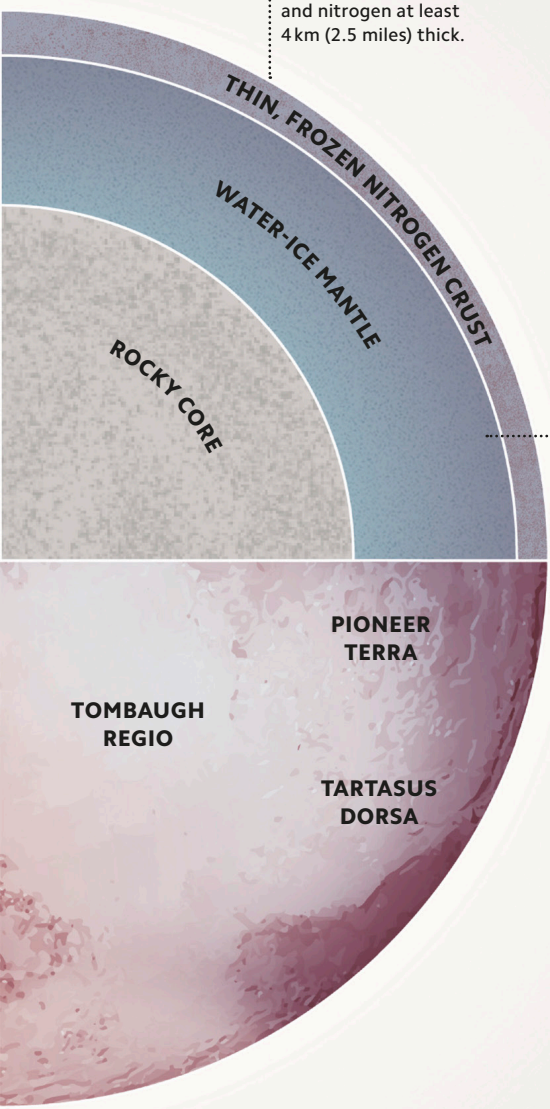
This large, bright surface feature spanning 1,590 km (990 miles) contains Sputnik Planitia, a deep nitrogen-ice basin dotted with towering water-ice mountains.

Light from the Sun takes more than five hours to reach Pluto.

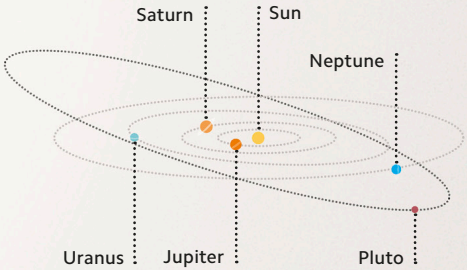
CRUST
The outer layer of Pluto is a thin, icy crust of methane and nitrogen at least 4 km (2.5 miles) thick.

Structure

Pluto's rocky core is unusually large, making up at least 60 per cent of the planet's mass. This core is encased in an icy mantle and a thin nitrogen crust. Winds in Pluto's thin nitrogen atmosphere blow in the opposite direction to its spin.



MANTLE
There may be a layer of liquid water at the bottom of Pluto's icy mantle.



Tilted orbit

Pluto's 248-year-long orbit is highly tilted and highly elliptical. As a result, at different times it is above or below the plane of the major planets, and its distance from the Sun varies greatly.



The Kuiper Belt is a flat, doughnut-shaped ring of icy objects that revolve around the Sun. With its inner edge close to the orbit of Neptune, the belt is thought to consist of debris dating from the early history of the Solar System. This material was originally closer to the Sun but has since drifted to its current colder, more distant position. There are trillions of objects in the Kuiper Belt, which is thought to be the source of most short-period comets as they orbit the Sun (see p.75). The Kuiper Belt is also home to the dwarf planets Haumea, Makemake, Eris, and Pluto.

SMALL WORLDS

Much like regular planets, dwarf planets are almost spherical bodies that orbit the Sun and were shaped by their own gravity. The key difference between dwarf planets and standard planets, as defined by the International Astronomical Union (IAU), is that dwarf planets have not cleared their orbital paths of other objects. The IAU only recognizes five dwarf planets, but astronomers believe that there may be hundreds more in the Kuiper Belt. Only one of the five official dwarf planets – Ceres – is found in the Asteroid Belt (see pp.54–55), while the rest are situated in the Kuiper Belt. Some dwarf planets have their own moons.

Eris
Diameter 2,326 km (1,445 miles)

Earth
Diameter 12,742 km (7,917 miles)

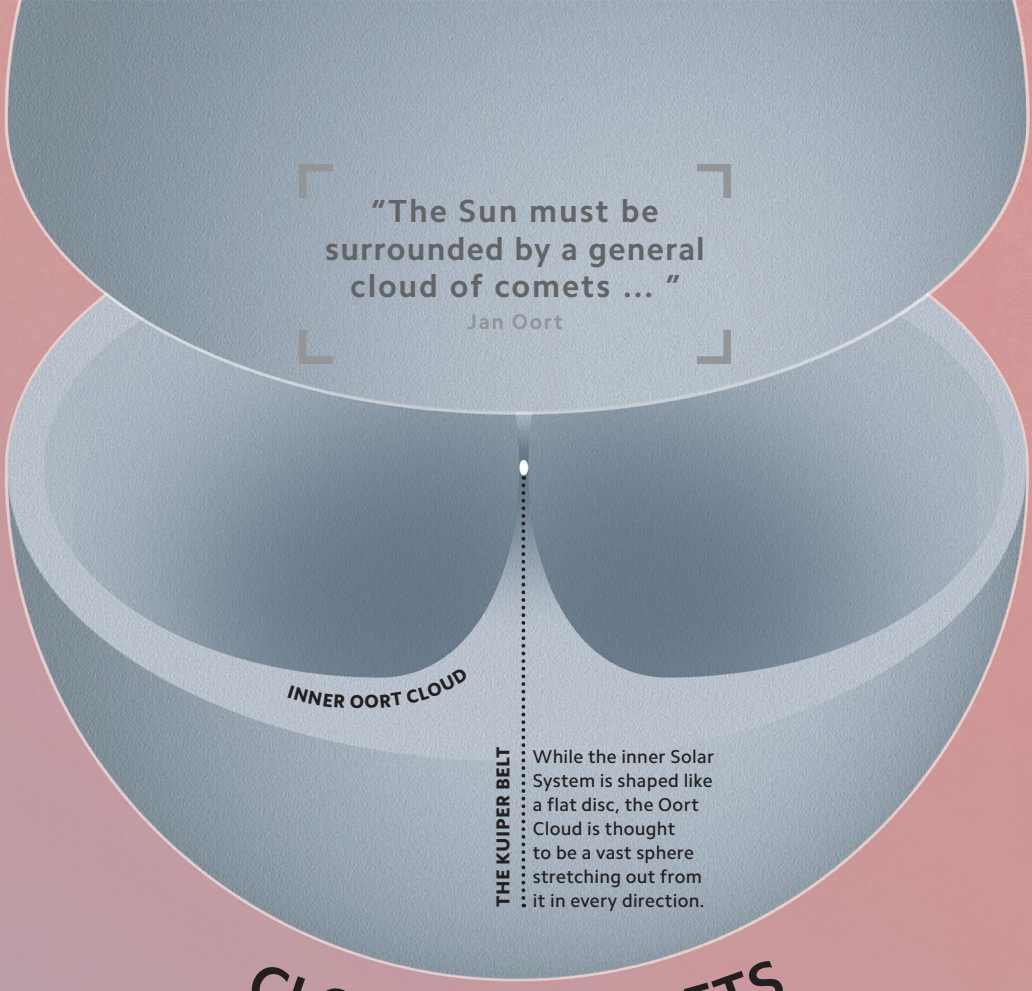
Pluto
Diameter 2,377 km (1,477 miles)

Haumea
Average diameter 2,200 km (1,400 miles)

Sunlight takes nine hours to reach Eris, the most distant dwarf planet.

Makemake
Diameter 1,430 km (890 miles)

Ceres
Diameter 952 km (592 miles)



CLOUD OF COMETS

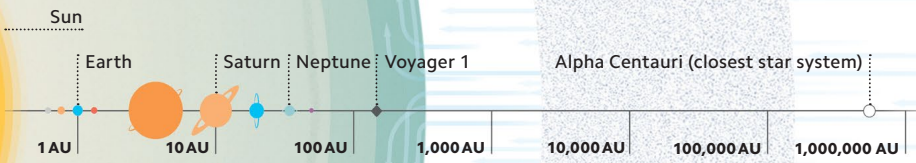
The Oort Cloud is a reservoir of small icy bodies believed to surround the Solar System, stretching far beyond the Kuiper Belt (see p.70). It is where most long-period comets come from and starts 300–750 billion km (190–470 billion miles) from the Sun. Its outer edge is 1.5–15 trillion km (0.9–9 trillion miles) from the Sun – or halfway to the nearest star. Oort Cloud objects orbit the Sun at all angles, unlike in the Kuiper Belt and Asteroid Belt, where most objects follow orbits close to the main plane of the Solar System.

THE OUTER LIMITS

The Sun constantly emits a flow of electrically charged particles, called the solar wind, which travels for three times the distance to Pluto. It extends through a region called the heliosphere until it collides with the interstellar medium – the thin gas between star systems – at the heliopause (the heliosphere's edge). Beyond, in interstellar space, lies the Oort Cloud.

TERMINATION SHOCK

The pressure of the interstellar medium slows the solar wind down abruptly at a boundary called the termination shock.



HELIOPAUSE

At the heliopause, the pressure from the solar wind is in balance with pressure from the interstellar medium.

BOW SHOCK

As the heliosphere moves through the interstellar medium, it produces a bow shock similar to that of a boat travelling through water.

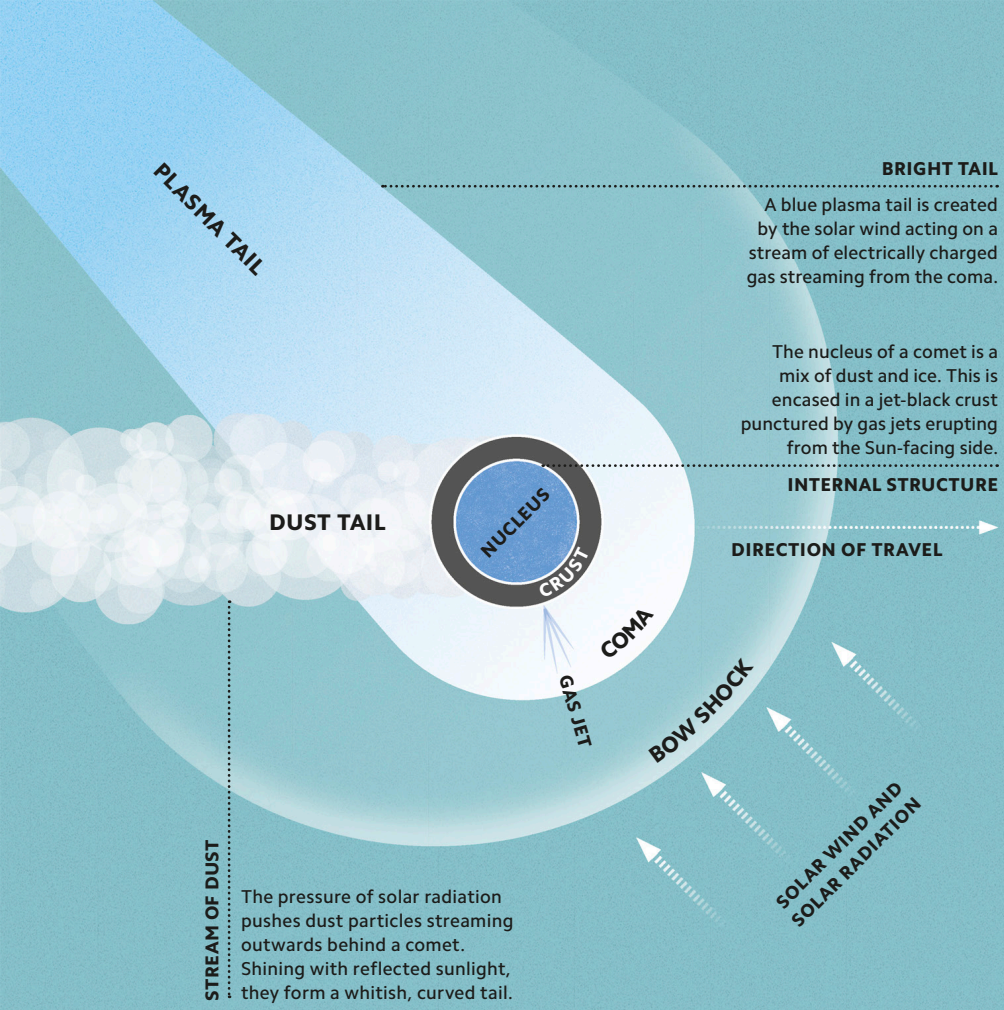
HELIOSPHERE

INTERSTELLAR MEDIUM

OORT CLOUD

Between star systems

The space between stars in a galaxy is called interstellar space. It is mostly filled with dust, gas, and cosmic rays. Stars form in the most dense regions of the interstellar medium.

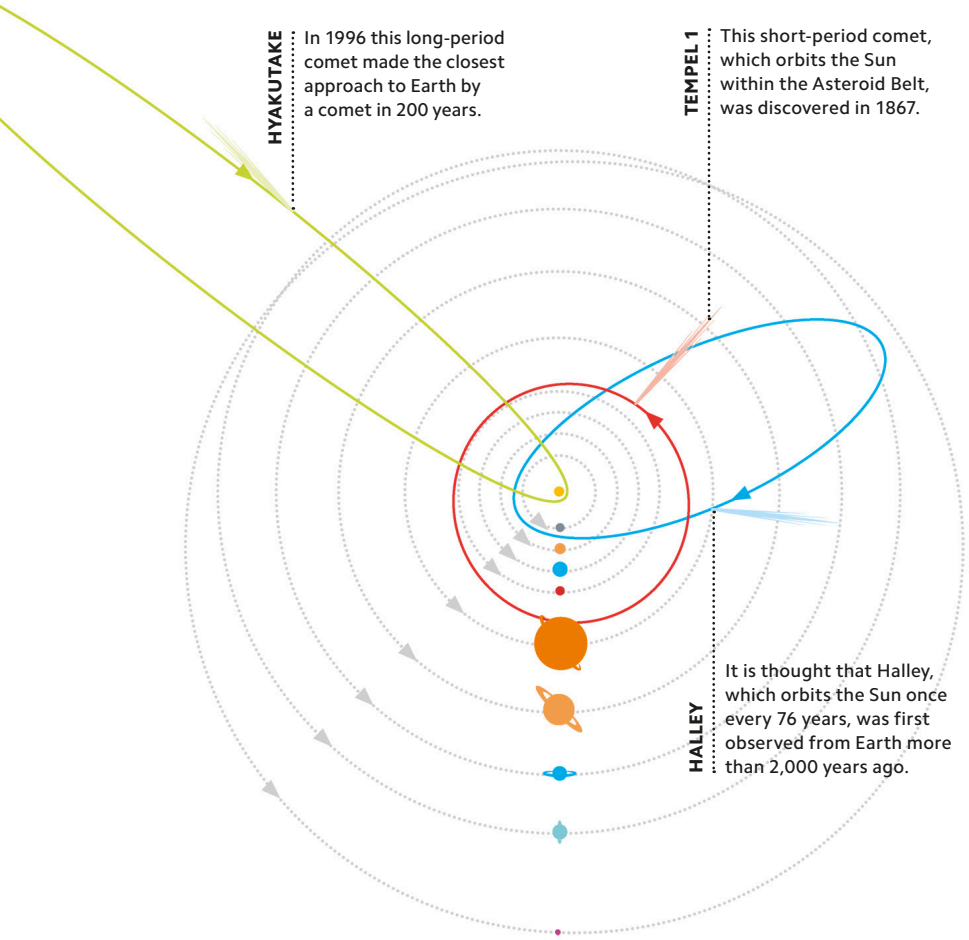


COSMIC SNOWBALLS

Comets are frozen bodies of dust and ice left over from the birth of the Solar System. Most are thought to come from the Oort Cloud (see p.72), although some come from the Kuiper Belt (see p.70). When a comet nears the Sun, solar heat vaporizes ice on its surface, creating a coma (atmosphere). A bow shock forms where the solar wind hits the coma. Interactions between the solar wind and the coma, along with solar radiation pressure on the coma, form two distinct tails.

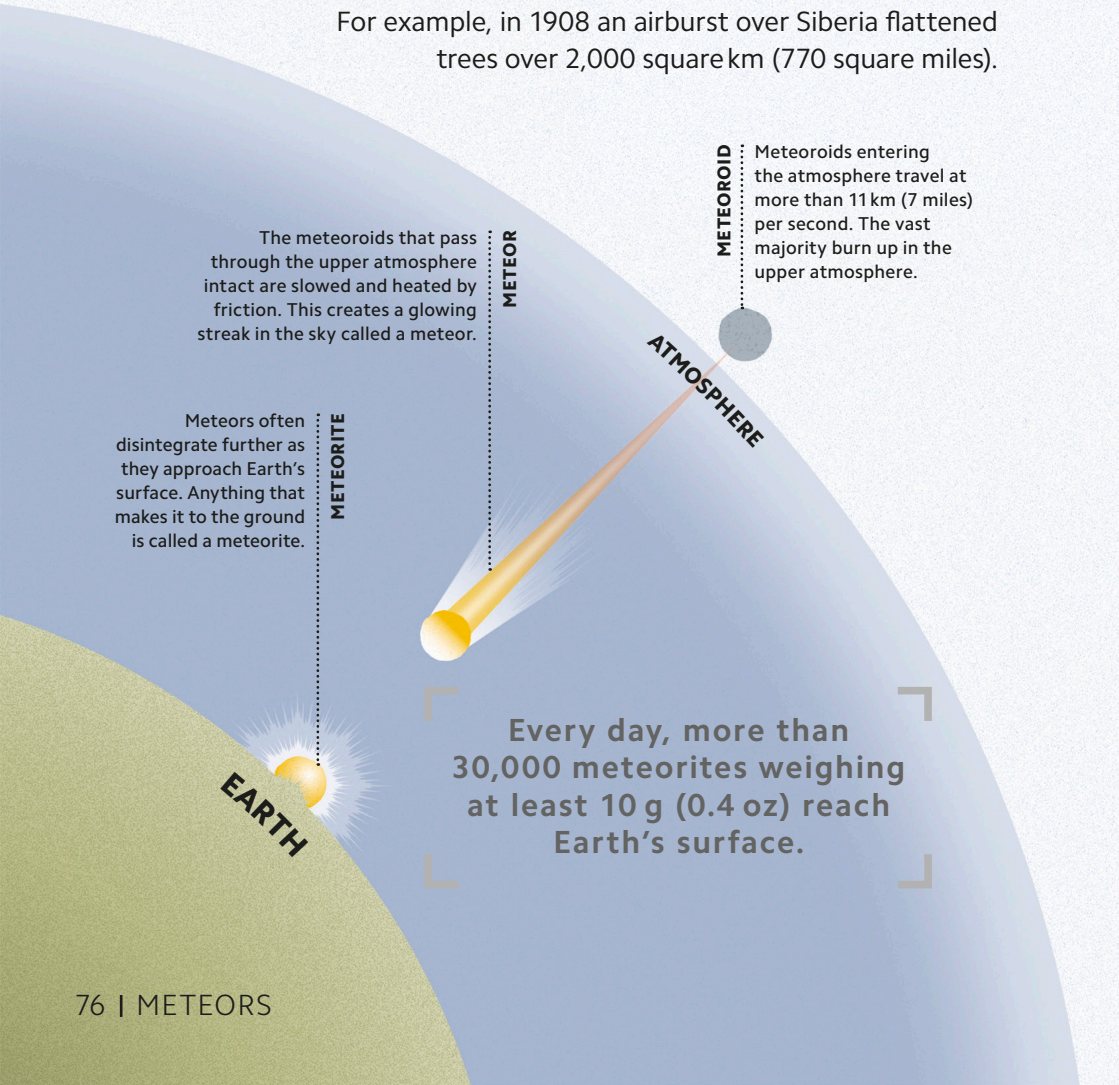
HOW COMETS ORBIT

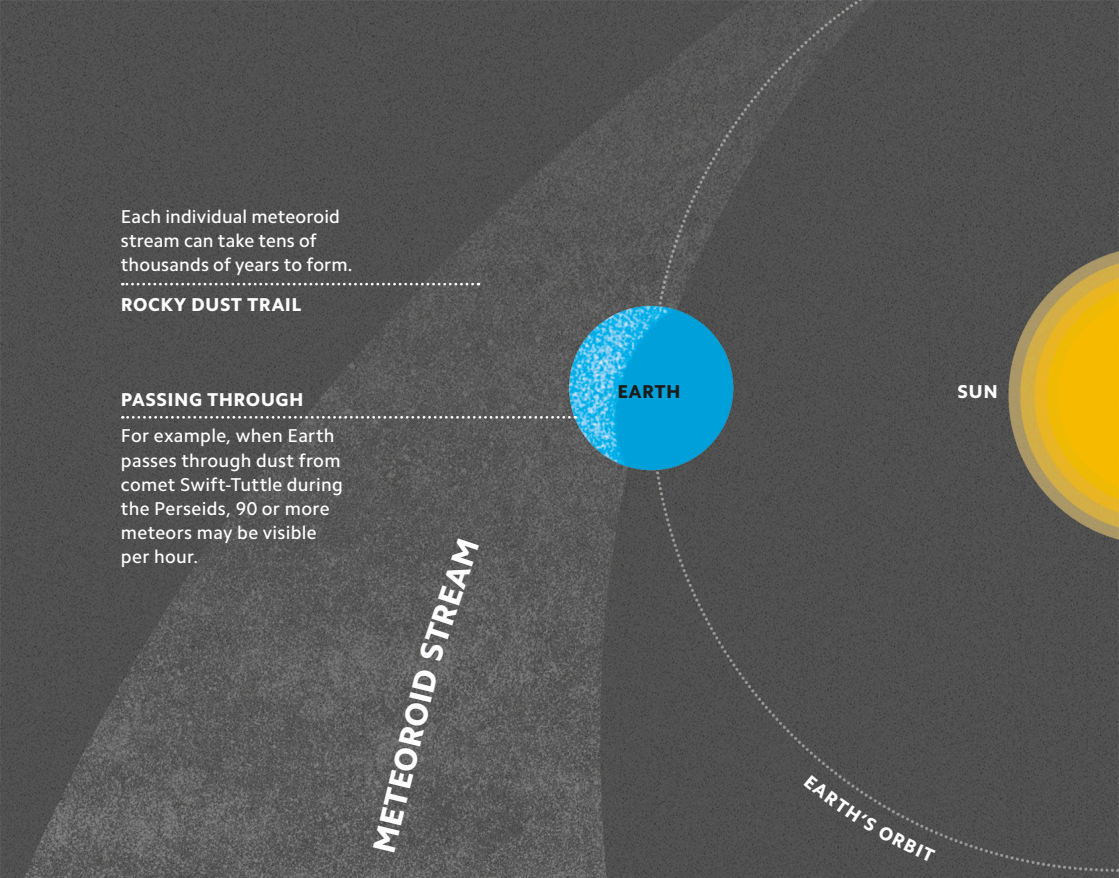
There are two main types of comet: long-period comets, which are thought to originate in the Oort Cloud and take more than 200 years to complete an orbit of the Sun; and short-period comets, which come from the Kuiper Belt and have an orbital period of less than 200 years. All comet orbits are elliptical (oval-shaped). Long-period comets often have orbits that are thin and elongated, while the orbits of short-period comets tend to be more circular.



STREAKS OF LIGHT

Meteors, or shooting stars, are flashes of light often visible from Earth's surface caused by stony or metallic objects called meteoroids entering Earth's atmosphere. Meteoroids are smaller than asteroids, and can be as small as a grain of sand. Those that are larger – around the size of a football – create much brighter flashes called fireballs that are visible for longer periods. Some meteoroids hit the atmosphere at such high speeds that they explode in mid-air, causing a meteor airburst. For example, in 1908 an airburst over Siberia flattened trees over 2,000 square km (770 square miles).





STREAMS OF DEBRIS

Comets and asteroids orbiting the Sun leave behind many streams of debris, mostly made up of rock and dust that breaks off in orbit. Some of these tiny objects, usually no larger than a pebble, lie in the path of Earth's orbit around the Sun. When Earth enters a stream of debris, more meteoroids than usual enter its atmosphere, causing what is known as a meteor shower. Meteor showers occur at roughly the same place in the sky at the same time every year, meaning that, although their intensity varies, they can be predictably observed from Earth's surface.

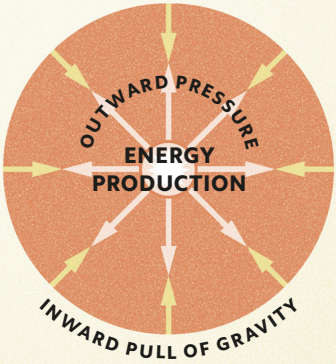
STAR GALA

SAND XIES

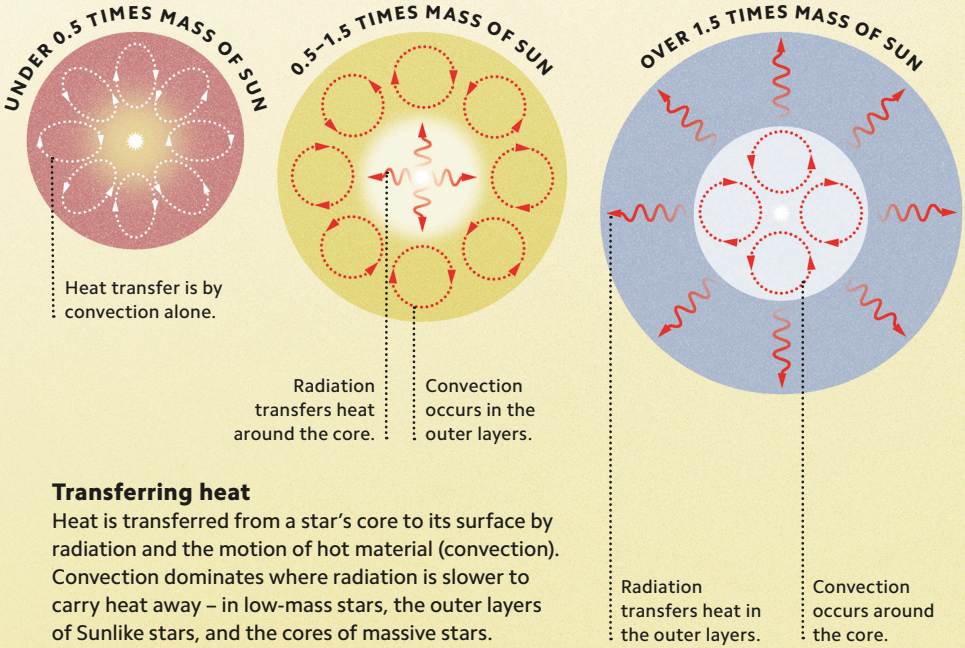
Despite its own great size, the Solar System is just a tiny part of a vast collection of stars, gas, and dust called the Milky Way galaxy – the Sun is just one of perhaps 400 billion stars in this assemblage. The Milky Way is a spiral galaxy, with several bright arms joined to its central nucleus, but galaxies can take various forms and they range widely in size. They can also be the focus of violent events. Collisions between galaxies are common. The black holes found at the centres of most galaxies can also drive highly energetic activity, emitting powerful blasts of radiation. Galaxies rarely occur in isolation but usually occur in groups called clusters that are, in turn, part of even larger superclusters.

LIGHTS IN THE SKY

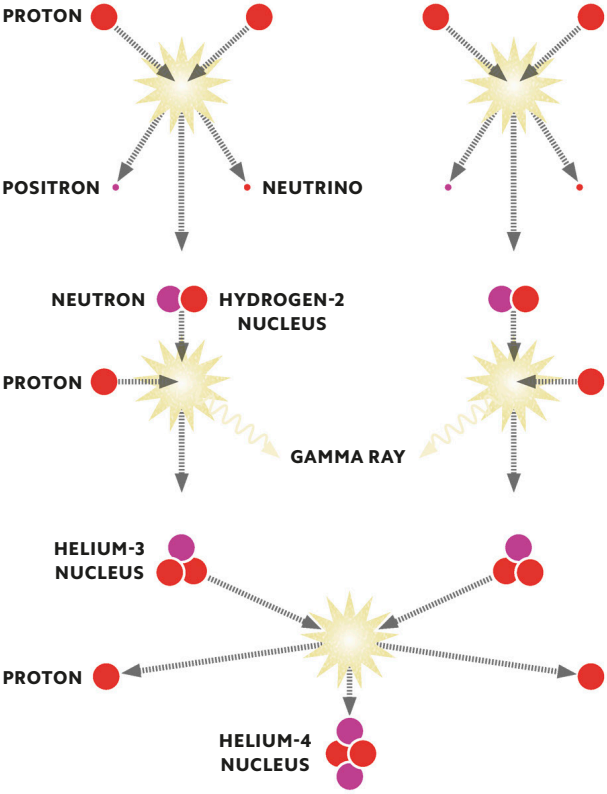
Each pinprick of light in the night sky is a distant sun – a massive ball of extremely hot, glowing gas. Nuclear fusion reactions in a star's core generate a huge amount of energy, which raises the temperature high enough to turn the star's gas into electrically charged plasma. Energy is transferred through the star's outer layers to the surface and emitted into space as heat and light.



Forces in stars
Stars would collapse under their own weight without the energy generated in their core. The pressure from expanding hot gas and radiation balances the inward pull of gravity.



Transferring heat
Heat is transferred from a star's core to its surface by radiation and the motion of hot material (convection). Convection dominates where radiation is slower to carry heat away – in low-mass stars, the outer layers of Sunlike stars, and the cores of massive stars.



THE PROTON-PROTON CHAIN

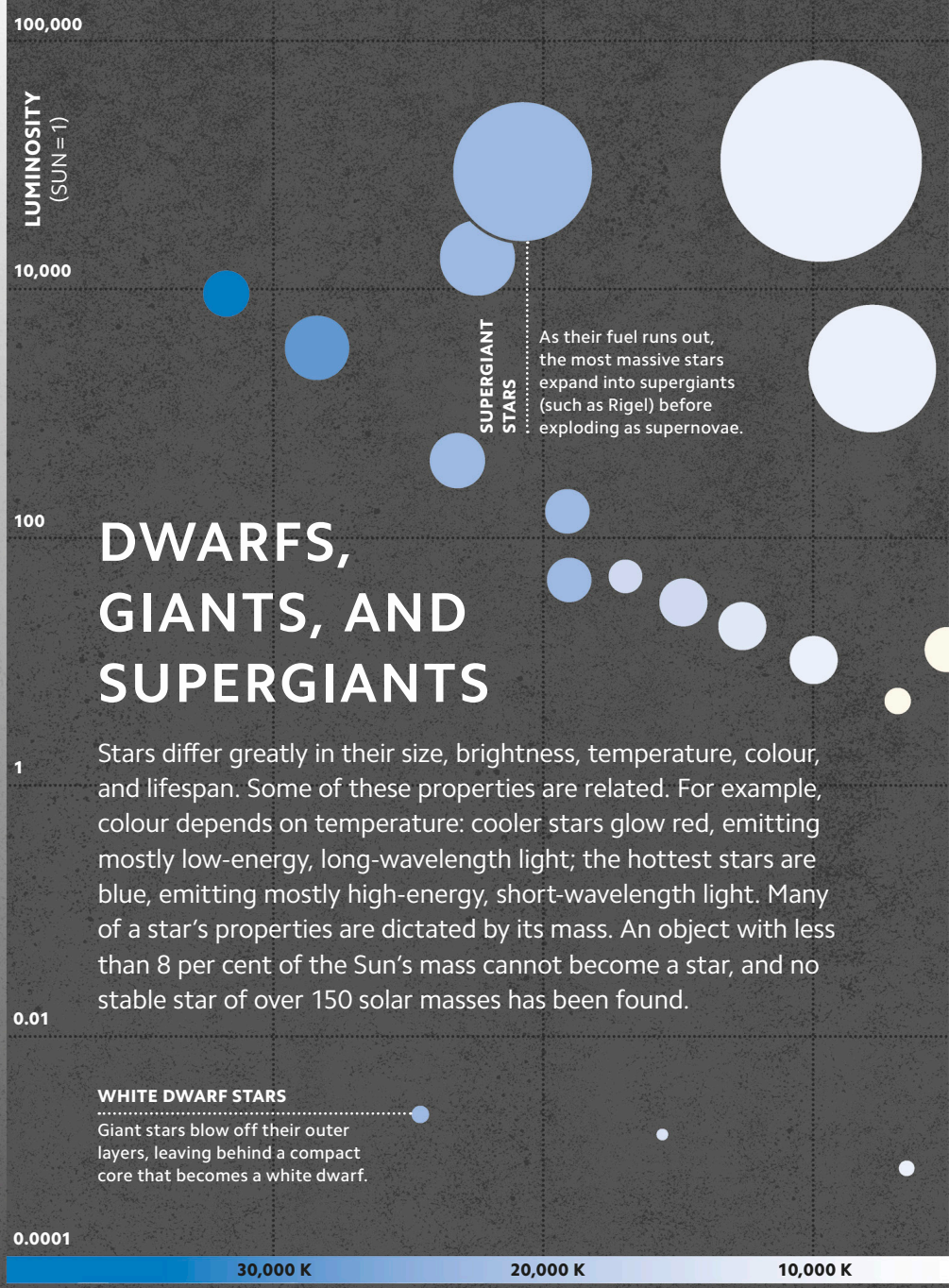
Protons fuse
Two protons fuse to form a hydrogen-2 nucleus (a proton and a neutron). A positron and a neutrino are released.

Energy is released
When another proton is encountered, a helium-3 nucleus is formed and energy is released in the form of a gamma ray.

Helium is produced
When two helium-3 nuclei collide, they fuse to form a single helium-4 nucleus, and two protons are released to carry on the chain reaction.

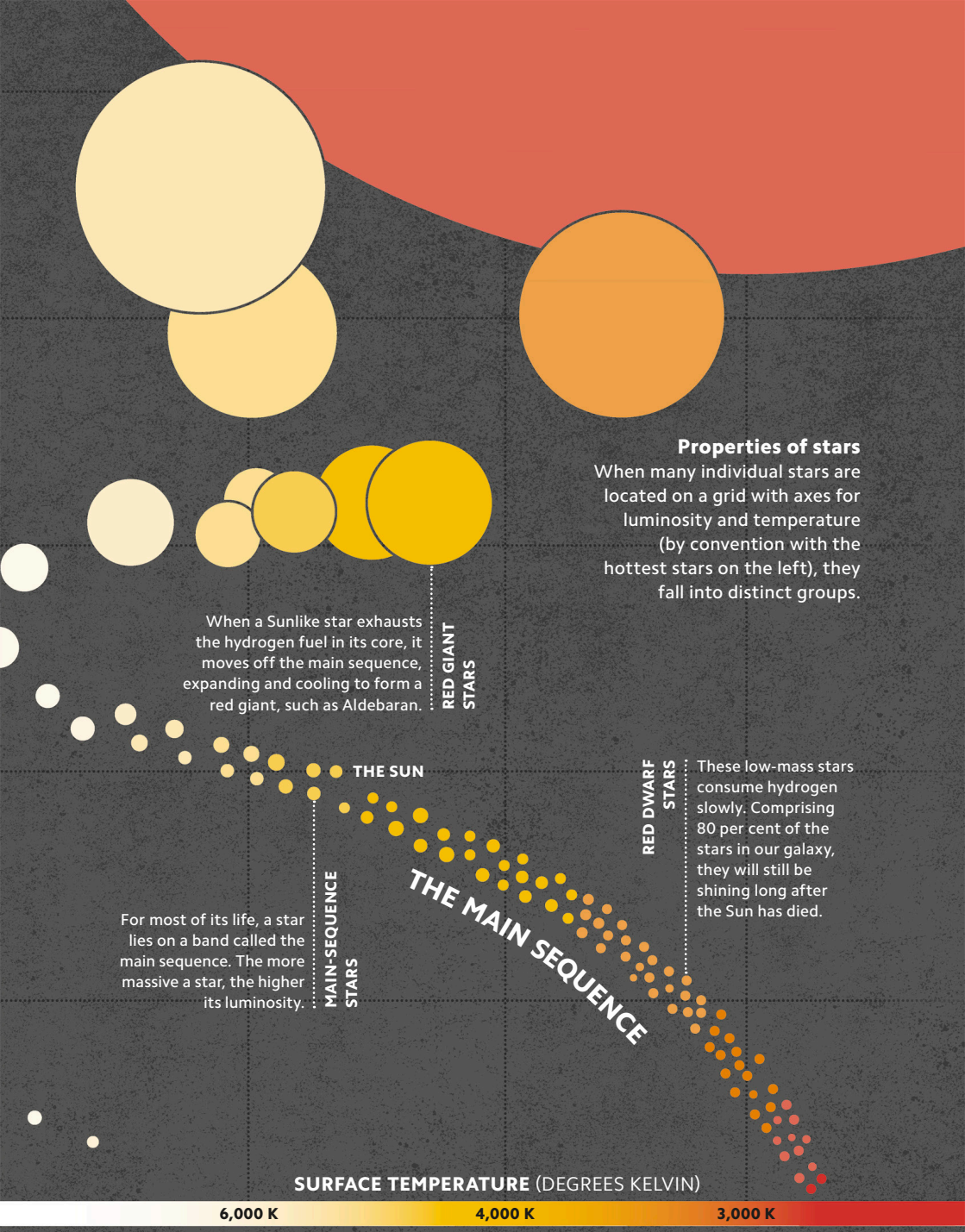
FROM MATTER TO ENERGY

The pressure and temperature in a star's core are high enough for hydrogen to be converted to helium by nuclear fusion. In a star like the Sun, this occurs through the proton-proton chain reaction. Energy is released when protons (hydrogen nuclei) combine to form heavier helium nuclei, with some protons left over to continue the reaction. In hotter, more massive stars, a more complex cycle converts hydrogen to helium using carbon, nitrogen, and oxygen as catalysts (the CNO cycle).



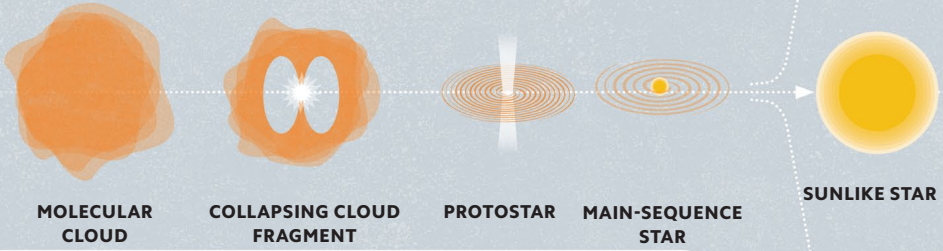
DWARFS, GIANTS, AND SUPERGIANTS

Stars differ greatly in their size, brightness, temperature, colour, and lifespan. Some of these properties are related. For example, colour depends on temperature: cooler stars glow red, emitting mostly low-energy, long-wavelength light; the hottest stars are blue, emitting mostly high-energy, short-wavelength light. Many of a star's properties are dictated by its mass. An object with less than 8 per cent of the Sun's mass cannot become a star, and no stable star of over 150 solar masses has been found.


Properties of stars
When many individual stars are located on a grid with axes for luminosity and temperature (by convention with the hottest stars on the left), they fall into distinct groups.
When a Sunlike star exhausts the hydrogen fuel in its core, it moves off the main sequence, expanding and cooling to form a red giant, such as Aldebaran.
RED GIANT STARS
THE SUN
RED DWARF STARS
These low-mass stars consume hydrogen slowly. Comprising 80 per cent of the stars in our galaxy, they will still be shining long after the Sun has died.
THE MAIN SEQUENCE
For most of its life, a star lies on a band called the main sequence. The more massive a star, the higher its luminosity.
MAIN-SEQUENCE STARS
SURFACE TEMPERATURE (DEGREES KELVIN)
6,000 K
4,000 K
3,000 K

STAR LIFE AND DEATH

All stars form in the same way from collapsing clouds of interstellar dust and gas, but once a star is born its evolution depends on its mass, which controls how the star uses up its fuel. A star may last from 10 million to 100 billion years. Stars like the Sun return much of their material back into space in their old age, seeding the next generation of stars.



MOLECULAR CLOUD

COLLAPSING CLOUD FRAGMENT

PROTOSTAR

MAIN-SEQUENCE STAR

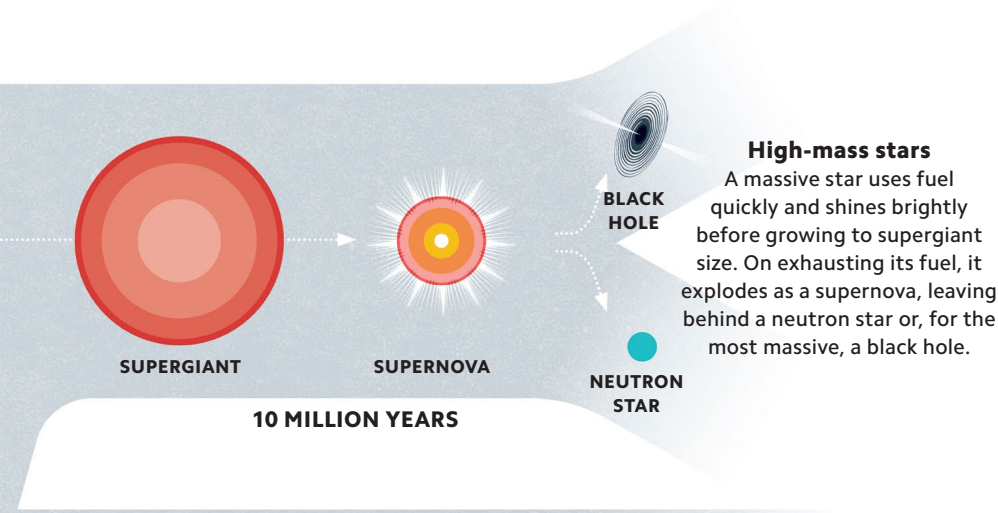
MASSIVE STAR

SUNLIKE STAR

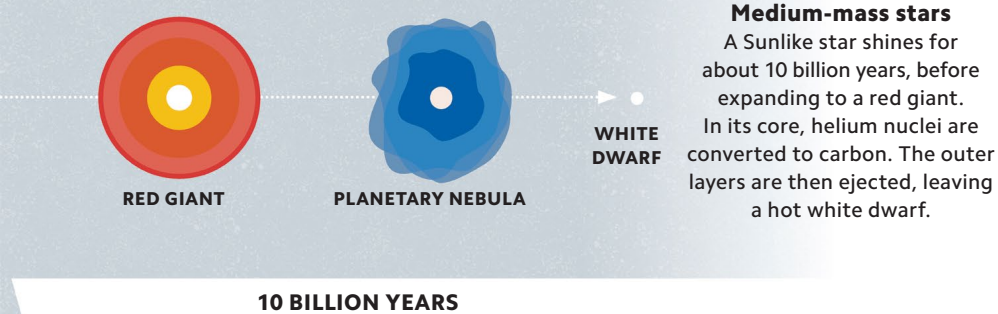
RED DWARF

Mass and lifespan
The more massive the star, the faster it uses up its fuel and the shorter its life. The least massive stars have yet to reach their theoretical ends as black dwarfs, since their lifespans are longer than the age of the Universe.

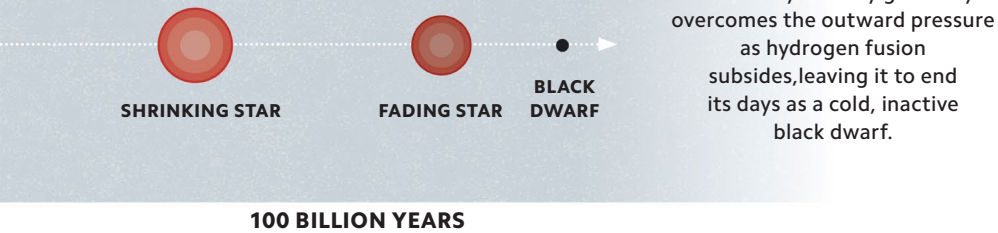
A star like the Sun takes about 50 million years to reach its stable main-sequence stage.



High-mass stars
A massive star uses fuel quickly and shines brightly before growing to supergiant size. On exhausting its fuel, it explodes as a supernova, leaving behind a neutron star or, for the most massive, a black hole.

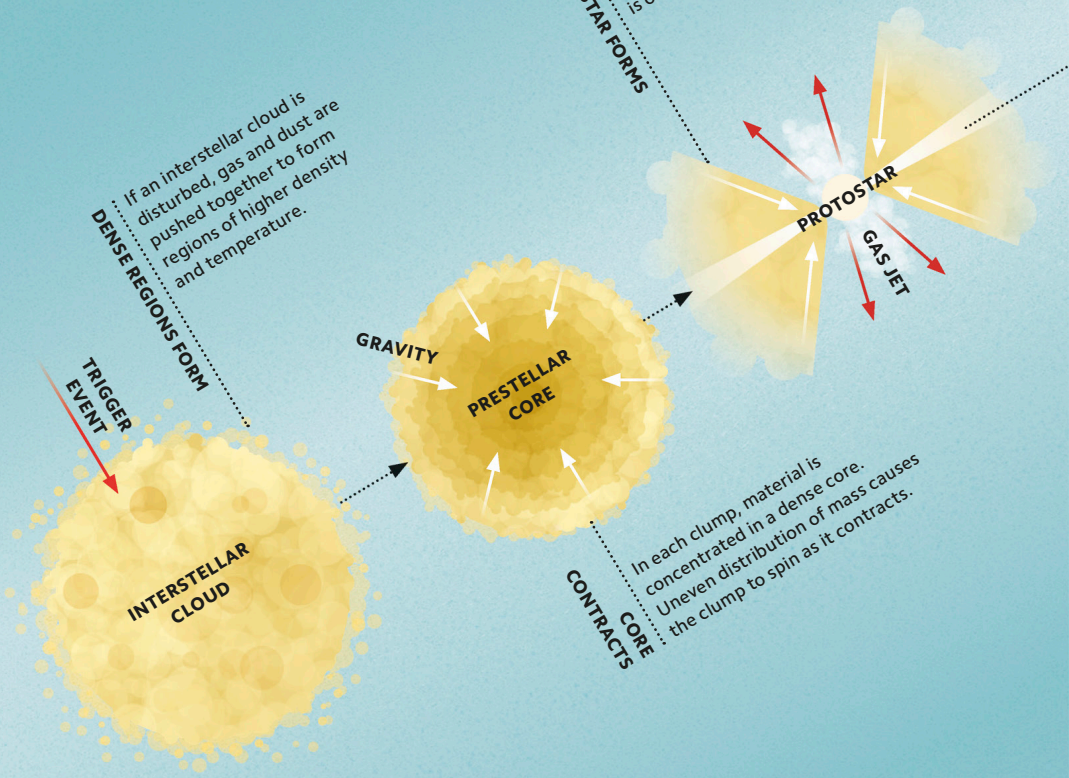


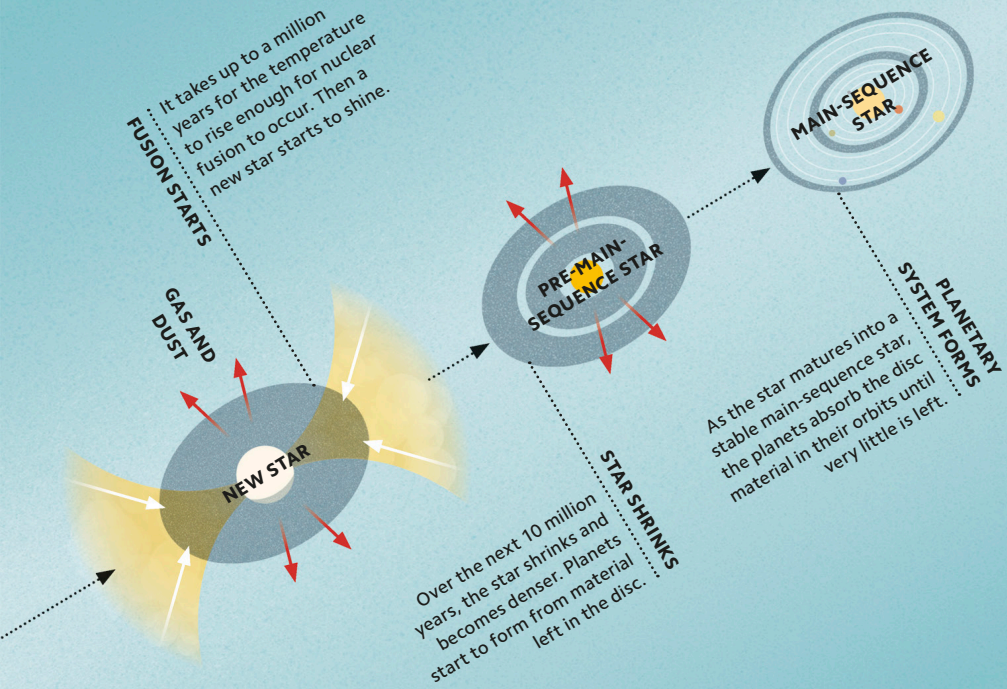
Medium-mass stars
A Sunlike star shines for about 10 billion years, before expanding to a red giant. In its core, helium nuclei are converted to carbon. The outer layers are then ejected, leaving a hot white dwarf.



Low-mass stars
A red dwarf consumes its fuel slowly. Gravity gradually overcomes the outward pressure as hydrogen fusion subsides,leaving it to end its days as a cold, inactive black dwarf.

The largest molecular clouds can be up to 600 light-years across – 140 times further than the distance to our nearest star.





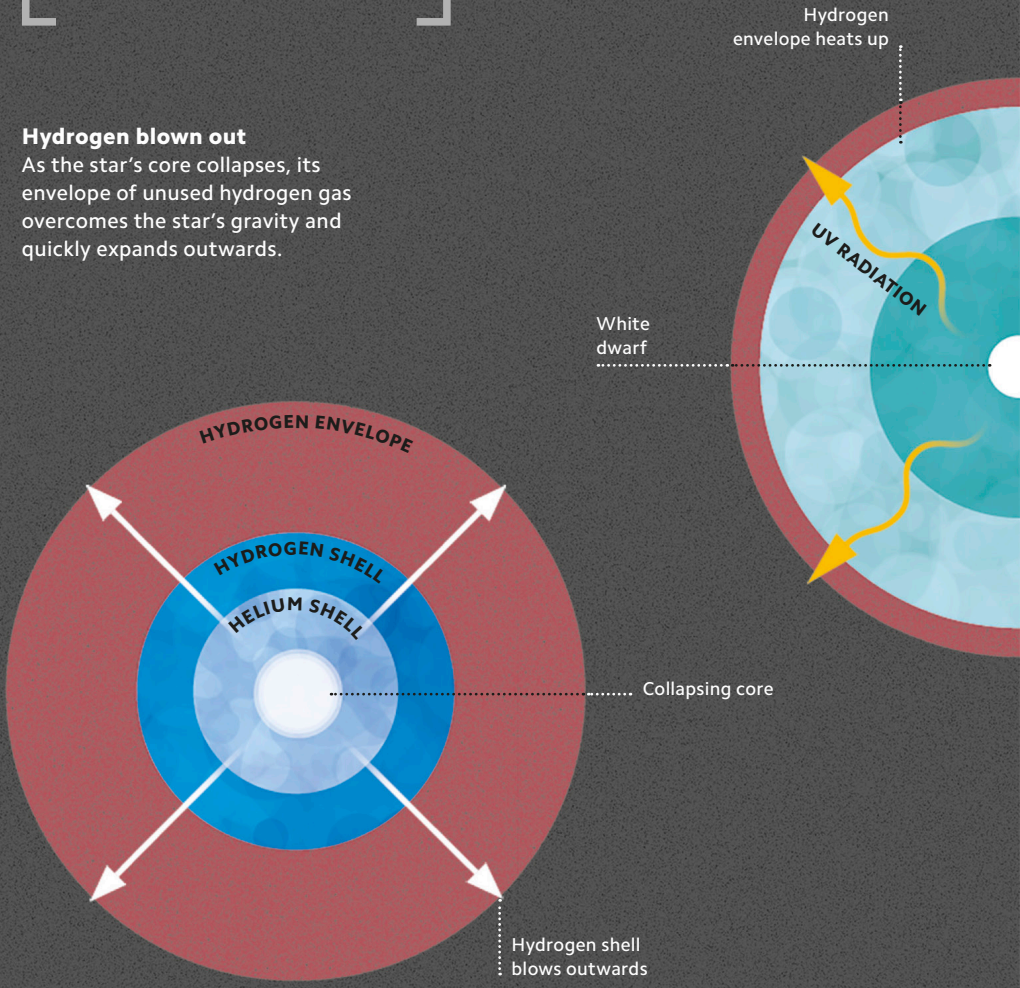
STARBIRTH

Stars are born from vast clouds of gas and dust, consisting mostly of molecules of hydrogen. A trigger event, such as the gravitational tug of a passing star or the shockwave from a nearby supernova explosion, increases the density in parts of the molecular cloud. When these clumps become dense and hot enough for nuclear fusion to occur, stars are born. As part of the same process, planets can form out of the disc of material rotating around the young star.

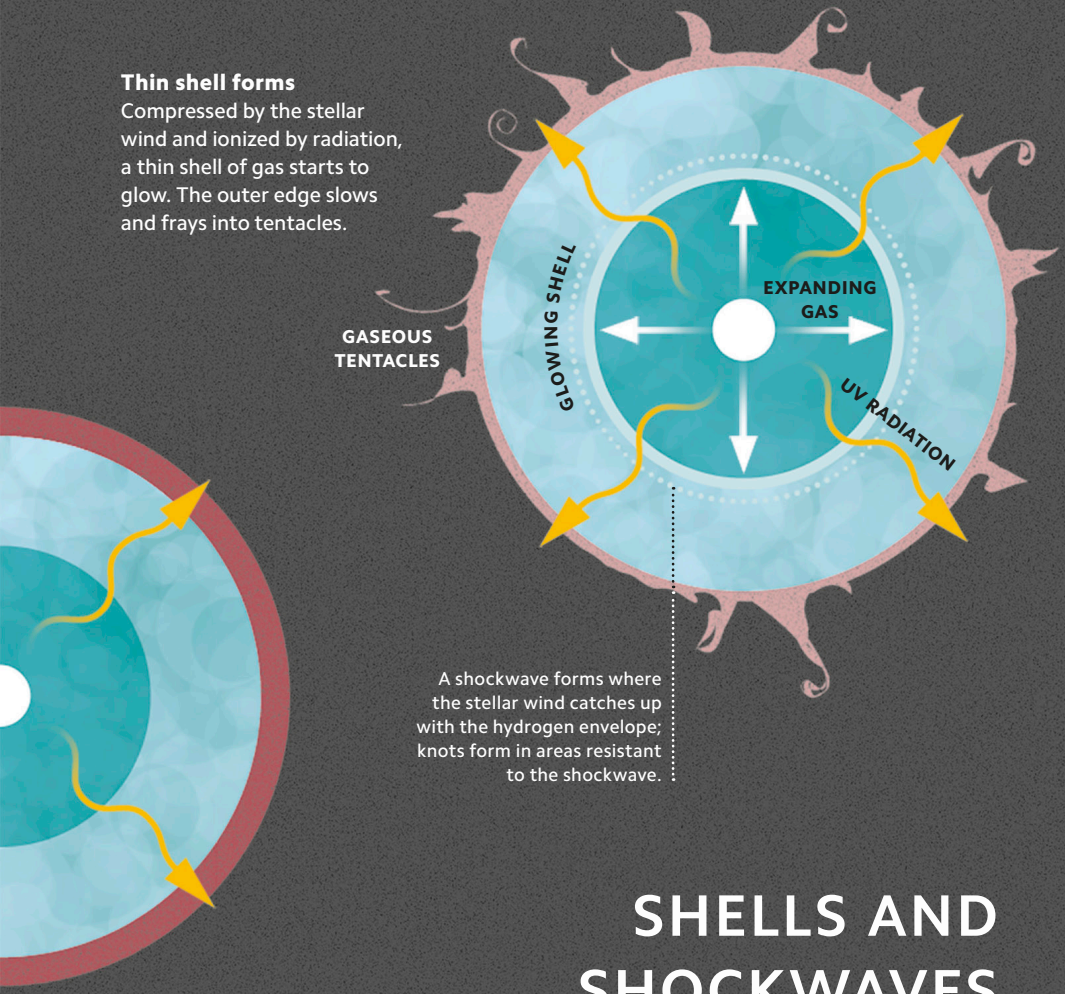
> Looking like planets in early telescopes, these clouds are called planetary nebulae, but they have nothing to do with planets.

Radiation emitted
The hot core is now a white dwarf, emitting a powerful stellar wind and intense ultraviolet (UV) radiation that heats the enriched hydrogen.

Hydrogen blown out
As the star's core collapses, its envelope of unused hydrogen gas overcomes the star's gravity and quickly expands outwards.

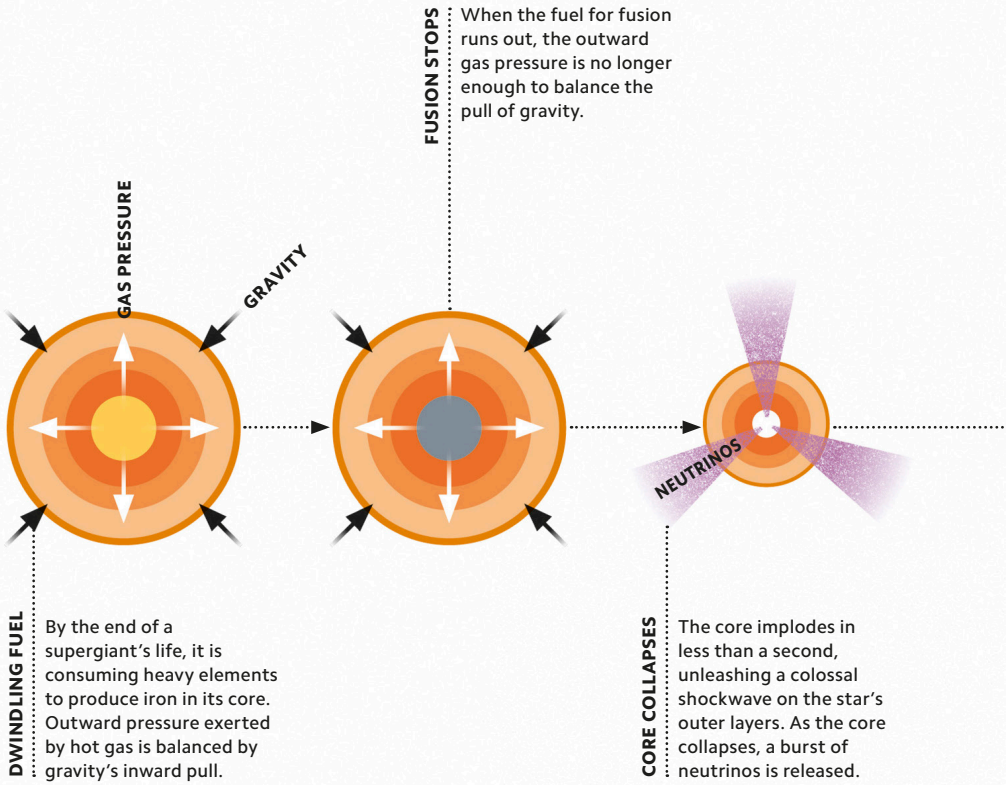


Thin shell forms
Compressed by the stellar wind and ionized by radiation, a thin shell of gas starts to glow. The outer edge slows and frays into tentacles.



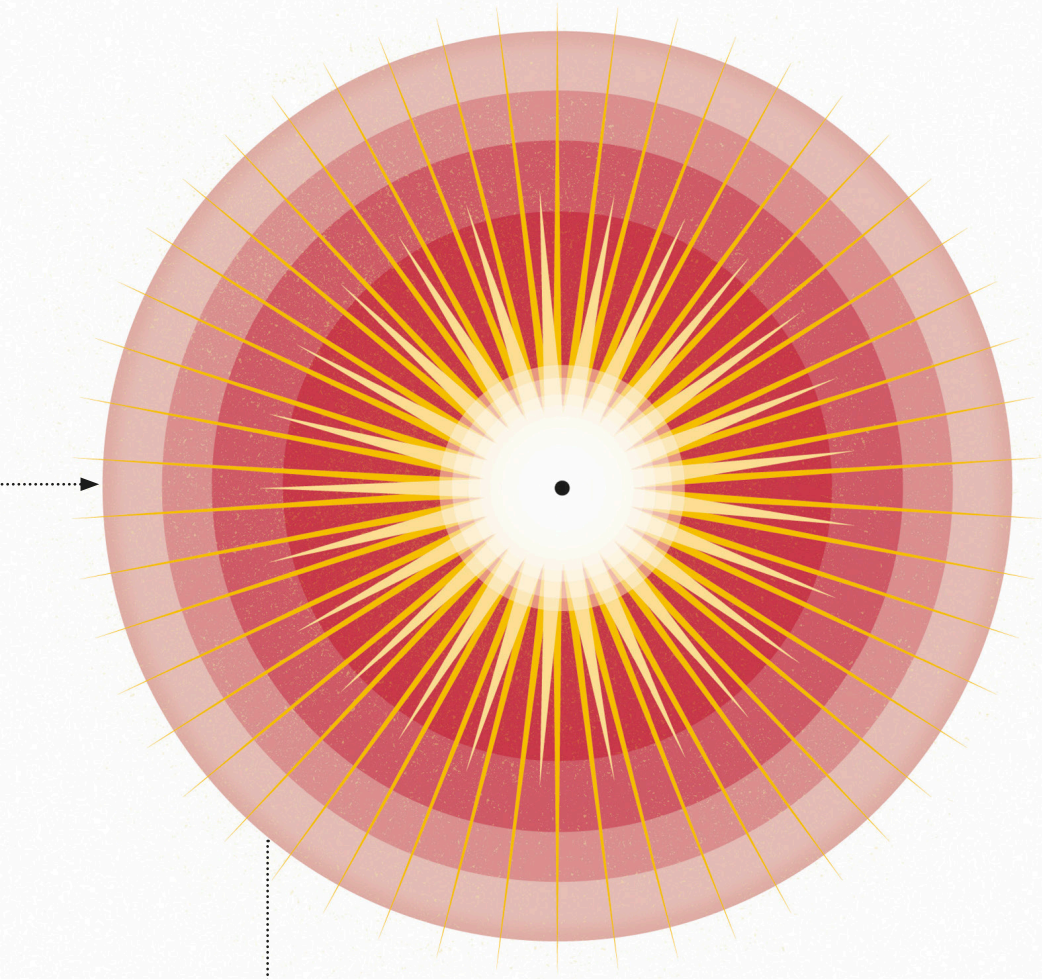
SHELLS AND SHOCKWAVES

When a red giant star reaches the end of its life, it ejects shells of gas and dust into space. Heated by radiation from the star's hot core and compressed by shockwaves from its stellar wind of charged particles, these shells of hydrogen gas, enriched with other elements produced inside the parent star, glow in various colours. Known as planetary nebulae, these beautiful objects are short lived and, after a few thousand years, disperse into space.



EXPLODING STARS

Stars can end in spectacularly bright explosions called supernovae. Some result when a massive star's core collapses suddenly after exhausting all its nuclear fuel. A supernova is also triggered when a white dwarf captures extra mass from a companion star or merges with another white dwarf. Briefly outshining an entire galaxy, a supernova usually fades within a few months.



STAR EXPLODES

The shockwave blasts apart the star's outer layers, which expand as an intensely bright cloud of gas – a type II supernova. The superdense remnant of the core remains.

A supernova can emit more light than several billion suns and outshine entire galaxies.

MAGNETIC FIELD

The powerful magnetic field funnels electromagnetic radiation out in jets from the magnetic poles.

RADIATION BEAM

CRUSHED STARS

When a star explodes as a supernova, the core collapses into an extremely hot, extremely dense object. If the star's core has between 1.2 and 2.2 times the mass of the Sun, the result is a neutron star with a temperature of up to 600,000° C (1,000,000° F) and a density approaching that of an atomic nucleus. Neutron stars are tiny – some only 10 km (6 miles) across – and have such strong gravitational fields that electrons and protons are crushed together to form neutrons – hence their name.

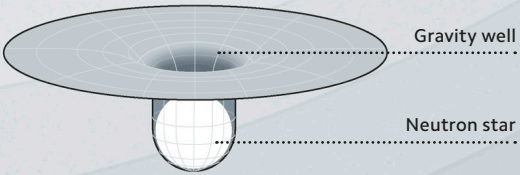
The contraction of a large mass to a small size means neutron stars spin rapidly (over 700 times per second in some cases), producing a magnetic field up to a million times stronger than Earth's.

NEUTRON STAR

RAPID SPIN

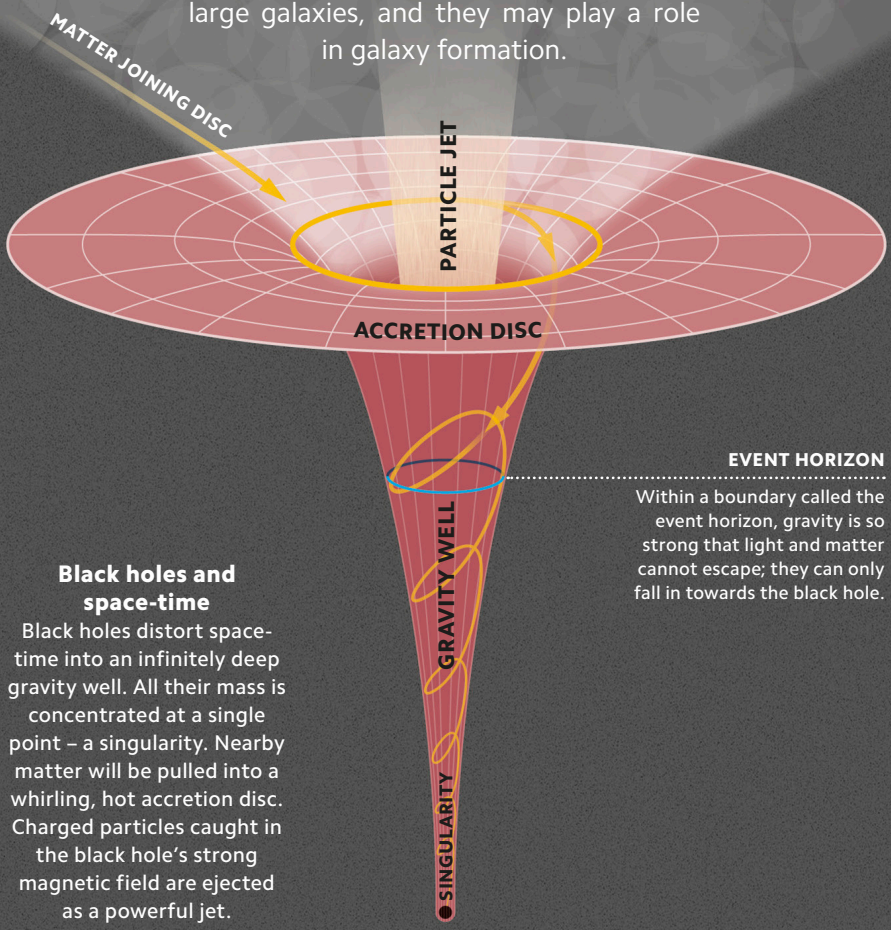
Space-time distortion

With its large mass squashed into a small space, a neutron star has a strong gravitational effect. On a space-time grid (see pp.26-27), this produces a steep-sided dent called a gravity well.



COLLAPSING TO INFINITY

If the remnant from a supernova explosion has more than about three times the mass of the Sun, it will collapse to form a stellar-mass black hole. Black holes are infinitely small and infinitely dense. Their gravity is so strong that no matter, or even light, can escape. Supermassive black holes, more than 100,000 times the mass of the Sun, are thought to lie at the centres of most large galaxies, and they may play a role in galaxy formation.

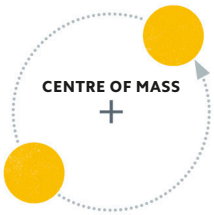


Black holes and space-time

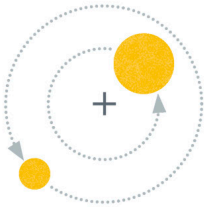
Black holes distort space-time into an infinitely deep gravity well. All their mass is concentrated at a single point – a singularity. Nearby matter will be pulled into a whirling, hot accretion disc. Charged particles caught in the black hole's strong magnetic field are ejected as a powerful jet.

EVENT HORIZON

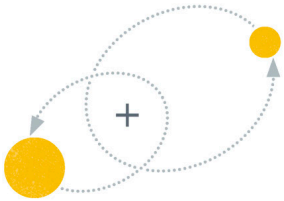
Within a boundary called the event horizon, gravity is so strong that light and matter cannot escape; they can only fall in towards the black hole.



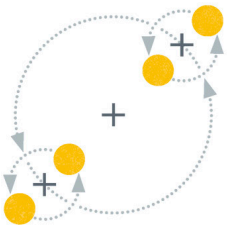
Equal mass
In a system of two stars on circular orbits, both orbit at the same distance from the balancing point between them (the centre of mass).



Unequal mass
If one of the stars in a binary system is more massive than the other, it will orbit closer to the pair's centre of mass.



Significant difference in mass
In most binary systems, the stars follow elliptical orbits in which the centre of mass lies at one focus of both ellipses.



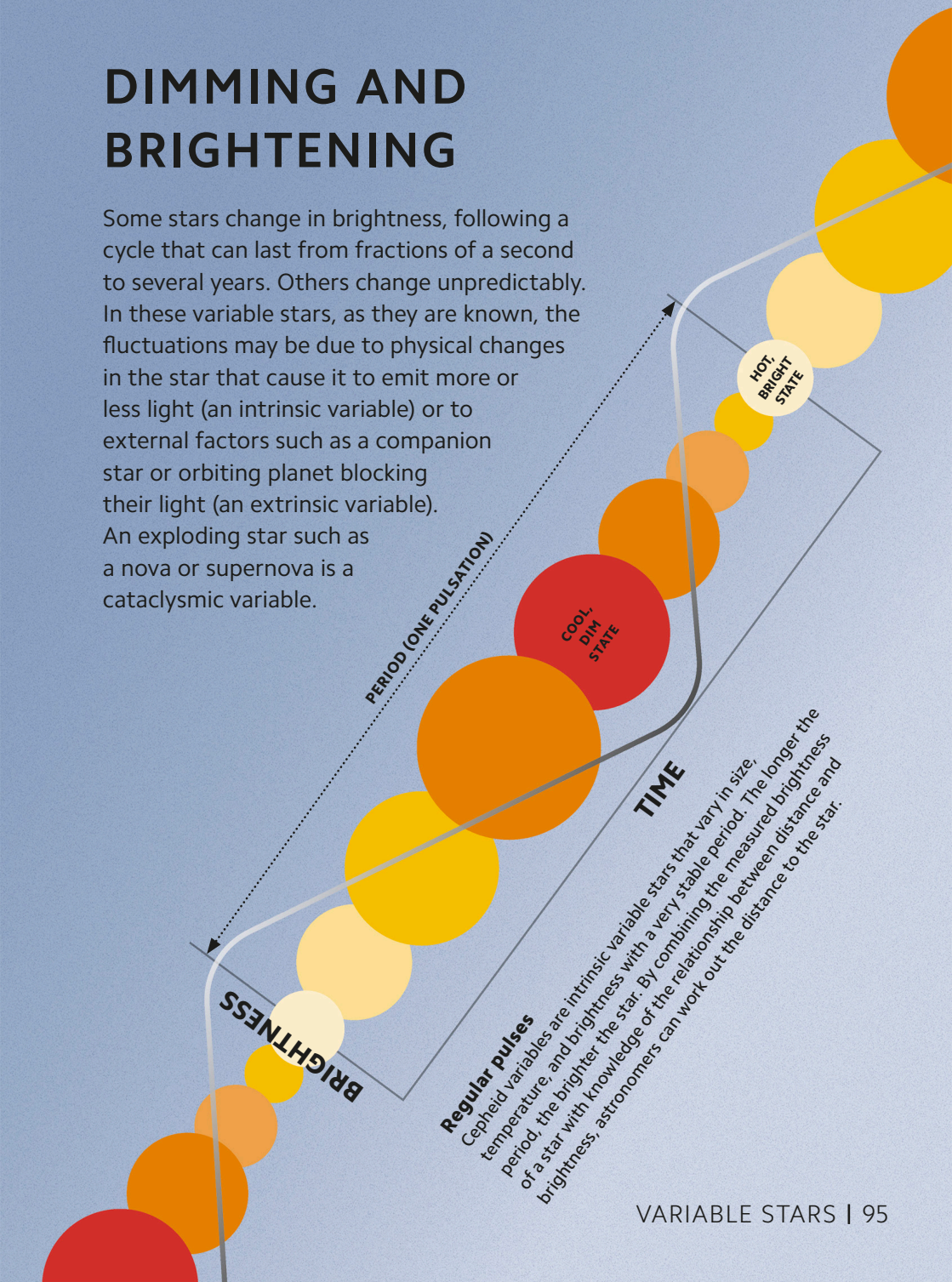
Double binary
In a double binary system, the centres of mass of two binary systems orbit around the common centre of mass of the four stars.

COMPANION STARS

Most stars are multiple stars – found in systems of two or more, bound together by gravity. A pair of stars is called a binary star, but some systems have as many as seven stars. If a large star is close enough to draw material from a smaller companion, the pair is called an interacting binary.

DIMMING AND BRIGHTENING

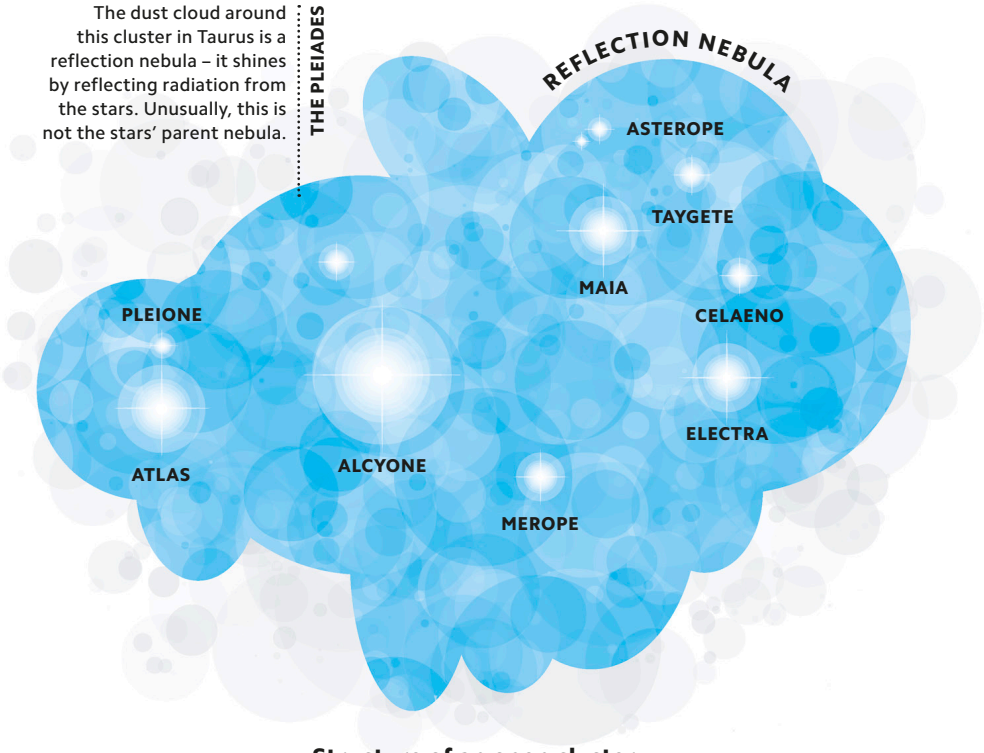
Some stars change in brightness, following a cycle that can last from fractions of a second to several years. Others change unpredictably. In these variable stars, as they are known, the fluctuations may be due to physical changes in the star that cause it to emit more or less light (an intrinsic variable) or to external factors such as a companion star or orbiting planet blocking their light (an extrinsic variable). An exploding star such as a nova or supernova is a cataclysmic variable.



SIBLING STARS

Stars that are born together stay together, at least initially, in groups called clusters. There are two main kinds of cluster: open and globular (see opposite). An open cluster is a loose group of stars formed from the same cloud of gas and dust. There may be a few hundred stars in the group: mostly young, hot blue stars less than a hundred million years old. There are over 2,000 open clusters in the galactic disc of the Milky Way (see p.103).

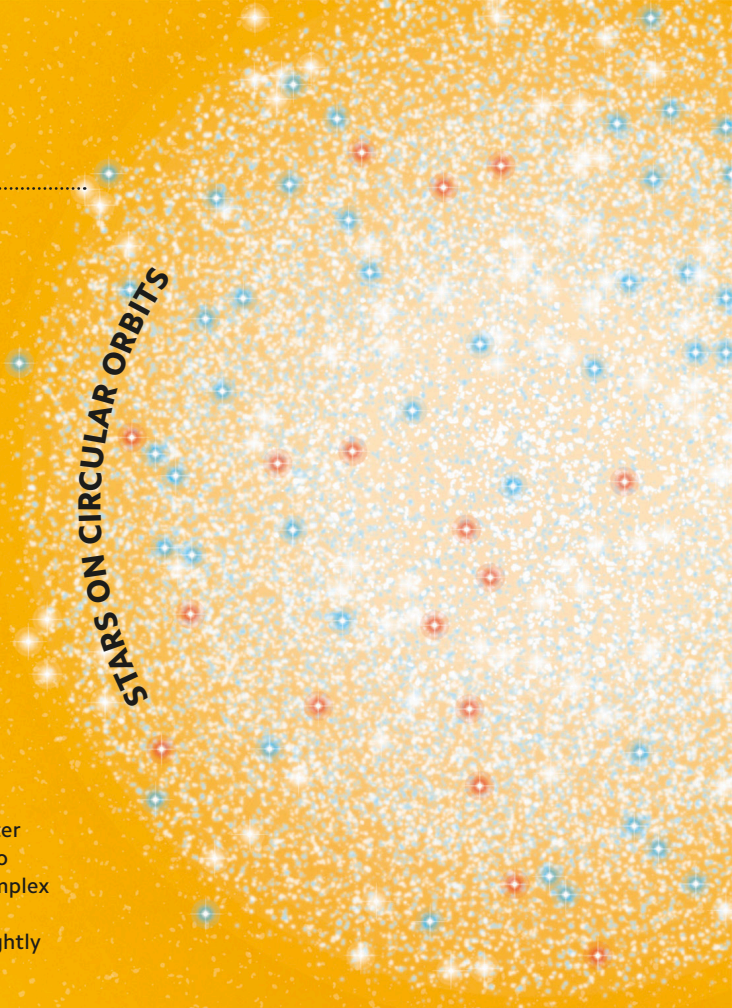
The dust cloud around this cluster in Taurus is a reflection nebula – it shines by reflecting radiation from the stars. Unusually, this is not the stars' parent nebula.



Structure of an open cluster
Open clusters are often found close to their parent clouds of gas and dust, but over a few hundred million years the stars move apart and the cluster disperses.

OMEGA CENTAURI

The stars in Omega Centauri have a greater range of ages than in most globular clusters, suggesting it may be the core of a dwarf galaxy captured by the Milky Way.



Structure of a globular cluster

The stars in a globular cluster are gravitationally bound to each other. They follow complex orbits within a spherical distribution that is more tightly packed towards the centre.

STAR CITIES

Compared with open clusters, globular clusters contain more stars – up to several million – and the stars are older: typically 10 billion years, almost as old as the Universe itself. Globular clusters are some of the first parts of a galaxy to form, perhaps before the disc. New star formation has largely ceased, all the raw material in the initial gas cloud having been used up. The Milky Way has at least 150 globular clusters, mostly outside the disc, in the halo.

PLANETS AROUND OTHER STARS

So far, over 4,000 planets have been found orbiting stars outside the Solar System, suggesting planet formation is relatively common. These extrasolar planets (exoplanets) range from gas giants twice the size of Jupiter to small, rocky, Mercury-sized worlds. Most are too distant to be directly observed from Earth. They have instead been detected by the effect they have on their parent star as they orbit it, such as causing the star to wobble or periodically dimming its light.

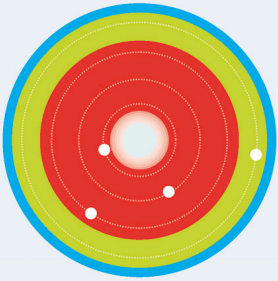
Neptunelike
Similar in size to Neptune or Uranus, Neptunian planets are likely to have atmospheres dominated by hydrogen and helium, and rocky cores.

Gas giant
The first exoplanets to be found were gas giants like Jupiter or Saturn, because they have the strongest gravitational effects on their parent stars.

Super-Earth
Planets that are larger than Earth, but smaller than Neptune, can be super-Earths if they are mostly rock or mini-Neptunes if they are mostly gas.

Terrestrial
A classic terrestrial planet is Earth-sized or smaller, composed mainly of rock, perhaps with an atmosphere or even oceans.

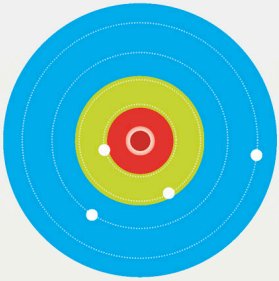
Hotter star
The location of a star's habitable zone depends on its temperature. If the Sun were hotter, there could be a habitable zone beyond Earth's orbit.



Sunlike star
In the Solar System, Earth's orbit lies comfortably within the Sun's habitable zone, but Venus is too hot, and Mars is too cold.



Cooler star
If the Sun were cooler, Mercury and Venus might be habitable but Earth might join Mars in the zone too cold for life.

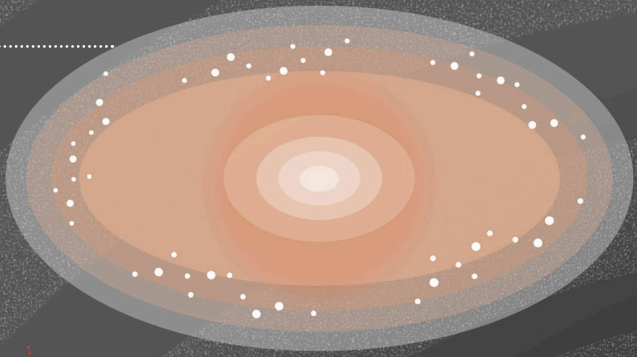


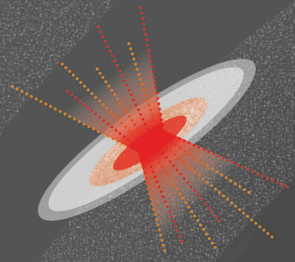
A PLACE TO LIVE

The search is now on for a second Earth – an exoplanet capable of supporting life. Such a planet would have to orbit within the habitable zone of its star, where it is warm enough for liquid water to exist but not so hot that all the water boils away. The atmospheres of planets in these zones will be examined for the presence of certain combinations of gases and the kind of chemical reactions that could be taking place, which might be signs of life.

ELLIPTICAL

Elliptical galaxies are featureless oval balls with no structure and contain mostly older stars. They range in size from small dwarfs to giant ellipticals.

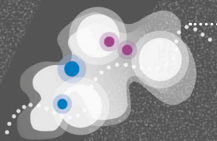




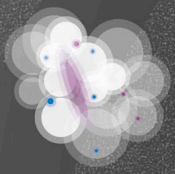
CIGAR GALAXY

IRREGULAR GALAXIES

The smallest galaxies are irregular. But larger galaxies can also have irregular shapes following a collision or near-miss with another galaxy. Three examples of irregular galaxies are shown here.



NGC 4449

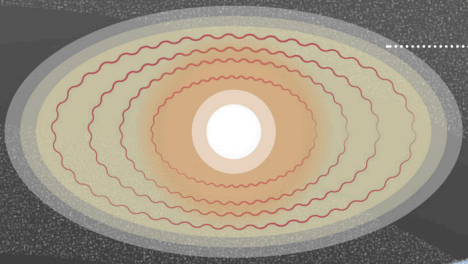


LARGE MAGELLANIC CLOUD

Types of galaxy

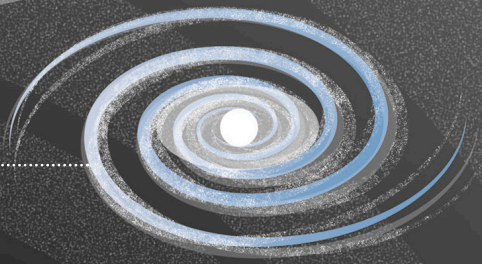
Galaxies are classified by shape, but a galaxy's classification may change as it evolves. Small galaxies can grow by colliding and merging, with structure emerging as the mass increases and the gravitational field becomes stronger.

> **"There are at least as many galaxies in our observable universe as there are stars in our galaxy."**
>
> Sir Martin Rees, cosmologist



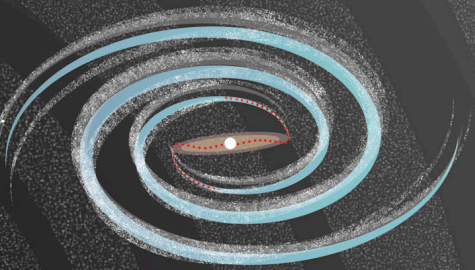
LENTICULAR

A lenticular galaxy is shaped like an optical lens. Some lenticulars may be spirals that have lost most of their gas and dust.



SPIRAL

Spiral galaxies have a dense nucleus of older stars, from which radiates a series of rotating arms of gas, dust, and newly formed stars.

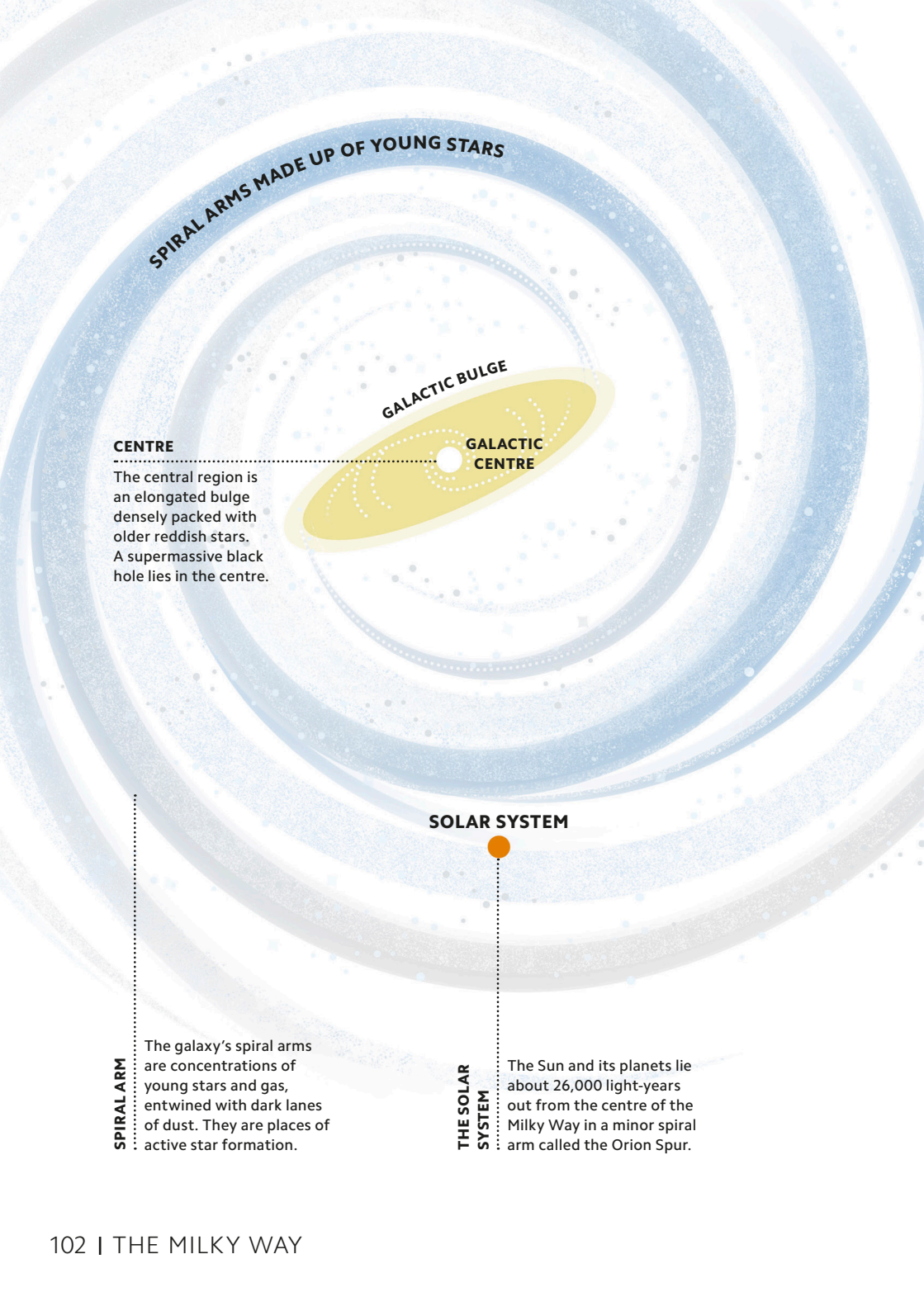


BARRED SPIRAL

In a barred spiral galaxy, the central bulge is elongated, with two dominant spiral arms extending from each end of the bar.

ISLANDS OF STARS

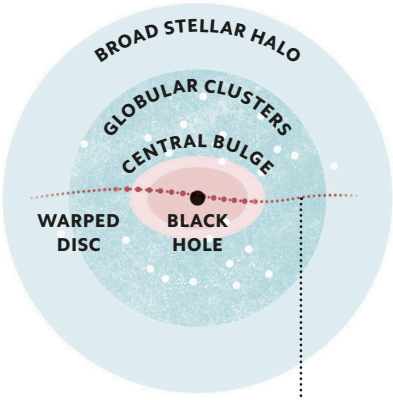
A galaxy is a large collection of stars, gas, and dust held together by gravity. The Universe contains about 2 trillion galaxies, ranging in size from a few thousand to more than 100,000 light-years across. They are the only places where matter is packed together tightly enough for stars to form. The smallest galaxies contain a few million stars and the largest have trillions, all orbiting around the galaxy's centre of mass.


SPIRAL ARMS MADE UP OF YOUNG STARS
GALACTIC BULGE
GALACTIC CENTRE
CENTRE
The central region is an elongated bulge densely packed with older reddish stars. A supermassive black hole lies in the centre.
SOLAR SYSTEM
SPIRAL ARM
The galaxy's spiral arms are concentrations of young stars and gas, entwined with dark lanes of dust. They are places of active star formation.
THE SOLAR SYSTEM
The Sun and its planets lie about 26,000 light-years out from the centre of the Milky Way in a minor spiral arm called the Orion Spur.

> "The Milky Way is nothing else but a mass of innumerable stars planted together in clusters."
>
> Galileo Galilei

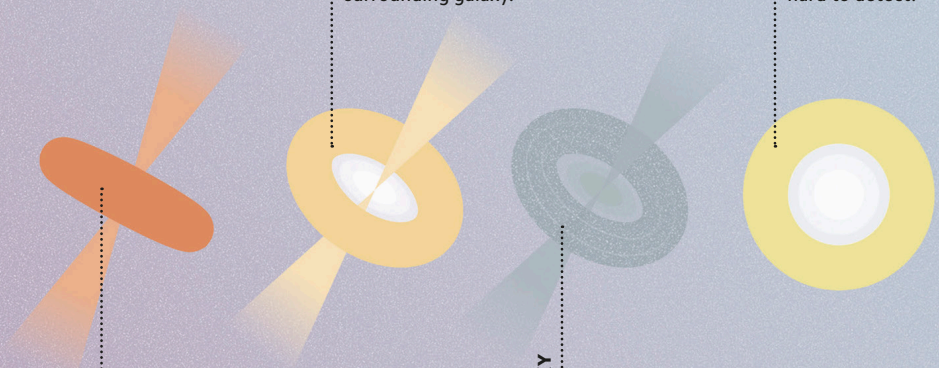
OUR GALAXY

The hazy band of starlight we see in the night sky is our galaxy viewed from within. The Milky Way is a barred spiral galaxy (see p.101) containing 100–400 billion stars. At its centre is an elongated bulge of old yellow stars, from which a number of spiral arms extend to form a thin disc of mostly young blue stars. All the galaxy's gas, dust, and stars are held in orbit around its centre by gravitational attraction.



Side view
Surrounding our galaxy's central bulge and bar is a spherical halo of older stars. Many of these are packed into about 150 globular clusters, which contain some of the galaxy's oldest stars.

GALACTIC DISC
A disc of stars, gas, and dust extends from the bulge. It varies in thickness and is warped at the edges after a collision with another, smaller galaxy.



QUASAR
If the dust ring is tilted towards Earth, we see the brilliant light of the accretion disc outshining the surrounding galaxy.

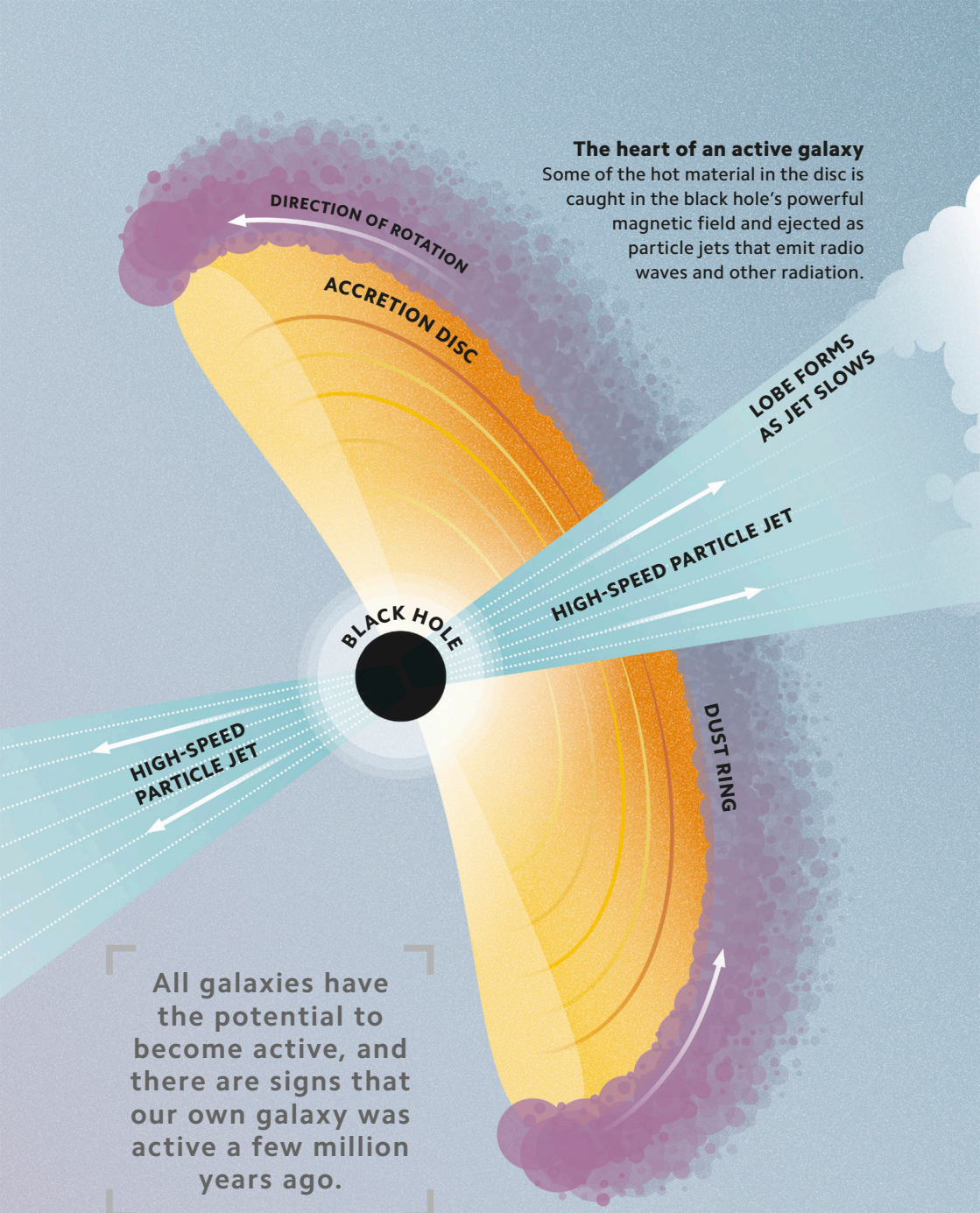
BLAZAR
We look straight down a blazar's radio jets into the nucleus, making the surrounding galaxy hard to detect.

RADIO GALAXY
The dust ring hides the bright accretion disc, so from Earth we see only the radio emissions from the particle jets.

SEYFERT GALAXY
The activity in a Seyfert galaxy is much weaker than in a quasar, allowing a clearer view of the surrounding galaxy.

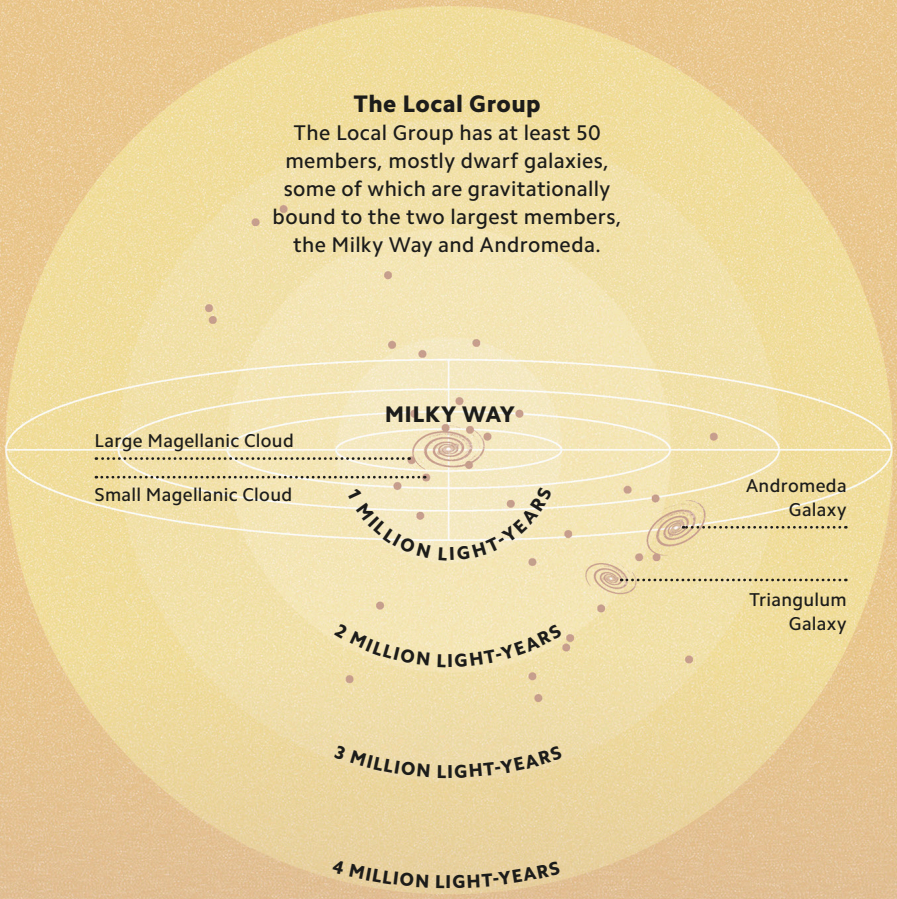
EXTREME ENERGY

Active galaxies are the most luminous long-lived objects in the sky. They emit unusual amounts of energy – more than can be accounted for by their stars alone. The source of this energy is an active galactic nucleus, where matter is torn apart as it falls into a black hole (see p.93), with some material being blasted out in a jet at very high speed. There are several types of active galaxy, which differ mainly in their angle of orientation towards Earth.



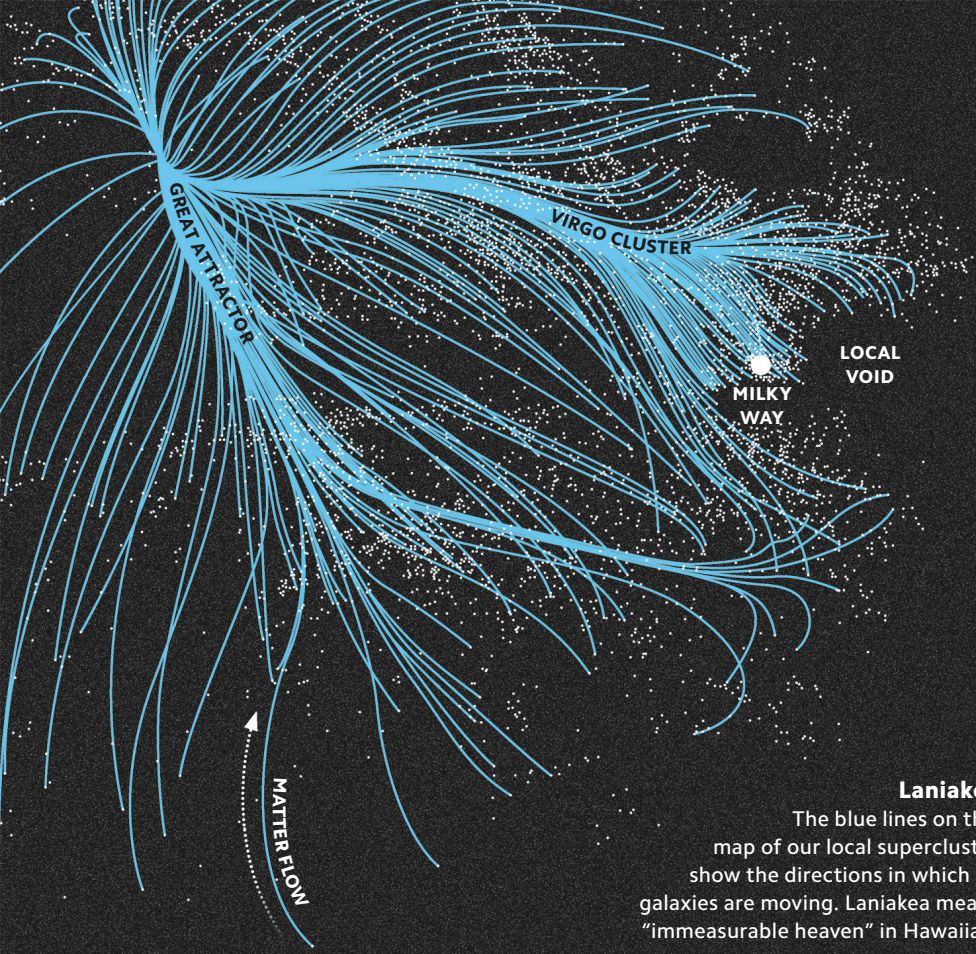
The heart of an active galaxy
Some of the hot material in the disc is caught in the black hole's powerful magnetic field and ejected as particle jets that emit radio waves and other radiation.

All galaxies have the potential to become active, and there are signs that our own galaxy was active a few million years ago.



GALAXIES FORM GROUPS

Drawn together by gravity, most galaxies are found in groups known as clusters, typically 5–15 million light-years across. The Milky Way belongs to a small, relatively young, sparse cluster known as the Local Group. As a cluster ages, its galaxies collide, merge, and grow larger, with giant elliptical galaxies emerging from a mix of different types of galaxy. Giant ellipticals are found densely packed around the centres of the oldest clusters.



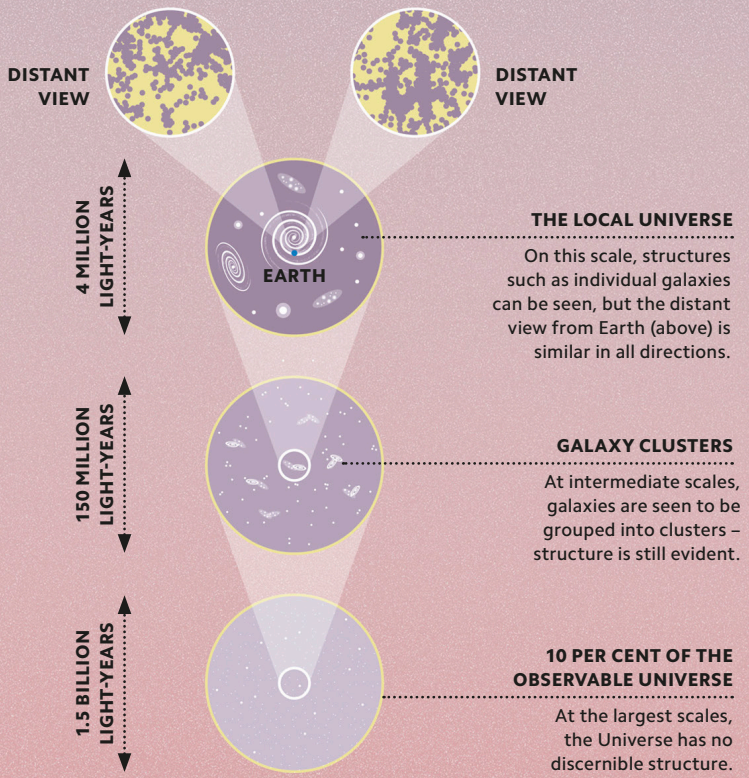
Laniakea
The blue lines on this map of our local supercluster show the directions in which its galaxies are moving. Laniakea means "immeasurable heaven" in Hawaiian.

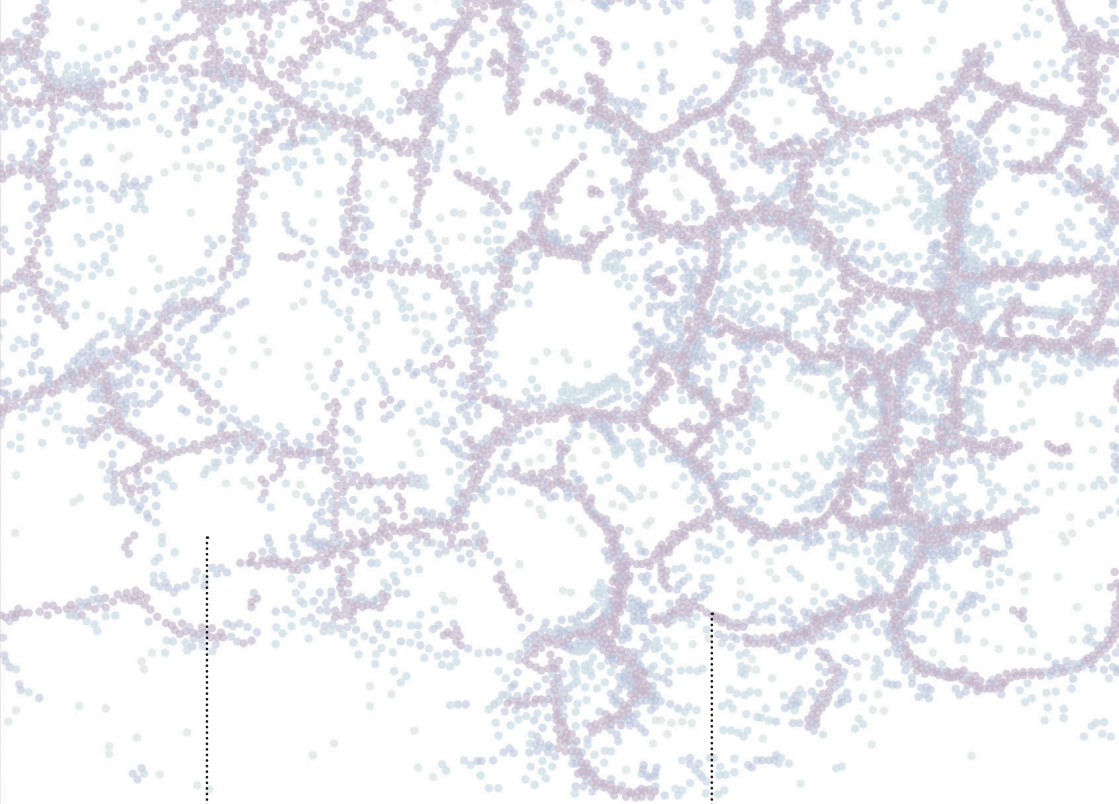
CLUSTERS OF CLUSTERS

Galaxy clusters are themselves parts of larger groups called superclusters. These are the largest structures in the Universe, arising from differences in the density of matter dating back to the earliest times in the Universe's history. The Local Group is, along with the nearby Virgo Cluster, on the edge of the Laniakea Supercluster of about 100,000 galaxies. The clusters in a supercluster are gradually moving towards local concentrations of matter – in our case, a region known as the Great Attractor.

THE MACROSCOPIC UNIVERSE

Although we can see structures ranging in scale from the Solar System up to galaxy clusters, when we expand our view to the Universe at larger scales, everything starts to look the same. The distribution of matter seems to be even (homogeneous) and the same in all directions (isotropic). This idea is called the cosmological principle. It naturally follows that the laws of physics that shape the Universe work in the same way everywhere and always have done.





VOIDS

Voids in the cosmic web are vast, dark spaces almost entirely empty of both ordinary matter and dark matter.

FILAMENTS

Galaxies and enormous clouds of hydrogen gas form long, thin wall- and sheetlike structures between the voids.

THE COSMIC WEB

The cosmological principle (see opposite) suggests that all parts of the Universe were once connected. It supports the theory that the Universe originated in the Big Bang and then expanded. The largest-scale structures we see today originate from energy fluctuations in the early Universe that have since spread out, leaving filaments of matter surrounding empty voids. This "cosmic web" has a foamy structure, with most galaxies found on the surfaces of the "bubbles".

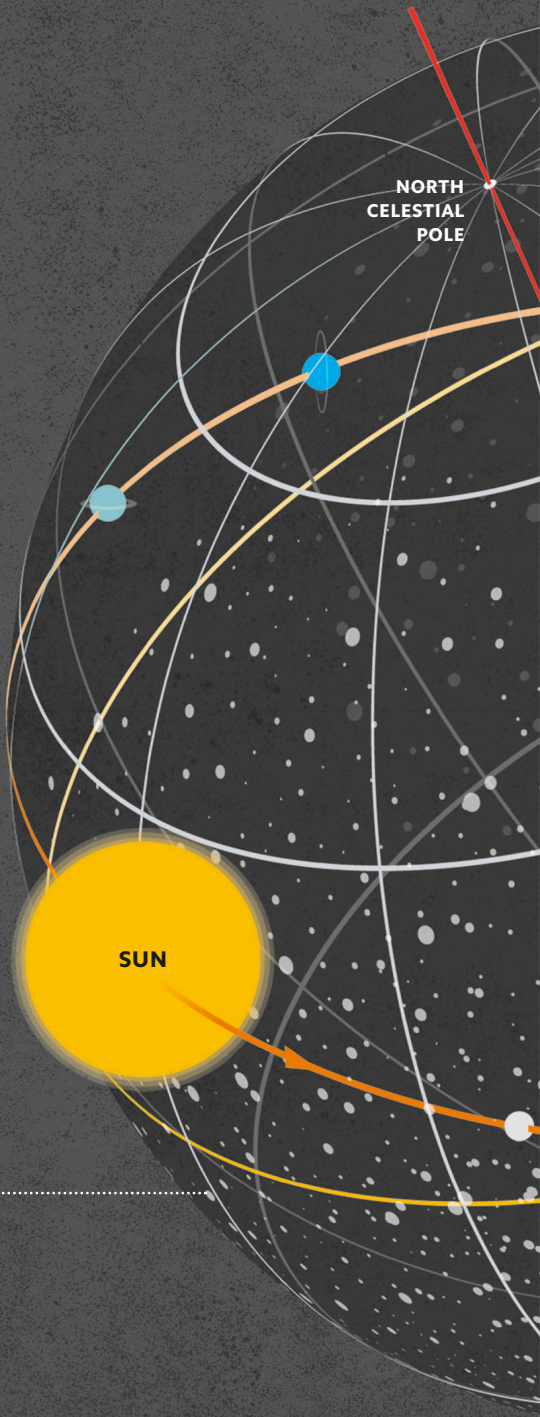
OBSERVING THE SKY

V I N G
G H T

Objects in the Universe are dispersed across the three dimensions of space. But from our Earth-centred viewpoint, they appear as points or smudges of light fixed to the backdrop of the night sky. As Earth turns on its axis and orbits the Sun, the stars and other distant objects seem to be in constant cyclical motion. The planets appear against the same backdrop, although moving independently of the stars. If you know when to look and choose your observing spot carefully, you can see a lot with the naked eye – the Moon, several planets, meteor showers, plenty of stars and constellation patterns, and even a few galaxies. Binoculars or an affordable telescope will reveal extra levels of detail.

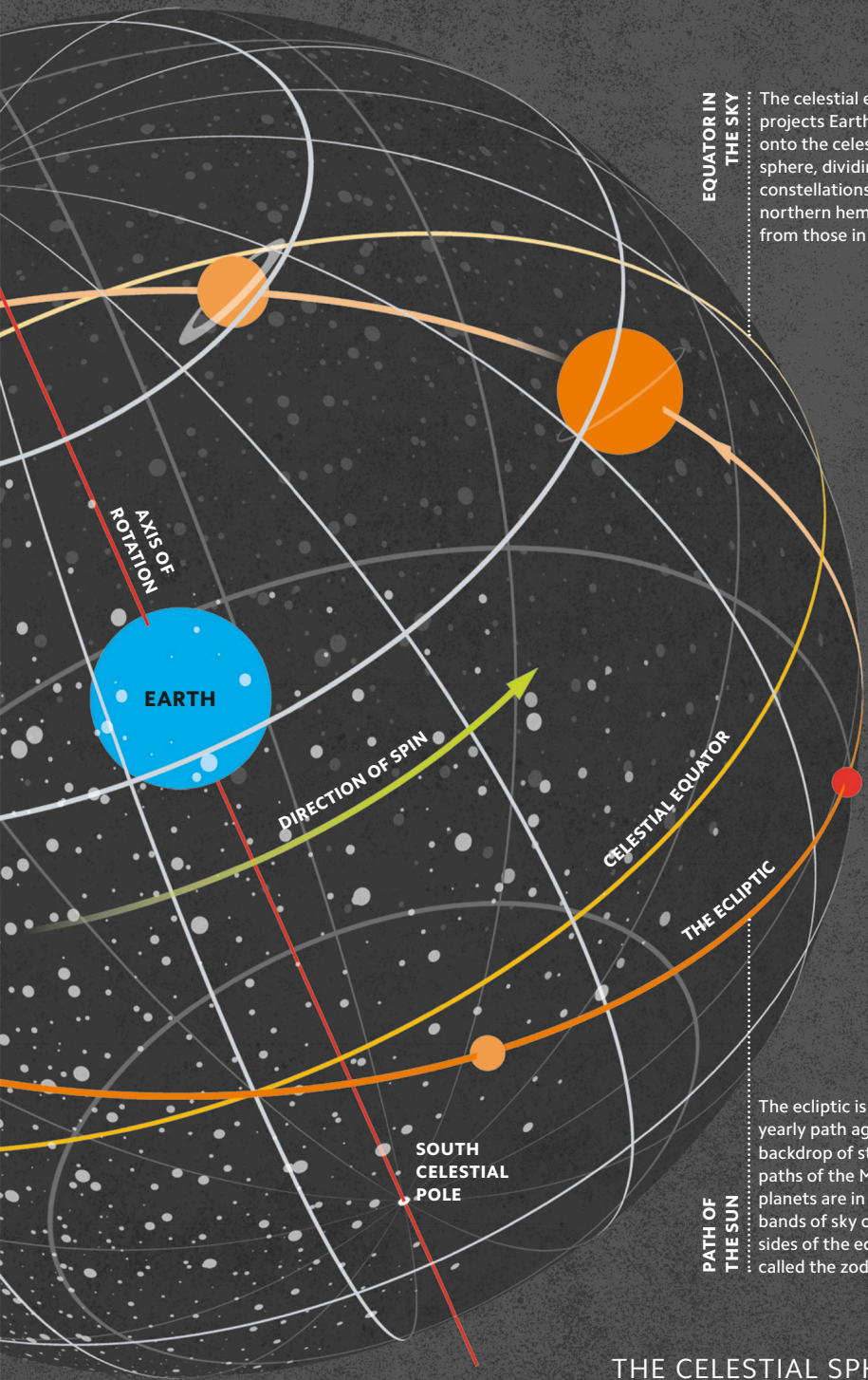
THE SKY AS A SPHERE

Because Earth spins on its axis, objects in our sky always seem to be moving. Additionally, because sky objects are all so far away, it is difficult to judge their distance from Earth relative to one another. These problems are well addressed by imagining the objects in our night sky fixed to the surface of a giant, glasslike sphere, all at the same distance, and with Earth at its centre. This model, known as the celestial sphere, helps us record the positions of celestial objects and make sense of our night sky. Most of the circular lines on the sphere are a projection of Earth's system of latitude and longitude.



FIXED STARS

In this model, the stars are placed on the surface of the sphere, and all at the same distance from Earth. They are fixed relative to one another.



EQUATOR IN THE SKY

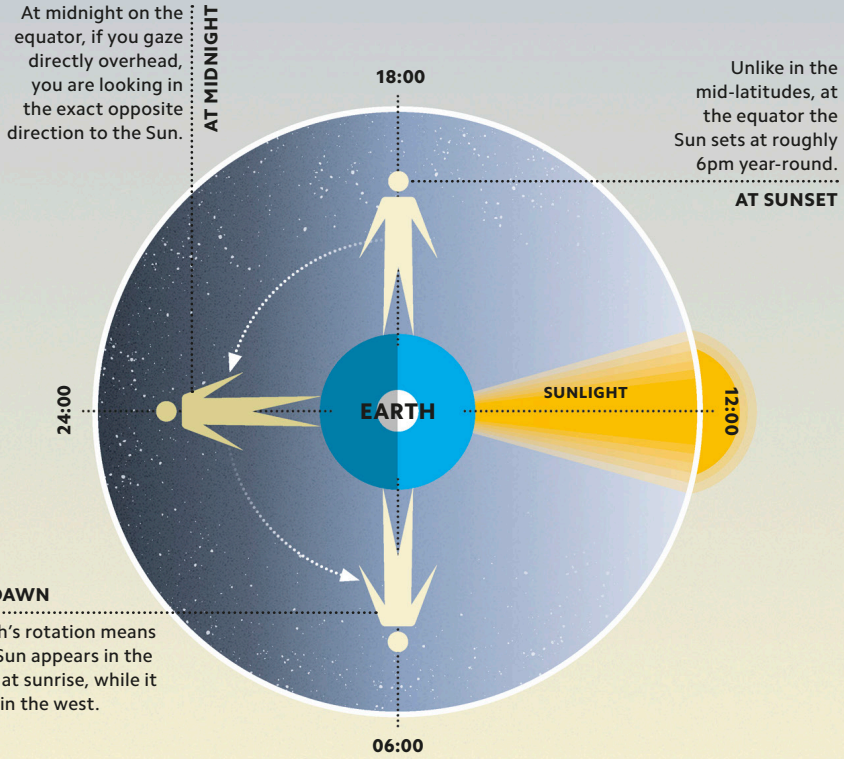
The celestial equator projects Earth's equator onto the celestial sphere, dividing constellations in the northern hemisphere from those in the south.

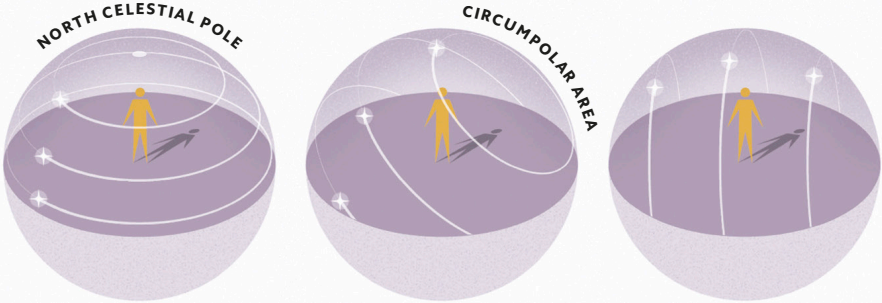
PATH OF THE SUN

The ecliptic is the Sun's yearly path against a backdrop of stars. The paths of the Moon and planets are in narrow bands of sky on both sides of the ecliptic, called the zodiac.

WHEN TO SEE STARS

Obscured by the Sun's glare, stars are invisible in the daytime sky. However, as the Sun sets, they gradually come into view, move across the sky, and then disappear in the morning (see opposite). Over a night, we see a constantly shifting portion of the celestial sphere (see pp.112–13). Our view also changes over a year as Earth orbits the Sun – different stars are visible at 10 pm in December compared with 10 pm in June. In a year, an observer at mid-latitudes, where most of us live, will see all of their own celestial hemisphere and part of the other.





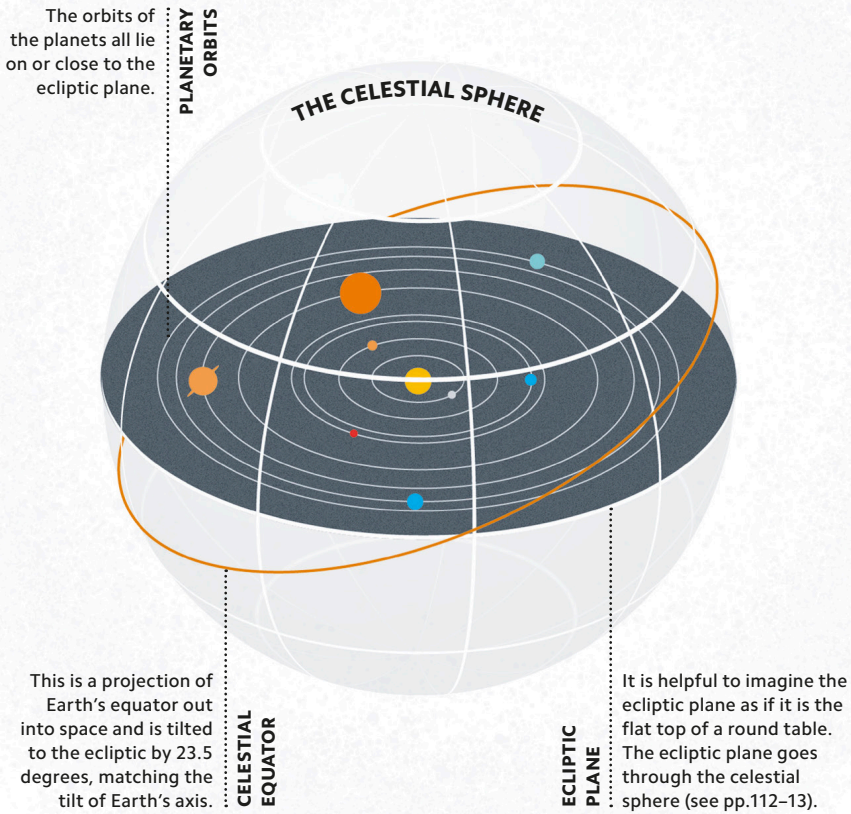
Observer at the north pole
At the north pole, an observer views the sky as if riding a carousel – the stars do not rise or set, but spin around at the same altitude.

Observer at mid-latitude
Over one night, stars rise in the east, cross the sky, then move down into the west. Some stars (circumpolar), however, are always visible.

Observer at the equator
Here, stars rise vertically somewhere along the eastern horizon and move over the sky, before setting vertically in the west.

WHERE IN THE WORLD

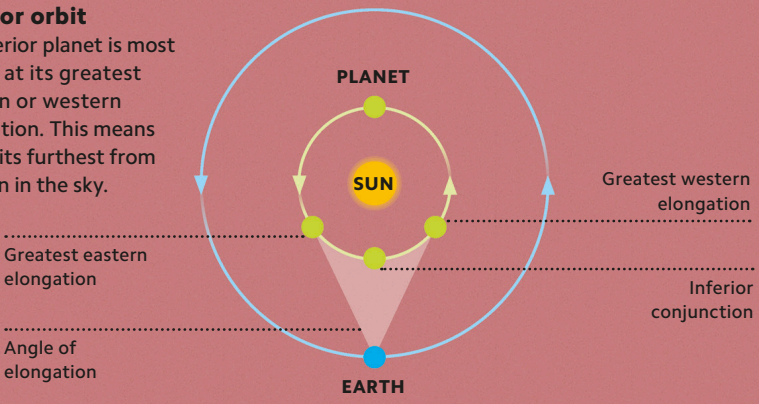
The effect of Earth's rotation on an observer's view of the night sky varies according to their location. In the mid-latitudes, stars appear to follow an arc across the sky, always rising in the east and setting in the west. The arc's route, however, depends on which hemisphere the observer is in. In the northern hemisphere, many stars appear to move up and reach a maximum height over the southern horizon, before descending and then setting. Conversely, in the southern hemisphere, many stars appear to rise towards a point above the northern horizon.



THE ORBITAL PLANE

The plane of Earth's orbit around the Sun is called the ecliptic plane. The major planets and many smaller bodies orbit on or close to it. This dates from the birth of the Solar System, when a spinning cloud of gas formed a flat disc dotted with planets and other objects. Locating the ecliptic is the first step to spotting the planets. To do this, follow the path of the Sun over the course of one day. Another guide is the Moon, which is always within 5 degrees of the ecliptic.

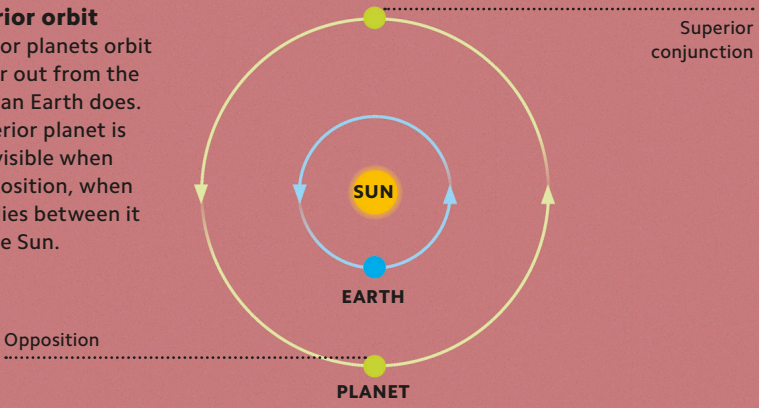
Inferior orbit
An inferior planet is most visible at its greatest eastern or western elongation. This means it is at its furthest from the Sun in the sky.

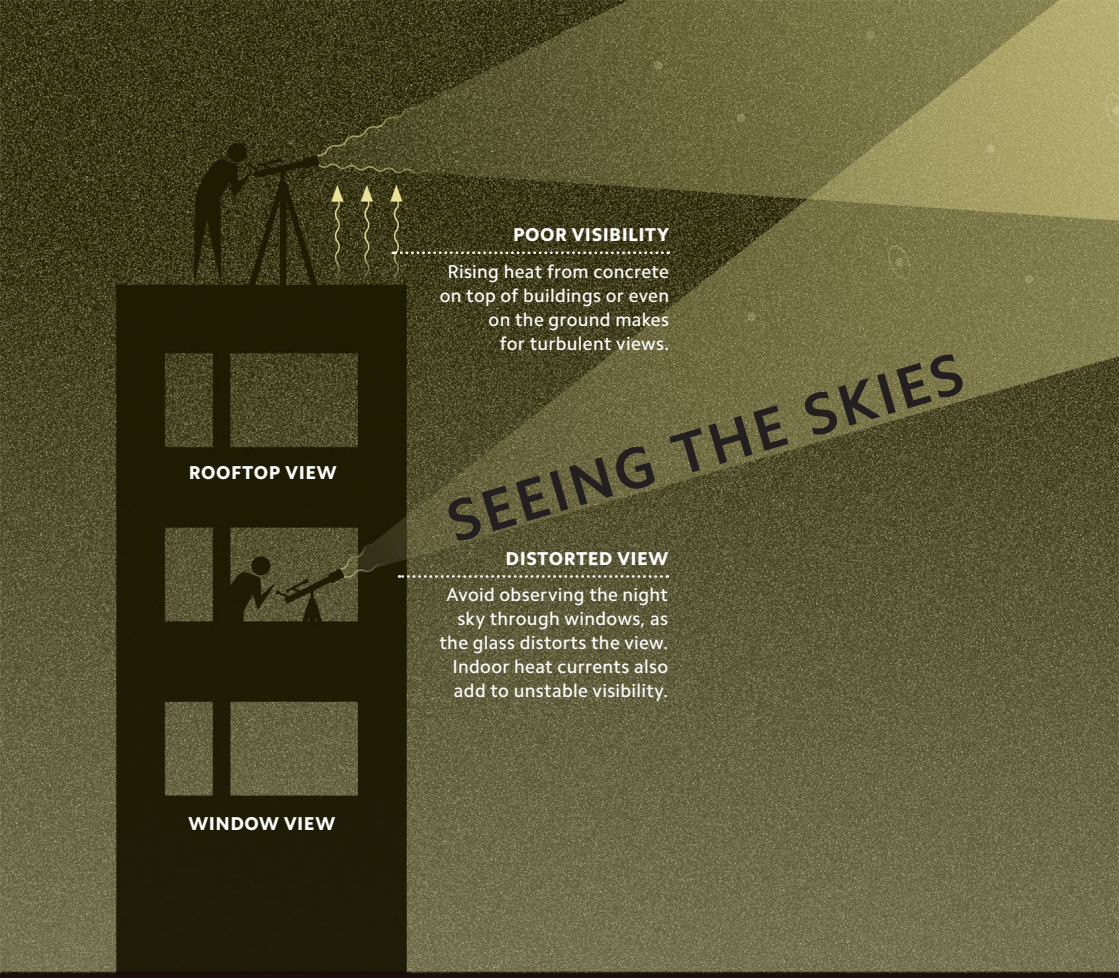


WHEN TO SPOT PLANETS

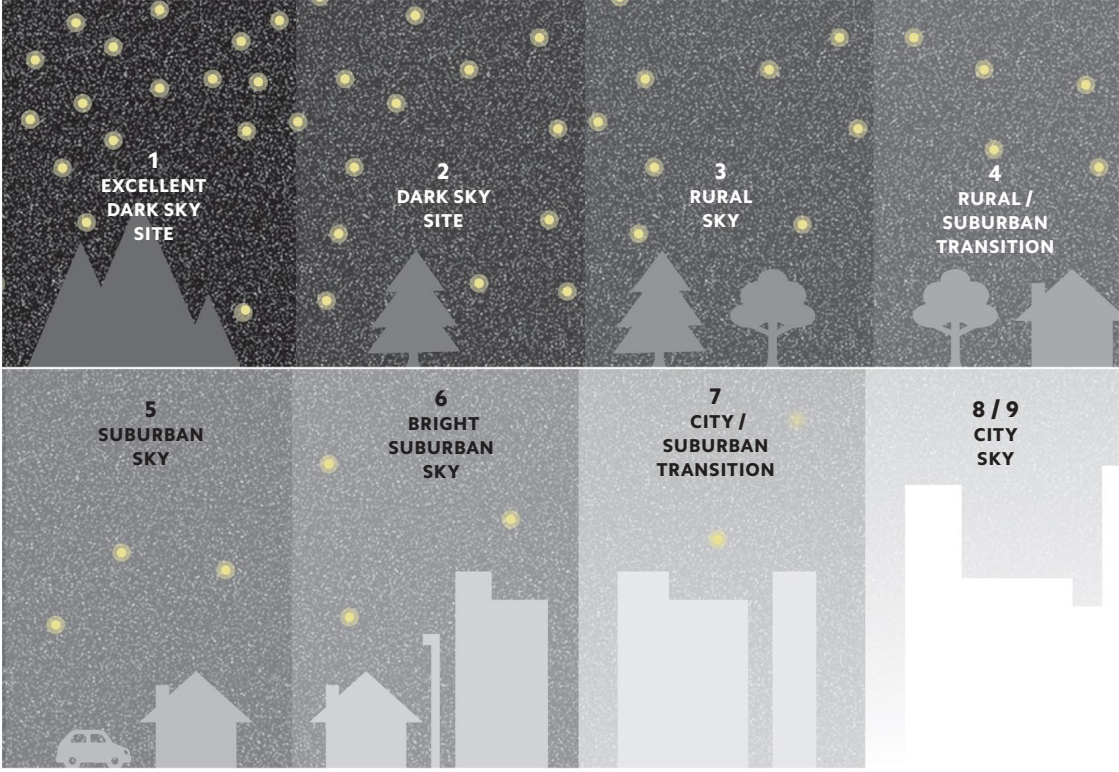
As planets orbit the Sun, their visibility from Earth changes. Mercury and Venus, called the inferior planets because they orbit closer to the Sun than Earth, are best seen at their maximum distances from the Sun in the sky. All other planets are most visible when directly opposite the Sun. No planet is visible when it passes behind the Sun as viewed from Earth.

Superior orbit
Superior planets orbit further out from the Sun than Earth does. A superior planet is most visible when at opposition, when Earth lies between it and the Sun.





To get the most out of stargazing, it pays to choose your observing site carefully. Look for an outdoor spot away from buildings, with their artificial light and warm air currents, which can blur images. If you are using binoculars or a telescope on a tripod, place it on solid ground to stop small vibrations from shaking the image. It can take half an hour or more for your eyes to fully adapt to the dark. As the air is often cold under clear skies, you might also need warm clothes.



LOST IN THE LIGHT

More than half of all people in the world live in urban areas, meaning light pollution is a problem for many stargazers, as it reduces the number of stars that can be seen in the night sky. This problem is at its worst in large cities, where a vast array of lighting, along with industrial pollution, can cause the night sky to take on an orange glow. Extremely heavy light pollution can even result in no stars being visible. In areas with near-zero light pollution, however, sky objects are numerous and fully visible.

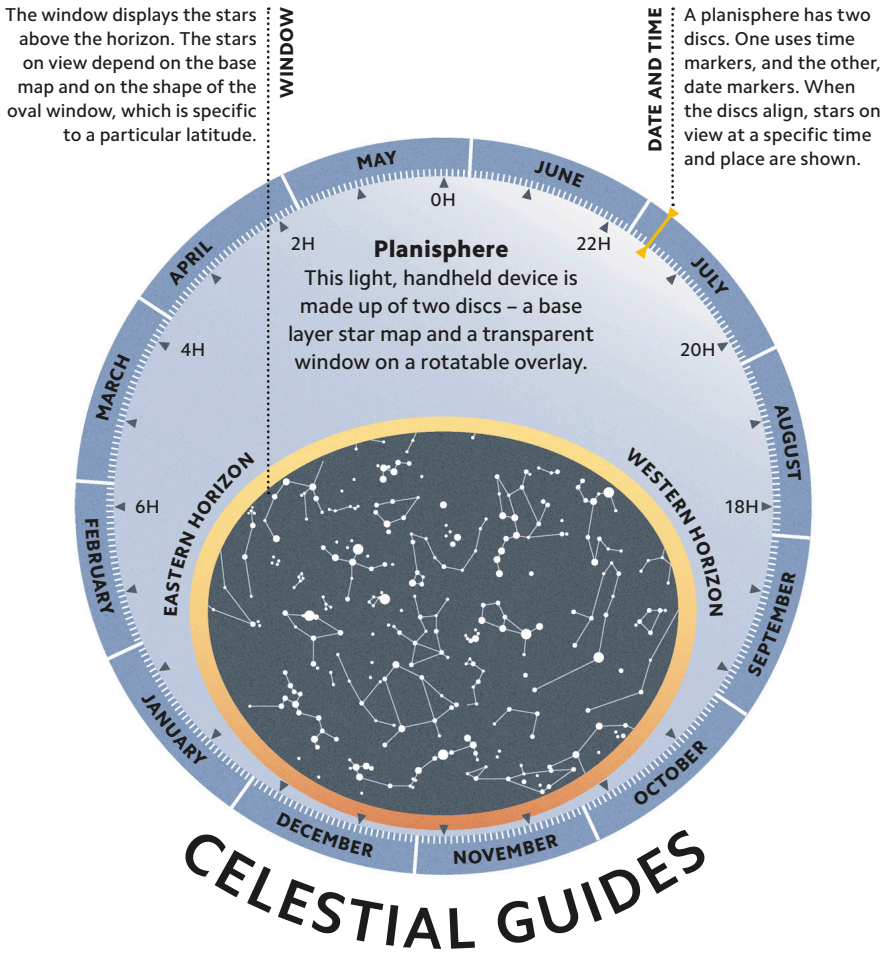
The Bortle scale
First published in 2001, the Bortle scale attempts to classify the darkness of a viewing location from 1 – the darkest sites onEarth – to 9 – the most light-polluted of all skies, as found in most cities.

COMPARING BRIGHTNESS

In astronomy, the term magnitude is used to mean brightness. The magnitude of a star as we see it is the result of a combination of its true brightness and how far away it is from Earth. As a result, the magnitude of stars is defined in two distinct ways: apparent magnitude, which refers to how bright a star appears from Earth; and absolute magnitude, which indicates a star's brightness from a set distance. On both types of magnitude scale, the smaller the number, the brighter the object, and the larger the number, the fainter the object.

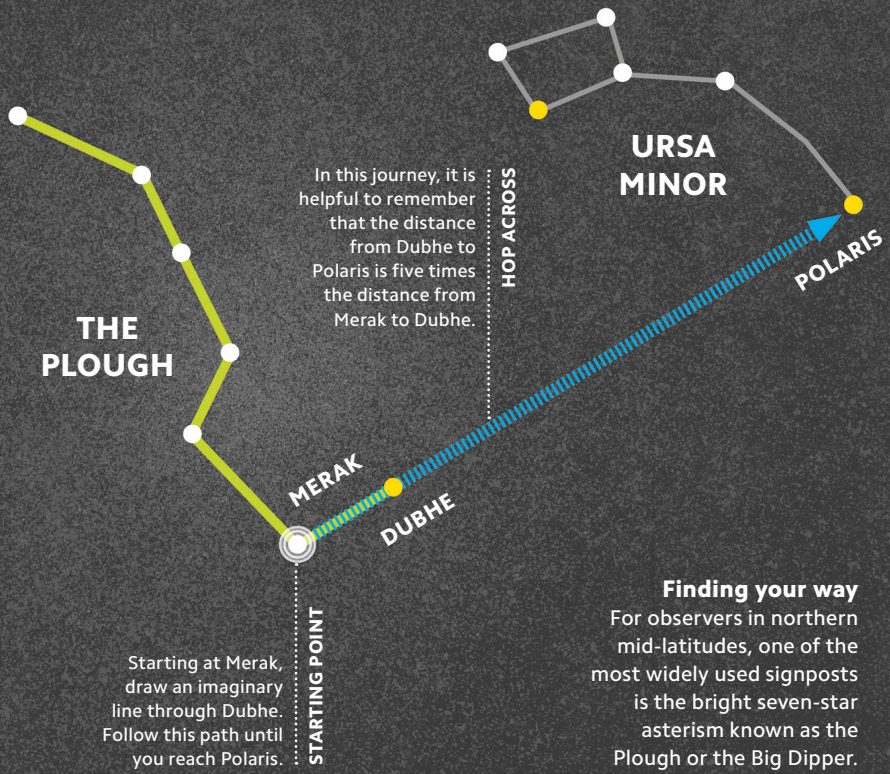


Apparent magnitude scale
The very brightest of stars have negative magnitude values. Each step in the scale represents an increase or decrease in brightness of 2.5 times. For comparison, this visual magnitude scale includes Venus, which often appears much brighter than any star.



CELESTIAL GUIDES

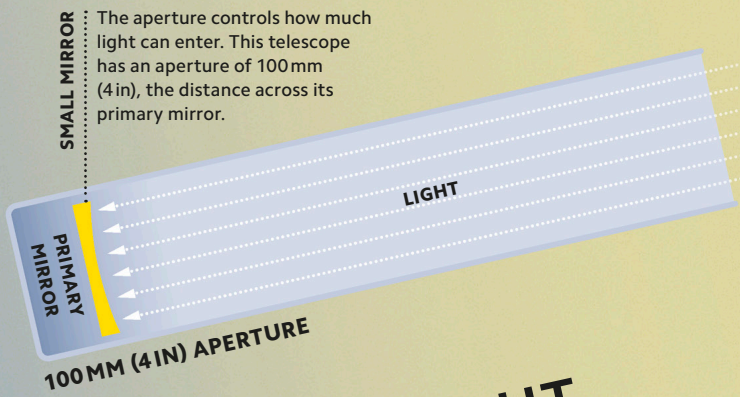
For thousands of years, observers of the night sky have been producing star charts. The purpose of these maps varied – from helping the observer identify the time of year to forming the foundation for mythological stories. From ancient times to the modern day, most star charts use bigger dots to depict brighter stars. This is an easy way to visualize the brightness of stars but not a direct representation of their scale, as all stars are same-sized single points in the night sky.



Finding your way
For observers in northern mid-latitudes, one of the most widely used signposts is the bright seven-star asterism known as the Plough or the Big Dipper.

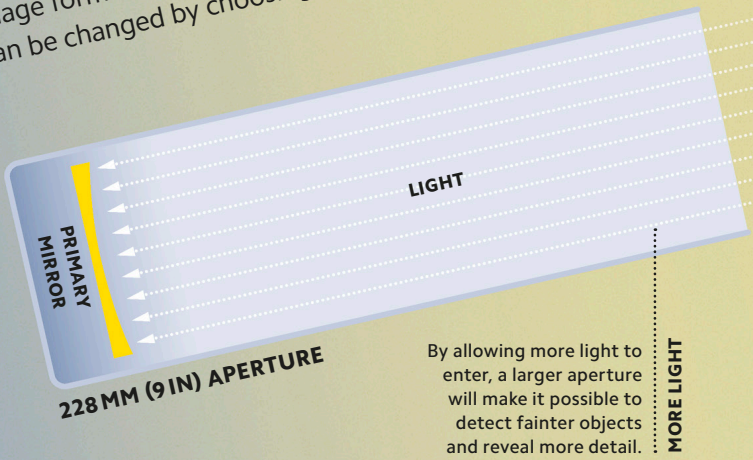
JOINING DOTS IN THE SKY

Navigating the vast network of constellations in the night sky might seem like a daunting task. A technique called starhopping, however, makes it much easier. Starhopping is the process of using brighter stars as signposts in order to find other stars located in different constellations. It involves tracing lines between stars in one star pattern in order to hop over to other stars in a different star pattern.



LETTING IN LIGHT

Optical instruments such as telescopes and binoculars enhance our views of distant objects because they are larger than the pupil of a human eye and collect more light than an eye can. Telescopes are usually described by the size of their aperture, which is the diameter of the primary mirror or objective lens (see pp.124–27). An eyepiece gives a magnified view of the image formed by the main mirror or lens. The magnification can be changed by choosing a different eyepiece.

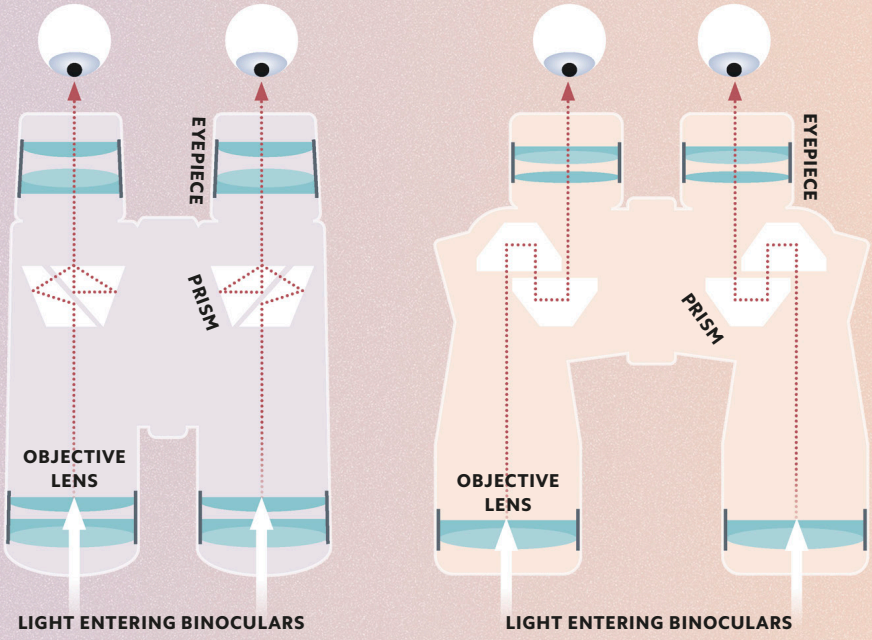


Roof prism

There are two main binocular designs. Roof prisms are lighter and more compact, but the complex light path results in narrower fields of view and less bright images.

Porro prism

With their off-set prisms, these classic binoculars are heavier and larger than the roof prism style. However, they have a wider field of view and better image clarity.



CHOOSING BINOCULARS

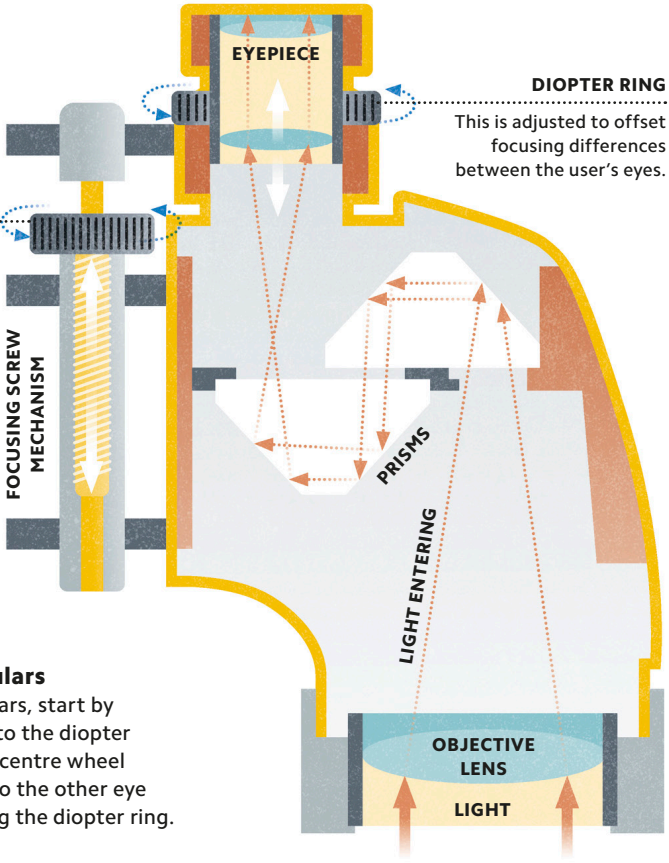
Using binoculars enhances many observations of the night sky. When choosing binoculars for astronomy, it is worth remembering that larger lenses will let in more light. However, larger styles are usually heavier and can be uncomfortable to hold for long periods. Many beginners choose 10 x 50 binoculars, which means they have a magnification of 10 x and objective lenses with a 50 mm (2 in) aperture.

STAYING FOCUSED

When using binoculars, first find a comfortable position. Holding binoculars for a long time can lead to aches in the neck and arms, but this can be counteracted by resting your elbows on a surface, sitting in a deckchair, or using a tripod. The two dials used to focus binoculars efficiently are the centre focus wheel and the diopter ring. The centre focus wheel modifies the focus of the binoculars for both eyes, while the latter corrects any differences between the eyes.

CENTRE FOCUS WHEEL

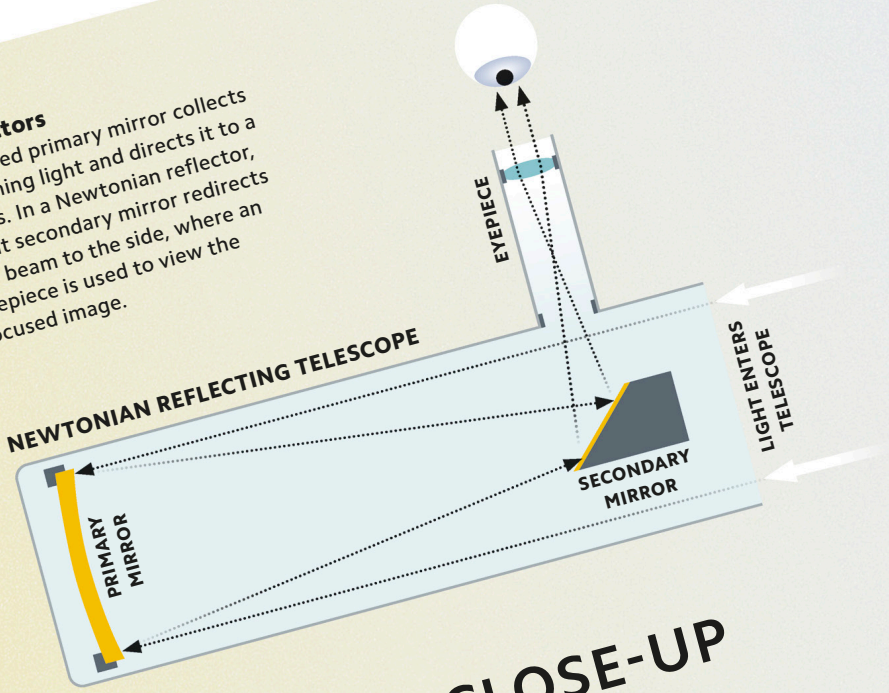
Once the focus of the binoculars has been set up, including the use of the diopter ring, this is the only dial you will need to turn to get crisp views of the sky.



How to use binoculars

To focus your binoculars, start by closing the eye close to the diopter ring before using the centre wheel to focus. Then swap to the other eye and focus it by turning the diopter ring.

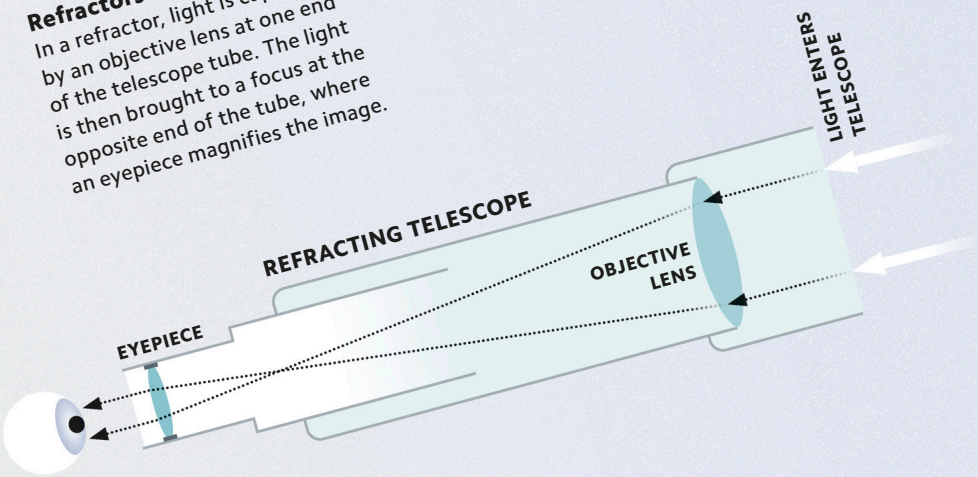
Reflectors
A curved primary mirror collects incoming light and directs it to a focus. In a Newtonian reflector, a flat secondary mirror redirects the beam to the side, where an eyepiece is used to view the focused image.



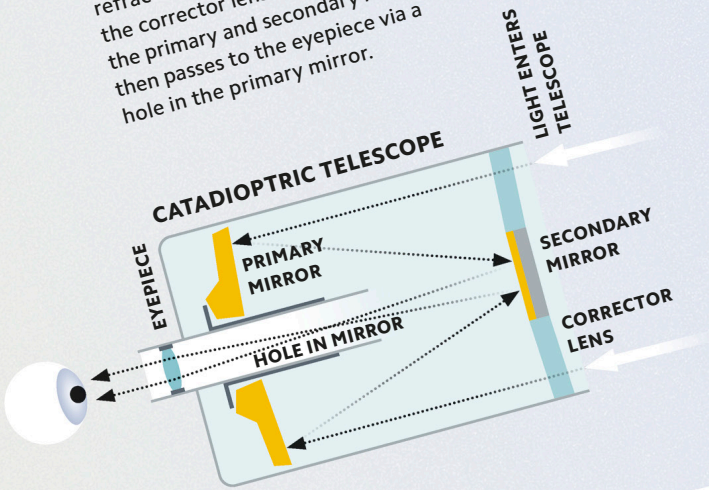
THE SKY IN CLOSE-UP

To get a detailed view of objects in space – including craters on the Moon, the rings of Saturn, double stars, clouds of dust, or distant galaxies – a telescope is needed. A telescope's size makes it far more effective than our eyes at capturing light. There are two basic designs: refractors and reflectors. Refractors, which use lenses to produce an image, appeared in the early 1600s, followed about 50 years later by reflectors, which use mirrors. Today, we also have catadioptrics, which combine lenses and mirrors.

Refractors
In a refractor, light is captured by an objective lens at one end of the telescope tube. The light is then brought to a focus at the opposite end of the tube, where an eyepiece magnifies the image.



Catadioptrics
This hybrid design is more compact than reflectors and refractors. Light enters through the corrector lens. It is reflected by the primary and secondary mirrors then passes to the eyepiece via a hole in the primary mirror.

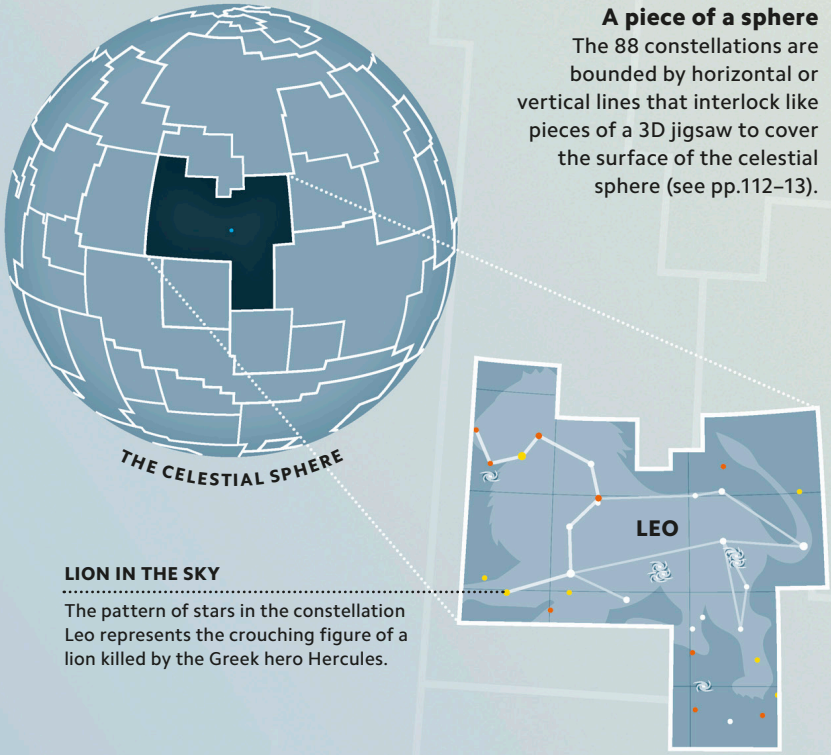


THE

CONSTEL

LATIONS

The human eye looks for patterns. The first constellations were groupings of stars that ancient peoples used for navigation, timekeeping, or storytelling. Astronomers today use a formal system in which constellations are areas of the sky that join up without gaps like the countries on a map of Earth. Constellations may no longer serve some of their original purposes. But many of the ancient names remain in the modern system, and the old patterns are still drawn on star charts, linking us to the astronomers of the past. Constellations are also a useful means of dividing the sky into easily recognizable sections, which can be used to identify the locations of stars and other celestial objects.



A piece of a sphere
The 88 constellations are bounded by horizontal or vertical lines that interlock like pieces of a 3D jigsaw to cover the surface of the celestial sphere (see pp.112–13).

LION IN THE SKY
The pattern of stars in the constellation Leo represents the crouching figure of a lion killed by the Greek hero Hercules.

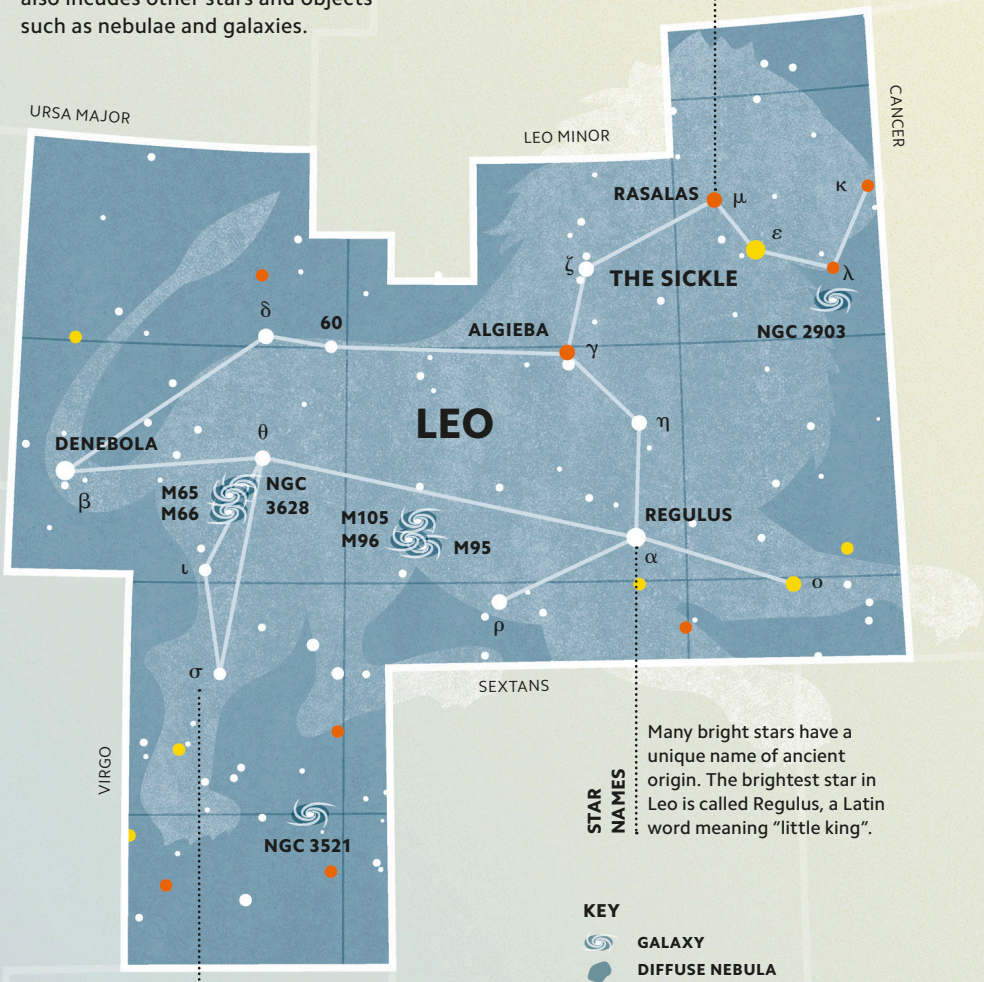
MAPPING THE SKY

"Constellation" comes from a Latin word meaning "group of stars", but in modern astronomy a constellation is an area of sky. Each such area includes two or more stars linked by imaginary lines to depict a shape, such as a mythical figure or an animal. Some of these patterns are so faint that finding them can be an adventure, but many are easily recognized by their bright stars. As Earth orbits the Sun, the constellations slowly move, so there is always something different to see.

Charting the constellations
Most star charts include the constellations. As well as the stars in its main pattern, a constellation also incudes other stars and objects such as nebulae and galaxies.

A few constellations also contain smaller patterns called asterisms. In Leo, six stars form an asterism called the Sickle.

ASTERISMS



Many bright stars have a unique name of ancient origin. The brightest star in Leo is called Regulus, a Latin word meaning "little king".

STAR NAMES

KEY

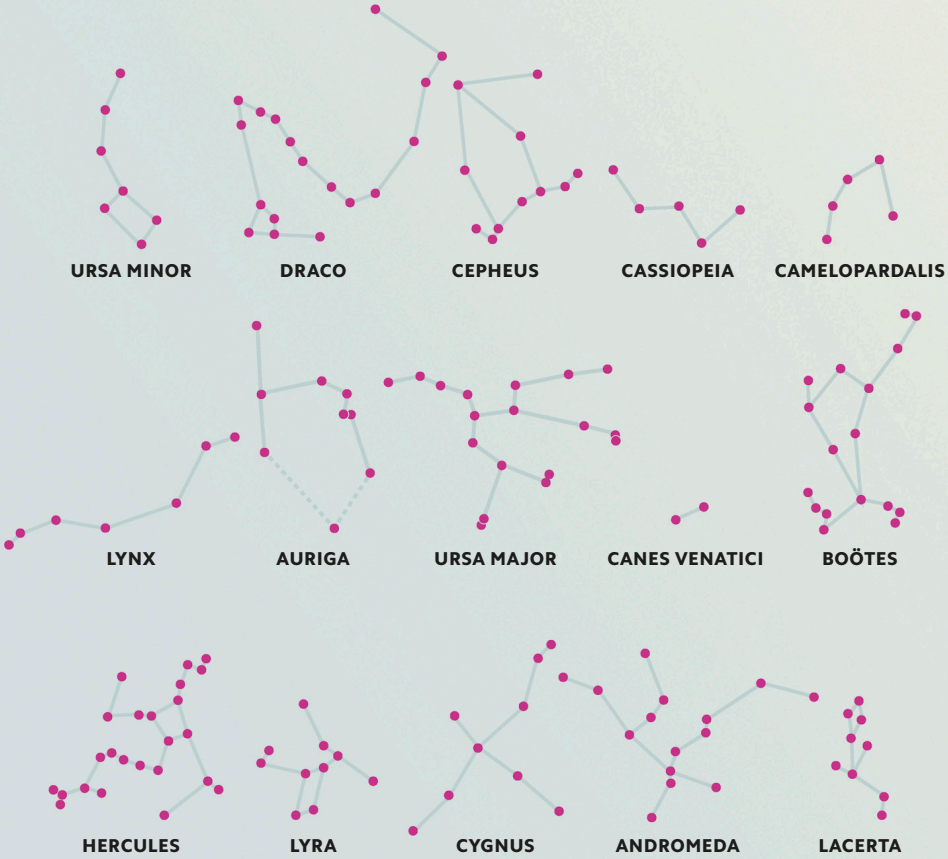
- GALAXY
- DIFFUSE NEBULA
- PLANETARY NEBULA OR SUPERNOVA REMNANT
- BLACK HOLE
- OPEN CLUSTER
- GLOBULAR CLUSTER

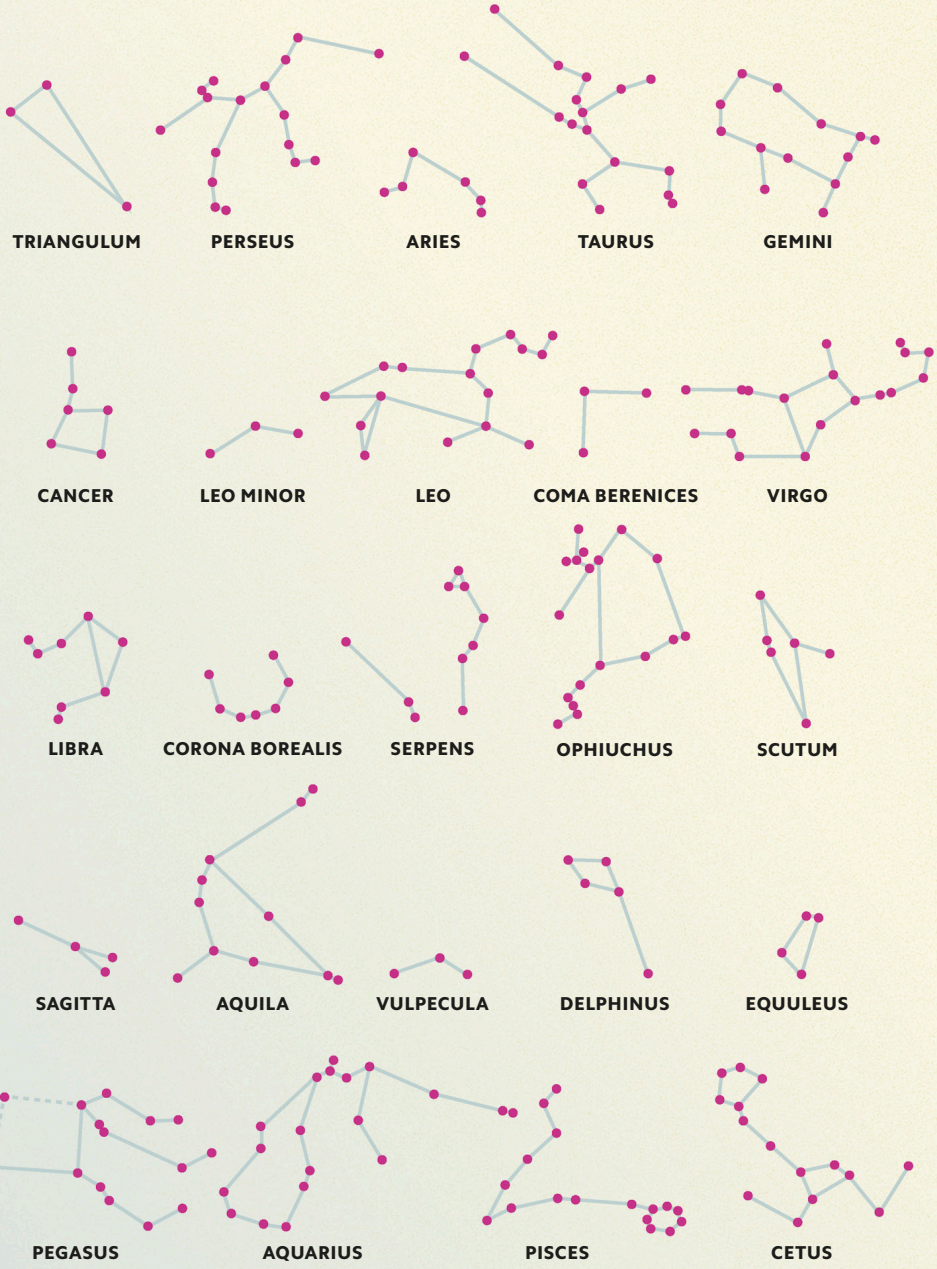
Some stars are given Greek letters. A constellation's brightest star is usually Alpha (α), the second brightest Beta (β), and so on.

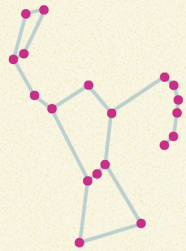
GREEK LETTERS

PATTERNS IN THE SKY

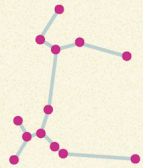
Most of the constellations we know today date from early recorded history and were invented by the Babylonians and the ancient Greeks. In 1922, a final list of 88 was formally codified by the International Astronomical Union (IAU). All are shown below, roughly in order from north to south.




TRIANGULUM
PERSEUS
ARIES
TAURUS
GEMINI
CANCER
LEO MINOR
LEO
COMA BERENICES
VIRGO
LIBRA
CORONA BOREALIS
SERPENS
OPHIUCHUS
SCUTUM
SAGITTA
AQUILA
VULPECULA
DELPHINUS
EQUULEUS
PEGASUS
AQUARIUS
PISCES
CETUS



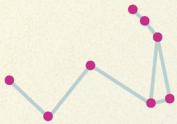
ORION



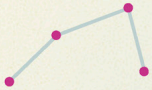
CANIS MAJOR



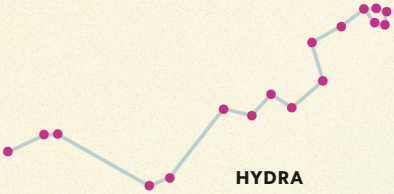
CANIS MINOR



MONOCEROS



ANTLIA



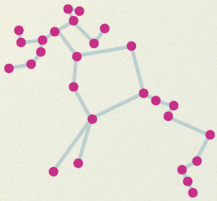
HYDRA



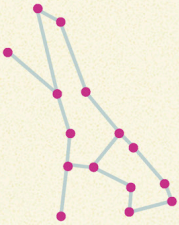
SEXTANS



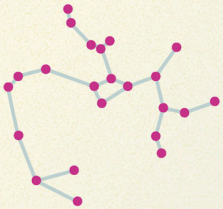
CRATER



CENTAURUS



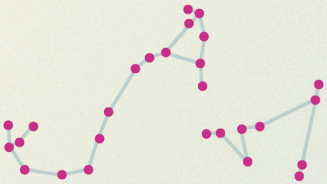
LUPUS



SAGITTARIUS



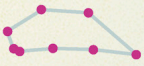
CORVUS



SCORPIUS CAPRICORNUS



MICROSCOPIUM



PISCIS AUSTRINUS



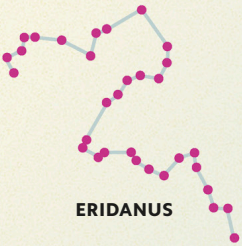
SCULPTOR



FORNAX



CAELUM



ERIDANUS



LEPUS



COLUMBA



PYXIS



PUPPIS



VELA



CARINA



CRUX



MUSCA



CIRCINUS



NORMA



TRIANGULUM AUSTRALE



ARA



CORONA AUSTRALIS



TELESCOPIUM



INDUS



GRUS



PHOENIX



TUCANA



HYDRUS



HOROLOGIUM



RETICULUM



PICTOR



DORADO



VOLANS



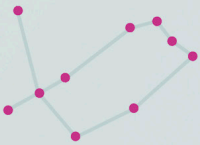
MENSA



CHAMAELEON



APUS



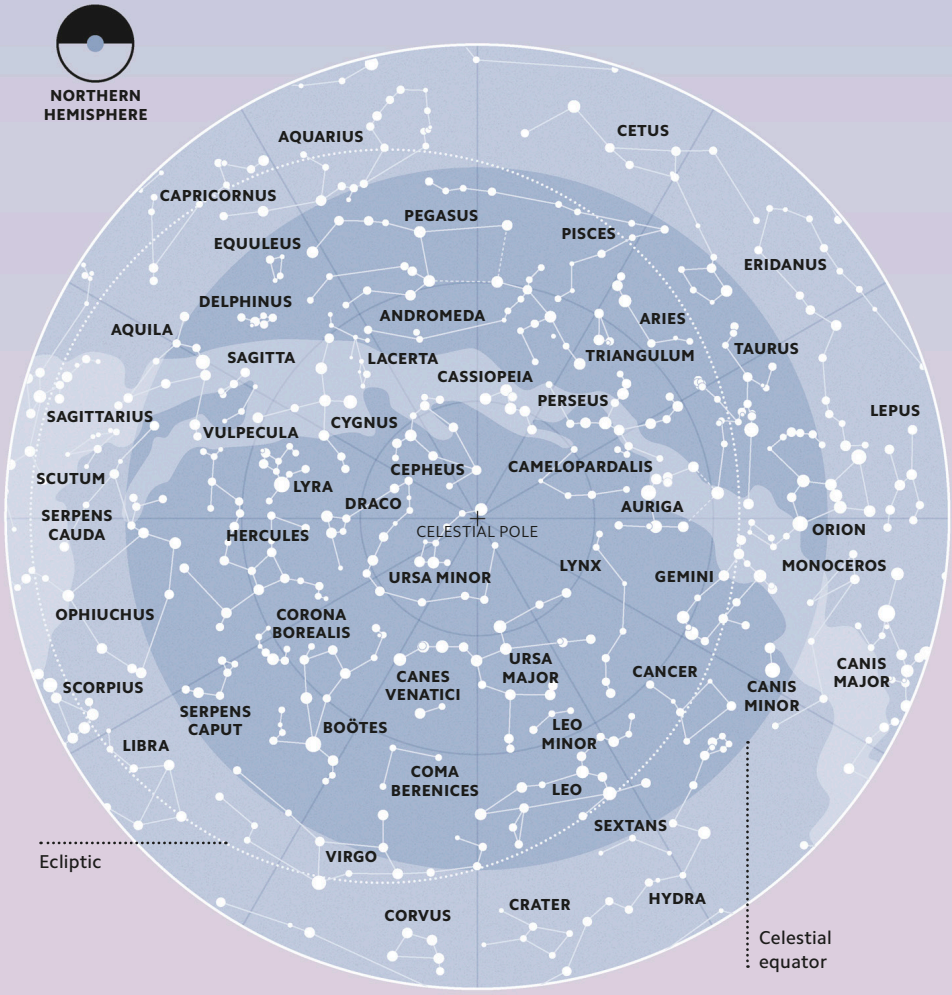
PAVO



OCTANS

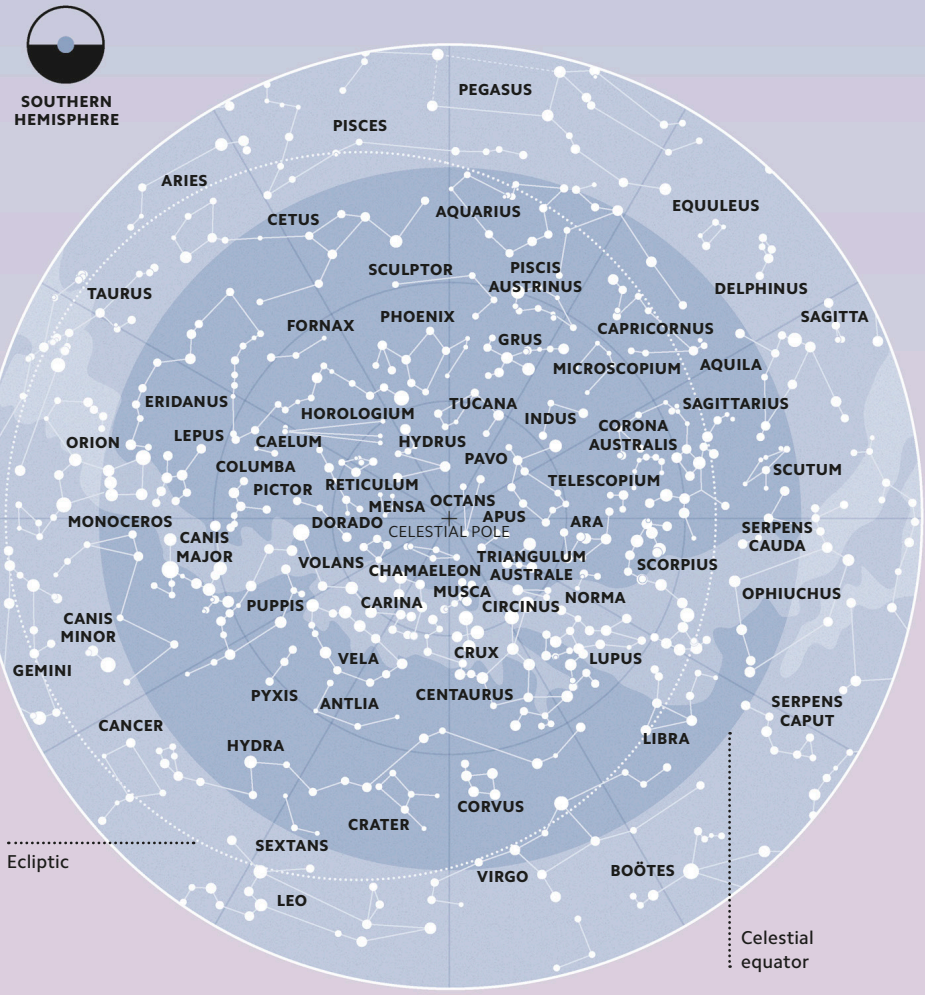
THE NORTHERN SKY

Most stars appear to move slowly around the sky at night. One star, however, sitting in Ursa Minor close to the north celestial pole, is near-motionless. This is Polaris, the North Star. It has been used by navigators to find north for centuries.



THE SOUTHERN SKY

In the southern hemisphere, near the south celestial pole, there is no bright star like Polaris to use as an easy navigational aid. However, the southern sky features the brightest constellations and the most brilliant sections of the Milky Way.



Dubhe is the westernmost star in the Plough asterism, which stretches east to Alkaid.

M101, or the Pinwheel Galaxy, is thought to be home to more than one trillion stars.

M101

ALCOR

ζ

MIZAR

ALIOTH

ε

ALKAID

η

δ

MEGREZ

THE PLOUGH

DUBHE

α

Early Arabic literature suggests that Mizar and Alcor, a famous double star, were used to test eyesight.

PHECDA

M108

MERAK

M109

γ

β

M97

M97 is also known as the Owl Nebula due to its two large dark patches, which look like an owl's eyes.

χ

ψ

TANIA BOREALIS

μ

λ

ν

ALULA AUSTRALIS

ξ

NORTHERN HEMISPHERE

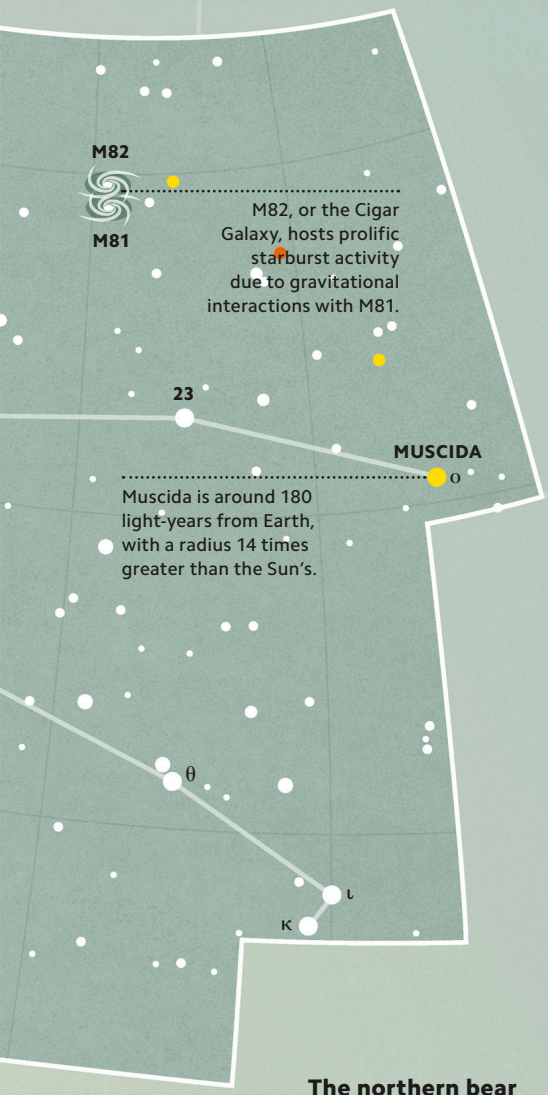


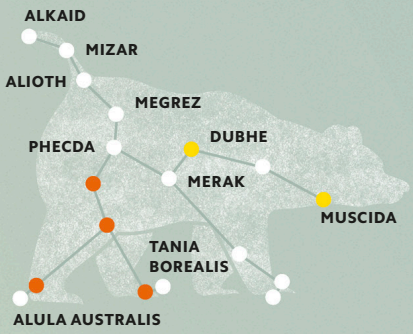
KEY DATA

Fully visible: *90°N–16°S*
Brightest star: *Alpha (α)*
Highest in sky at 10pm: *February–May*

THE GREAT BEAR

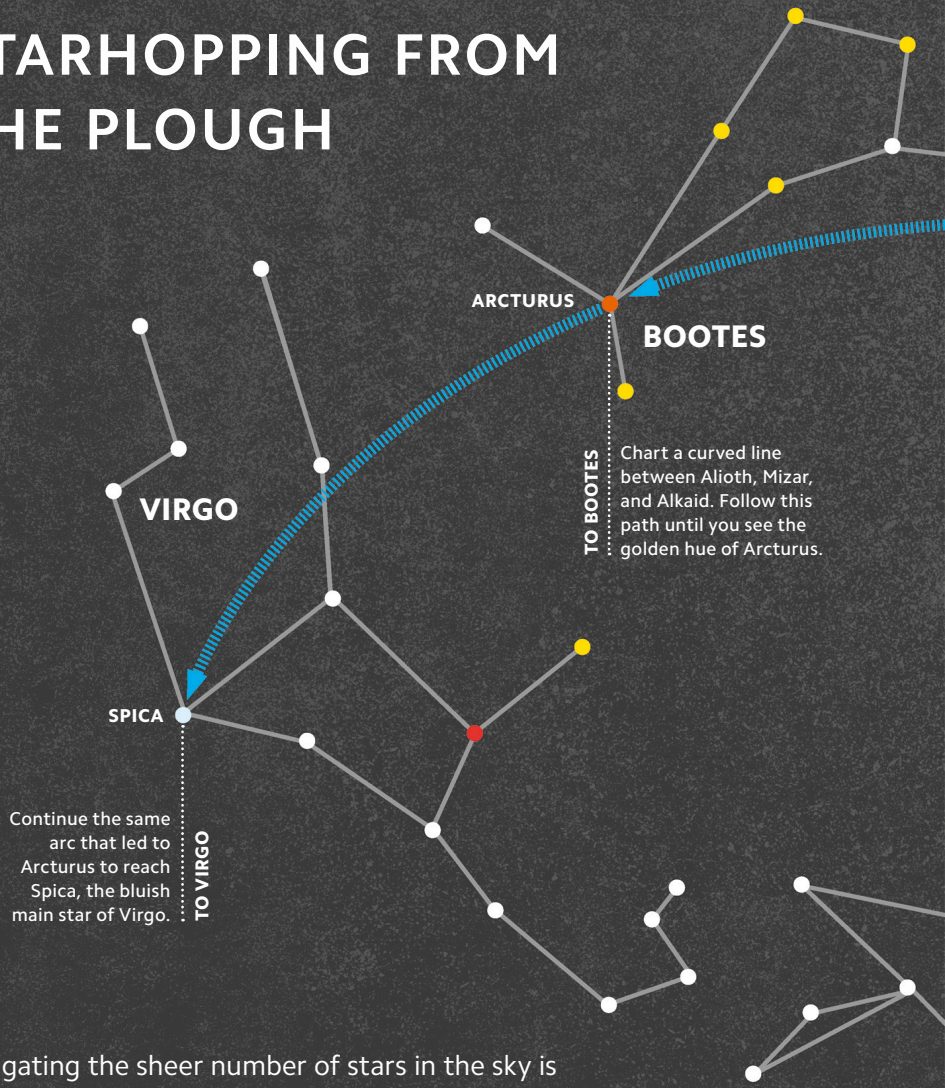
Ursa Major is the third largest constellation in the night sky. Covering a vast area of the northern skies, its most recognizable star group, or asterism, is known as the Plough, or the Big Dipper. Ursa Major features in many distinct mythologies, such as those of ancient Greece, China, and North America. The constellation is also referred to both in the Old Testament of the Bible and in Homer's *Iliad*.



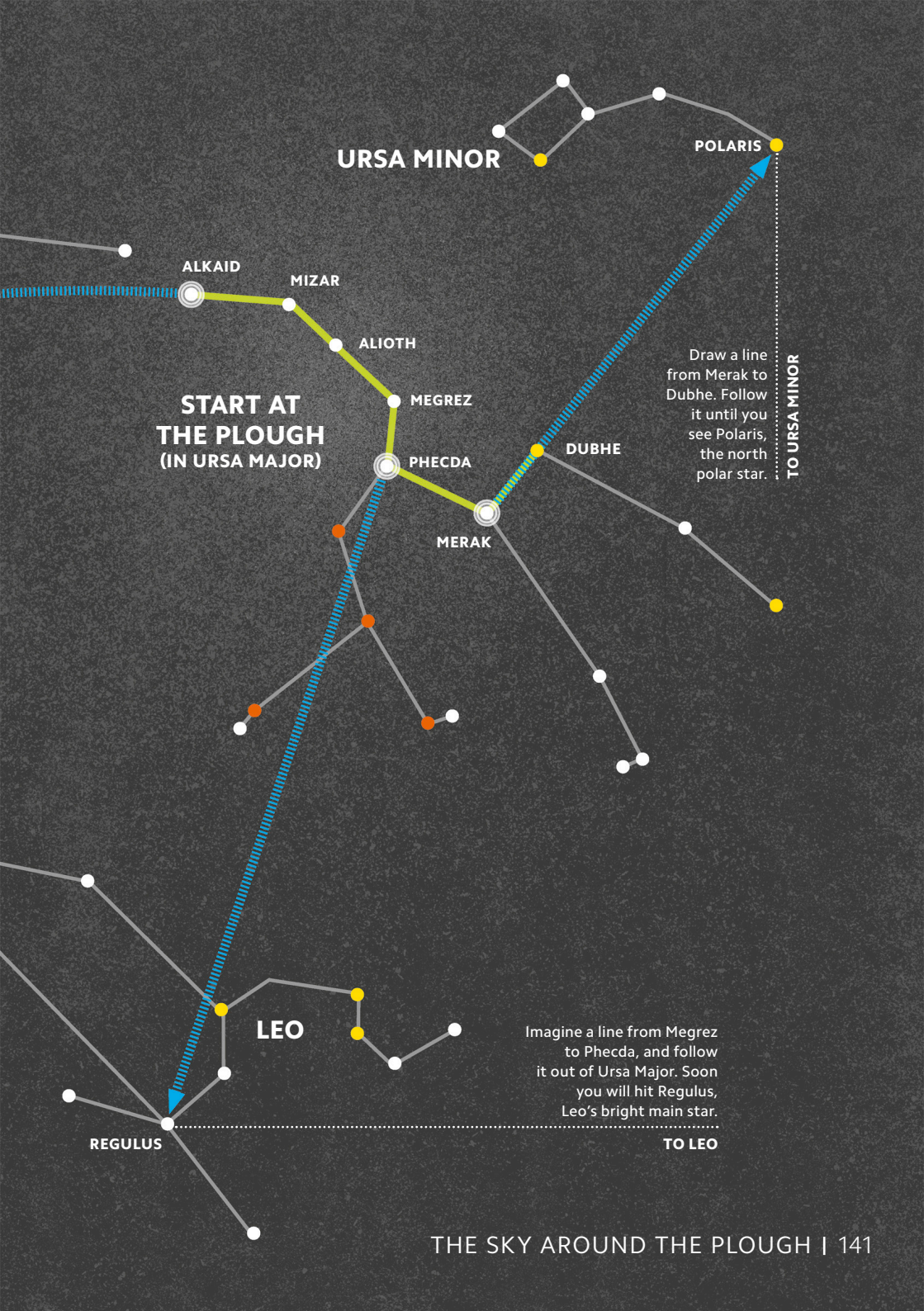


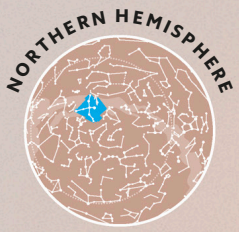
The northern bear
The bear in the sky stretches from Alkaid (η), its tail, through Phecda (γ), Merak (β), and Dubhe (α), to Muscida (ο), its head.

STARHOPPING FROM THE PLOUGH

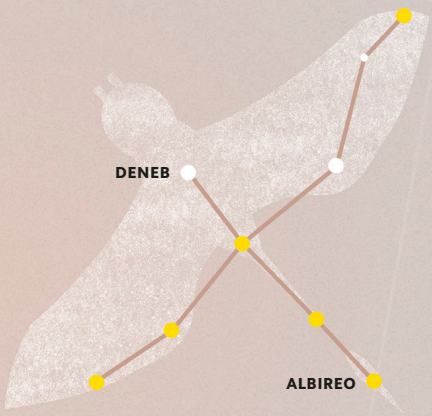


Navigating the sheer number of stars in the sky is made easier when you start from easy-to-spot star groups. These act as signposts to other stars. In the northern hemisphere, one such starting point is the Plough. This asterism includes several bright stars in a recognizable pattern and is always visible at night.


URSA MINOR
POLARIS
ALKAID
MIZAR
ALIOTH
MEGREZ
START AT
THE PLOUGH
(IN URSA MAJOR)
PHECDA
DUBHE
MERAK
Draw a line
from Merak to
Dubhe. Follow
it until you
see Polaris,
the north
polar star.
TO URSA MINOR
LEO
Imagine a line from Megrez
to Phecda, and follow
it out of Ursa Major. Soon
you will hit Regulus,
Leo's bright main star.
REGULUS
TO LEO



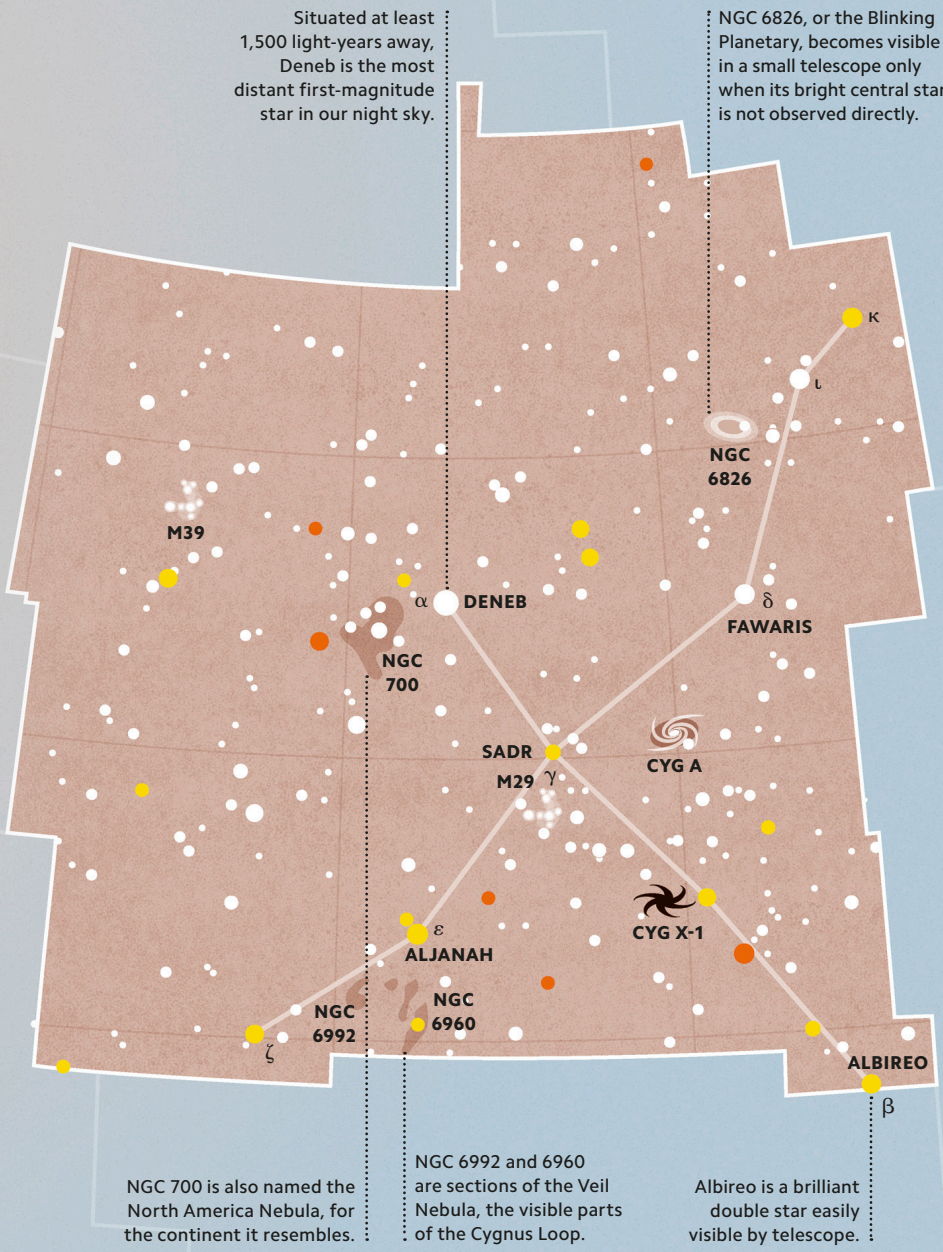
KEY DATA
Fully visible: *90°N–29°S*
Brightest star: *Deneb (α)*
Highest in sky at 10pm:
August–September



In flight at night
Cygnus, the swan, stretches from Deneb (α), at its tail, to Albireo (β), at its head. Between the two, on both sides, stretch the swan's wings.

THE SWAN

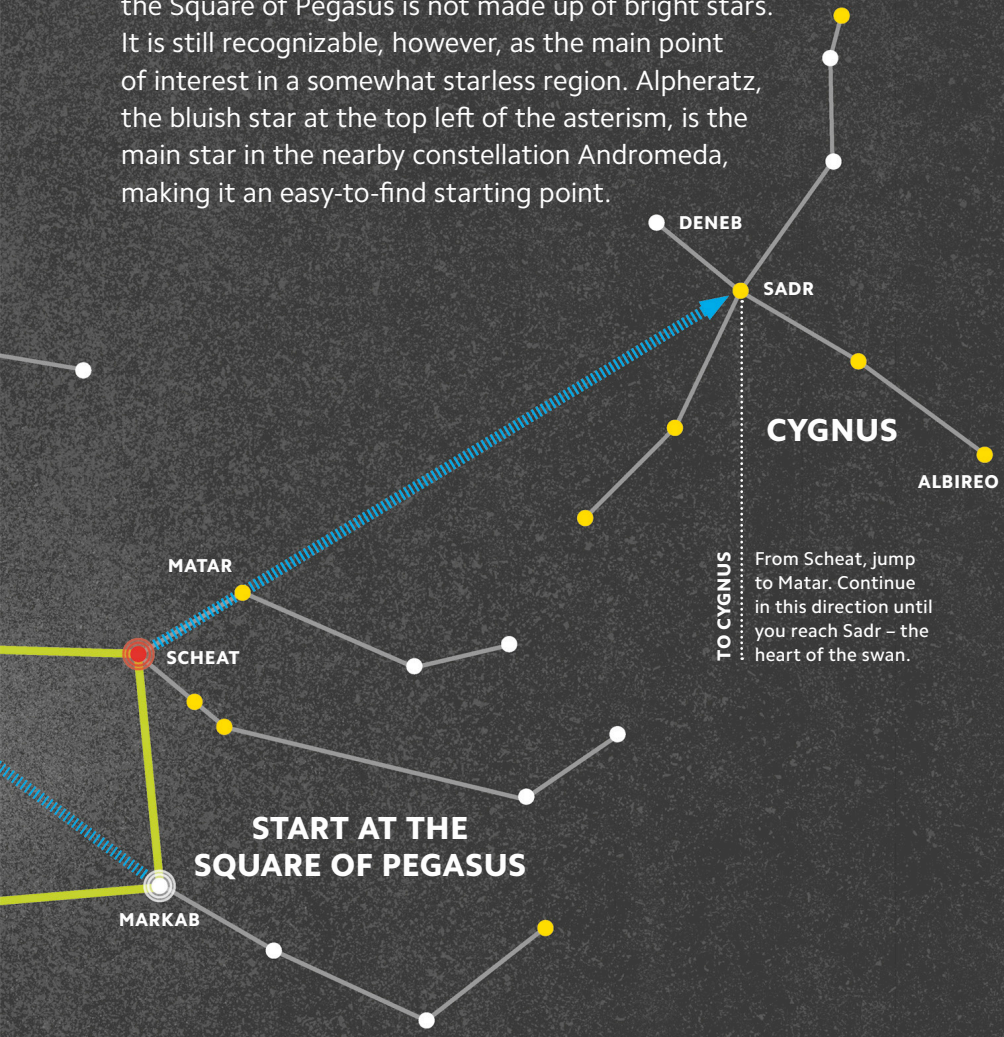
As well as depicting a swan in flight across the Milky Way, the main stars in Cygnus also form an asterism called the Northern Cross. With low light pollution, binoculars will reveal a myriad of star clusters and clouds of gas and dust. Cygnus is significant in the mythological traditions of ancient Greece, ancient Rome, China, and several Polynesian islands. Notable objects within the constellation include Cygnus X-1, the first recorded black hole; the Cygnus Loop, a remnant of a 10,000-year-old supernova; and NGC 700, the North America Nebula.

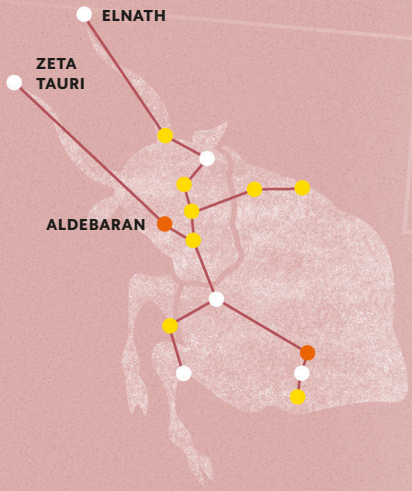

Situated at least 1,500 light-years away, Deneb is the most distant first-magnitude star in our night sky.
NGC 6826, or the Blinking Planetary, becomes visible in a small telescope only when its bright central star is not observed directly.
κ
ι
NGC 6826
M39
α DENEB
δ FAWARIS
NGC 700
SADR
M29 γ
CYG A
CYG X-1
ε ALJANAH
NGC 6960
NGC 6992
ζ
ALBIREO
β
NGC 700 is also named the North America Nebula, for the continent it resembles.
NGC 6992 and 6960 are sections of the Veil Nebula, the visible parts of the Cygnus Loop.
Albireo is a brilliant double star easily visible by telescope.


TO PERSEUS
Draw a line from Markab to Alpheratz. Continue out of Andromeda until you see Mirphak in Perseus.
Jump from Algenib to Alpheratz. Follow this path onwards until you see Caph.
TO CASSIOPEIA
MIRPHAK
CAPH
CASSIOPEIA
PERSEUS
ANDROMEDA
ARIES
HAMAL
ALPHERATZ
TO ARIES
Trace a line from Scheat to Alpheratz. Continue on this path until you arrive at Hamal, the brightest star in Aries.
ALGENIB

STARHOPPING FROM THE SQUARE OF PEGASUS

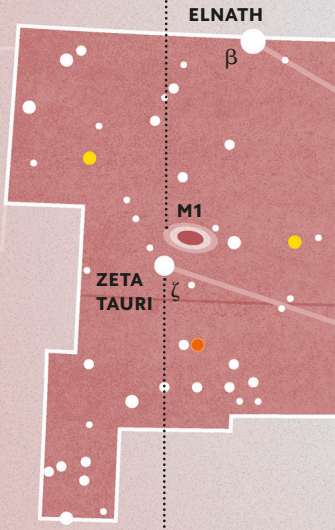
From September to November, the night sky contains a simple four-star shape useful for hopping to several constellations. Unlike other popular starting points, the Square of Pegasus is not made up of bright stars. It is still recognizable, however, as the main point of interest in a somewhat starless region. Alpheratz, the bluish star at the top left of the asterism, is the main star in the nearby constellation Andromeda, making it an easy-to-find starting point.





Great bull of heaven
Only the front half of the bull is shown in Taurus. Zeta Tauri (ζ) and Elnath (β) mark the tips of the bull's horns, and Aldebaran (α) its eye. Below, other stars make up its front legs and torso.

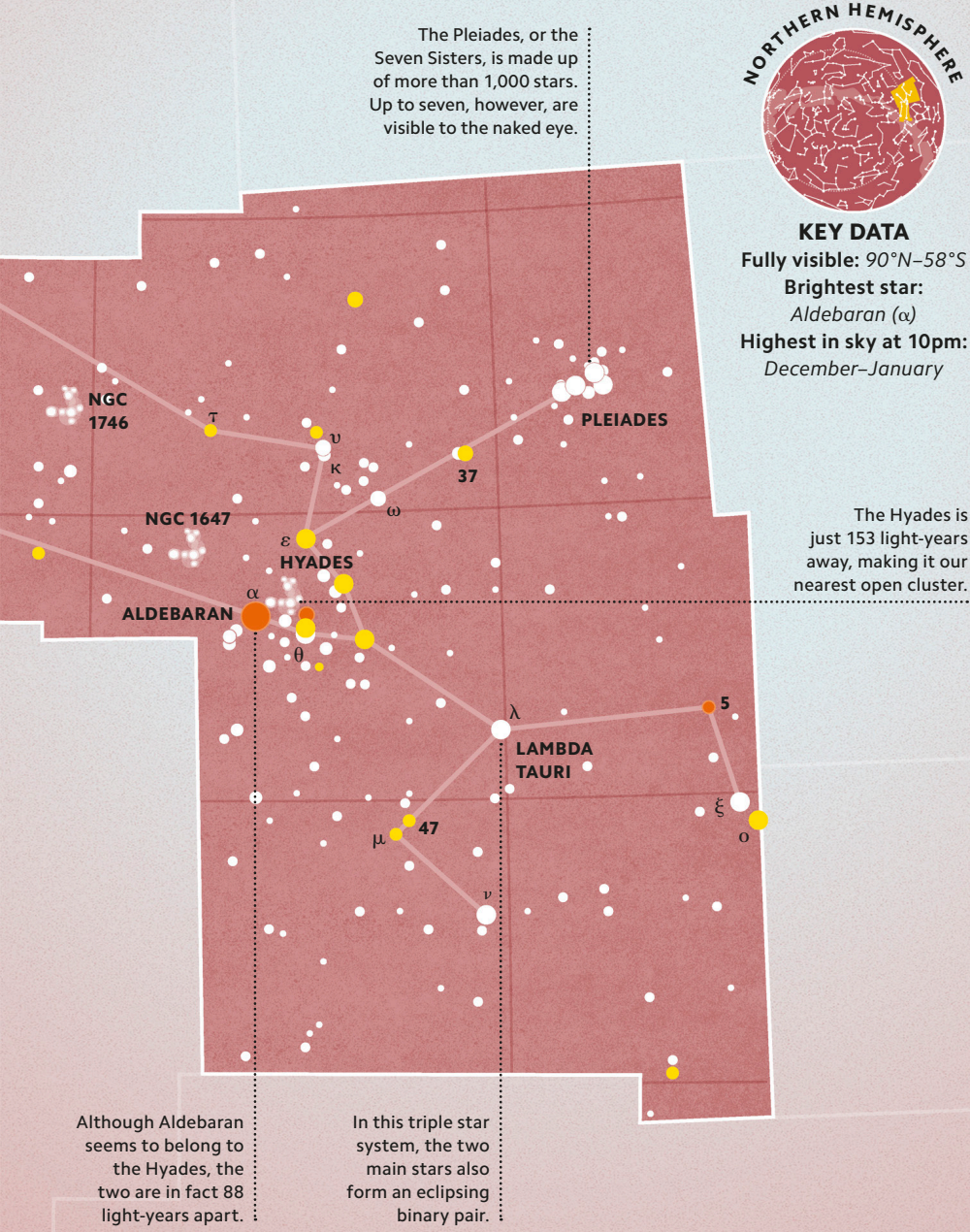
M1, or the Crab Nebula, is a supernova remnant discovered by English astronomer John Bevis in 1731.



Zeta Tauri, a binary star, is roughly 440 light-years away.

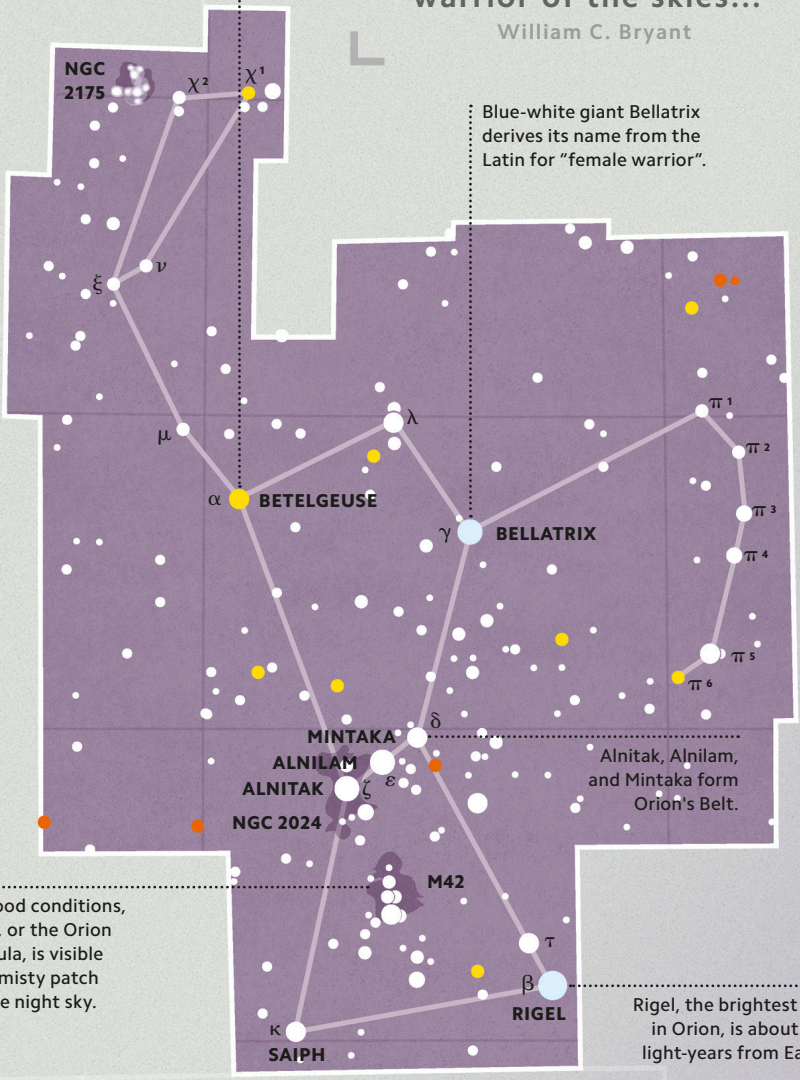
THE RAGING BULL

Situated in the northern sky, the constellation Taurus has been identified with bull myths across many ancient civilizations. From 4000 BCE, the sight of the Sun rising through Taurus marked the arrival of the spring equinox in Mesopotamia. Taurus also contains the Hyades, a well-studied star cluster some 153 light-years away. Several of its stars are in the process of evolving into giant stars. Like the Hyades, the nearby Pleiades cluster is also visible to the naked eye.


NORTHERN HEMISPHERE
KEY DATA
Fully visible: 90°N–58°S
Brightest star:
Aldebaran (α)
Highest in sky at 10pm:
December–January
The Pleiades, or the Seven Sisters, is made up of more than 1,000 stars. Up to seven, however, are visible to the naked eye.
The Hyades is just 153 light-years away, making it our nearest open cluster.
Although Aldebaran seems to belong to the Hyades, the two are in fact 88 light-years apart.
In this triple star system, the two main stars also form an eclipsing binary pair.
NGC 1746
τ
υ
κ
37
PLEIADES
ω
NGC 1647
ε
HYADES
α
ALDEBARAN
θ
λ
5
LAMBDA TAURI
ξ
ο
μ
47
ν

When Betelgeuse explodes, it will release more energy than the Sun will produce in its lifetime.

> "I have seen the great Orion ... that large-limbed warrior of the skies..."
>
> William C. Bryant

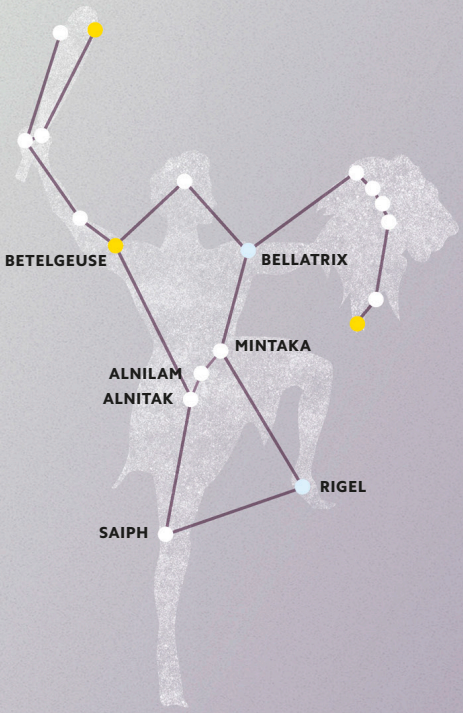


Blue-white giant Bellatrix derives its name from the Latin for "female warrior".

Alnitak, Alnilam, and Mintaka form Orion's Belt.

In good conditions, M42, or the Orion Nebula, is visible as a misty patch in the night sky.

Rigel, the brightest star in Orion, is about 860 light-years from Earth.



Belted warrior
Betelgeuse (α) and Bellatrix (γ) mark Orion's shoulders. Below his belt lie Saiph (κ), which marks his knee, and Rigel (β), his foot.



KEY DATA
Fully visible: *79°N–67°S*
Brightest star: *Rigel (β)*
Highest in sky at 10pm: *December–January*

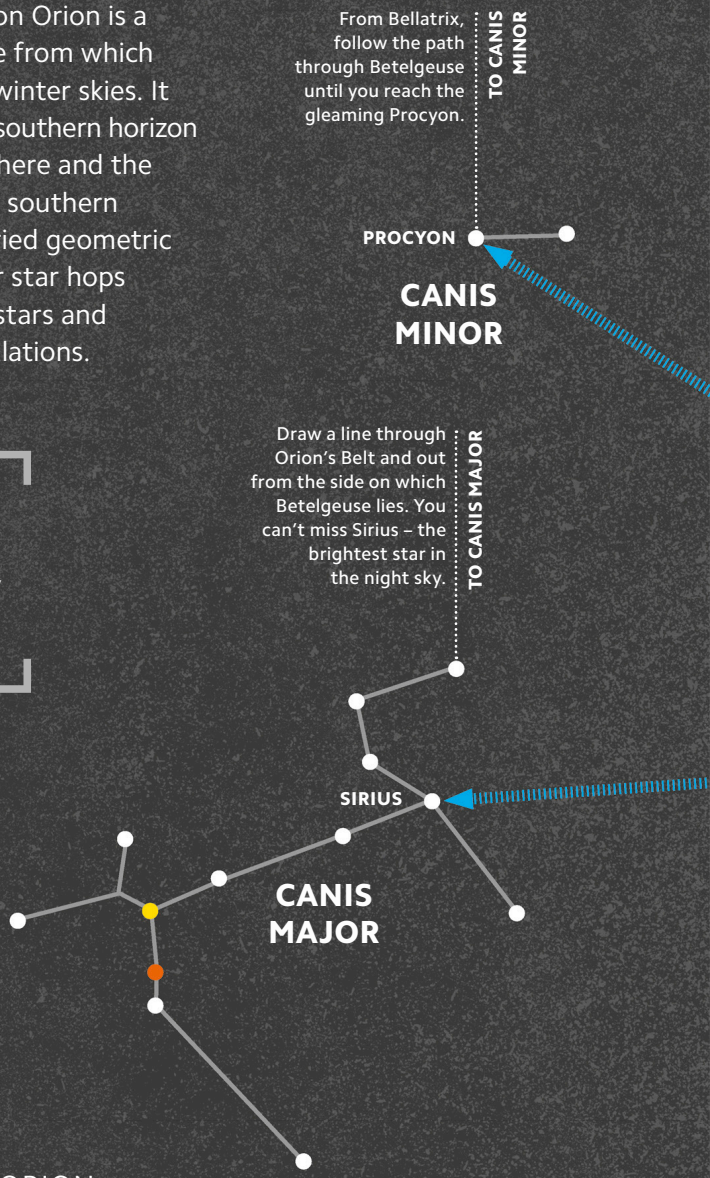
THE HUNTER

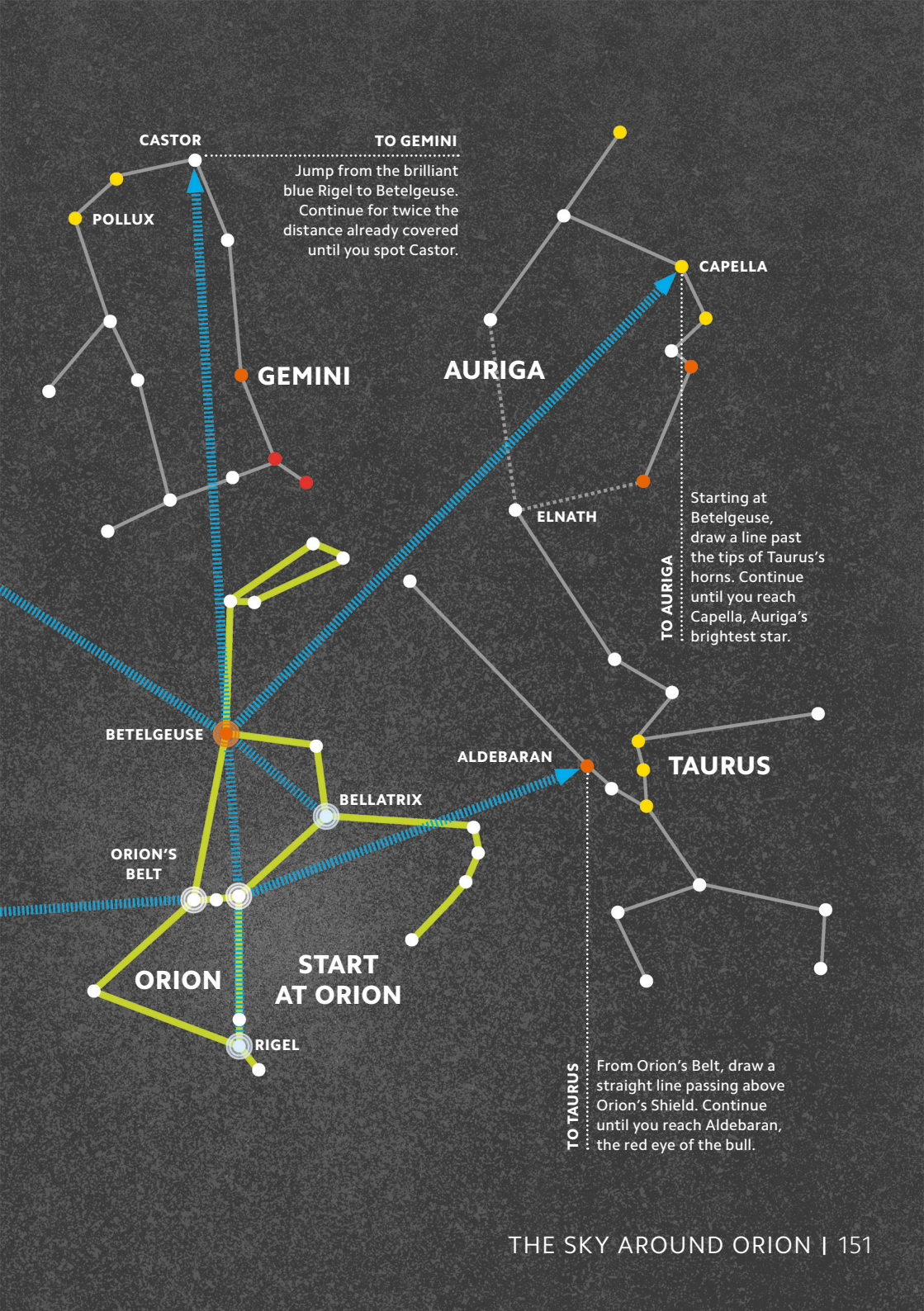
With more bright stars than any other constellation, Orion is the centrepiece of the night sky from January through to March. Its brightest stars are Betelgeuse, with its glowing red hue, and the blue supergiant Rigel. Other notable features of the Hunter's pattern include Orion's Belt, a three-star asterism, and Orion's Sword, which hangs from his belt and includes M42, or the Orion Nebula, itself filled with hundreds of young stars.

STARHOPPING FROM ORION

The brilliant constellation Orion is a strong choice for a base from which to starhop around the winter skies. It is easily seen above the southern horizon in the northern hemisphere and the northern horizon in the southern hemisphere. Orion's varied geometric configurations allow for star hops to many nearby bright stars and their associated constellations.

The Milky Way runs from Canis Major to Auriga, passing the left side of Orion.

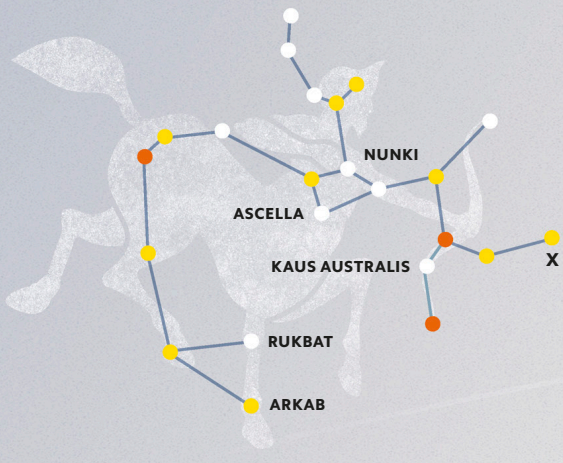



CASTOR
POLLUX
GEMINI
TO GEMINI
Jump from the brilliant blue Rigel to Betelgeuse. Continue for twice the distance already covered until you spot Castor.
AURIGA
CAPELLA
ELNATH
TO AURIGA
Starting at Betelgeuse, draw a line past the tips of Taurus's horns. Continue until you reach Capella, Auriga's brightest star.
BETELGEUSE
ALDEBARAN
TAURUS
BELLATRIX
ORION'S BELT
ORION
START AT ORION
RIGEL
TO TAURUS
From Orion's Belt, draw a straight line passing above Orion's Shield. Continue until you reach Aldebaran, the red eye of the bull.

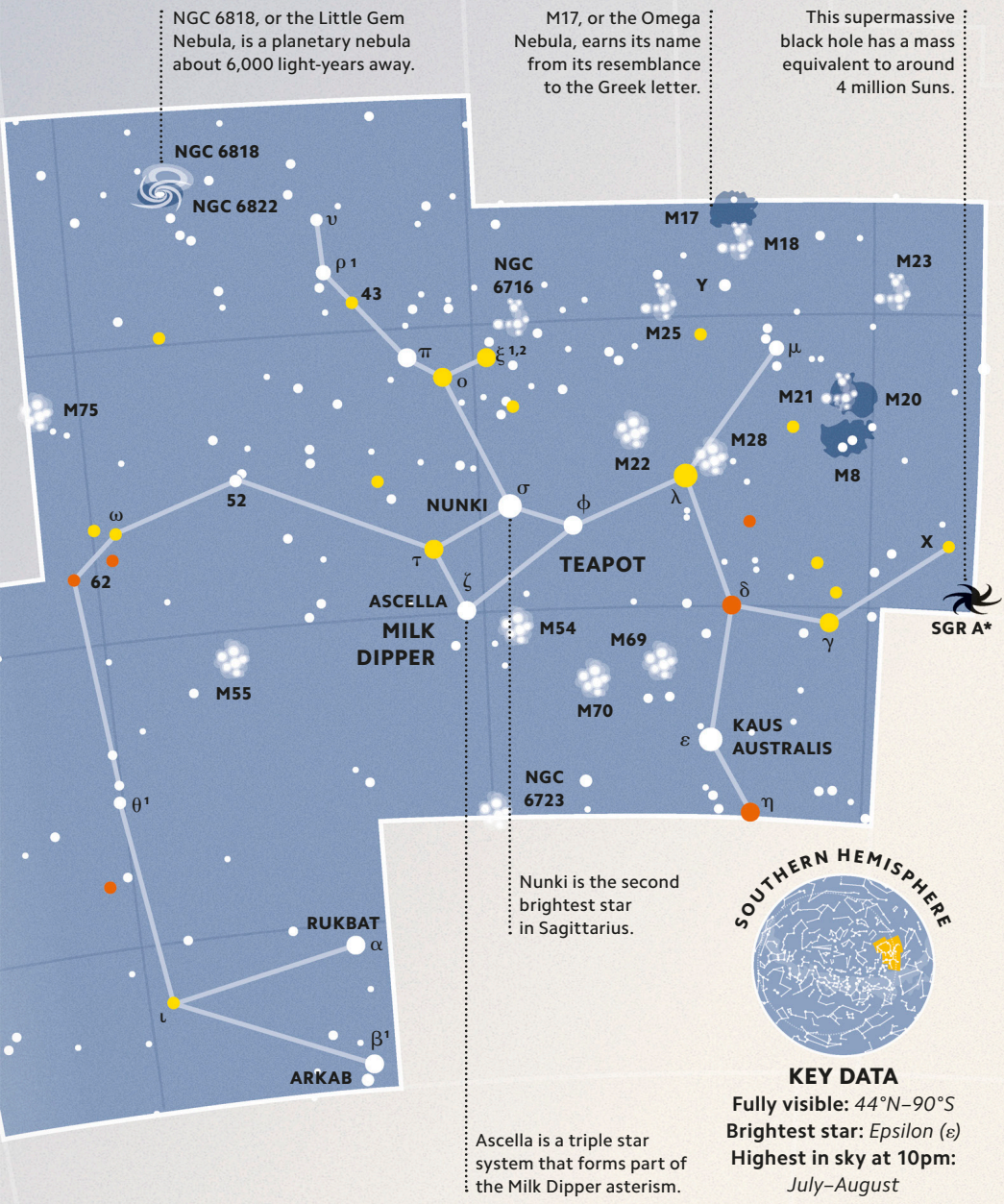
THE ARCHER

Since as early as the 11th century BCE, the constellation pattern of Sagittarius has been identified with a centaur figure (half-man, half-horse) drawing a bow. Looking up at the centaur means gazing into the very centre of the Milky Way, which is also the location of the supermassive black hole Sagittarius A* (SGR A*). This glowing stretch of sky is strewn with dark and bright nebulae, globular clusters, and open star clusters. In the centre, eight bright stars form an asterism known as the Teapot, the handle of which is sometimes referred to as the Milk Dipper.

The bright clouds of M8, or the Lagoon Nebula, extend for three times the width of a full Moon.

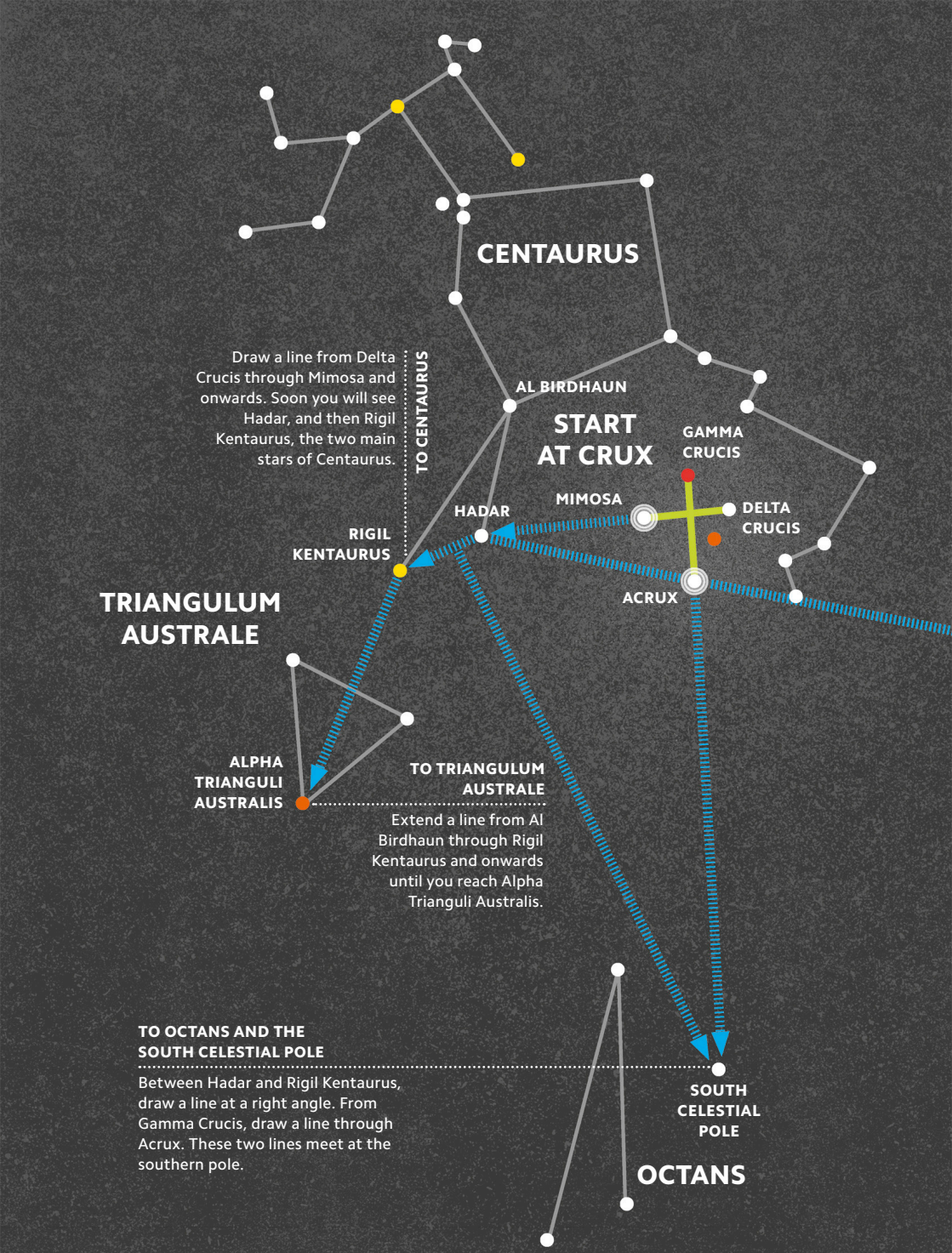


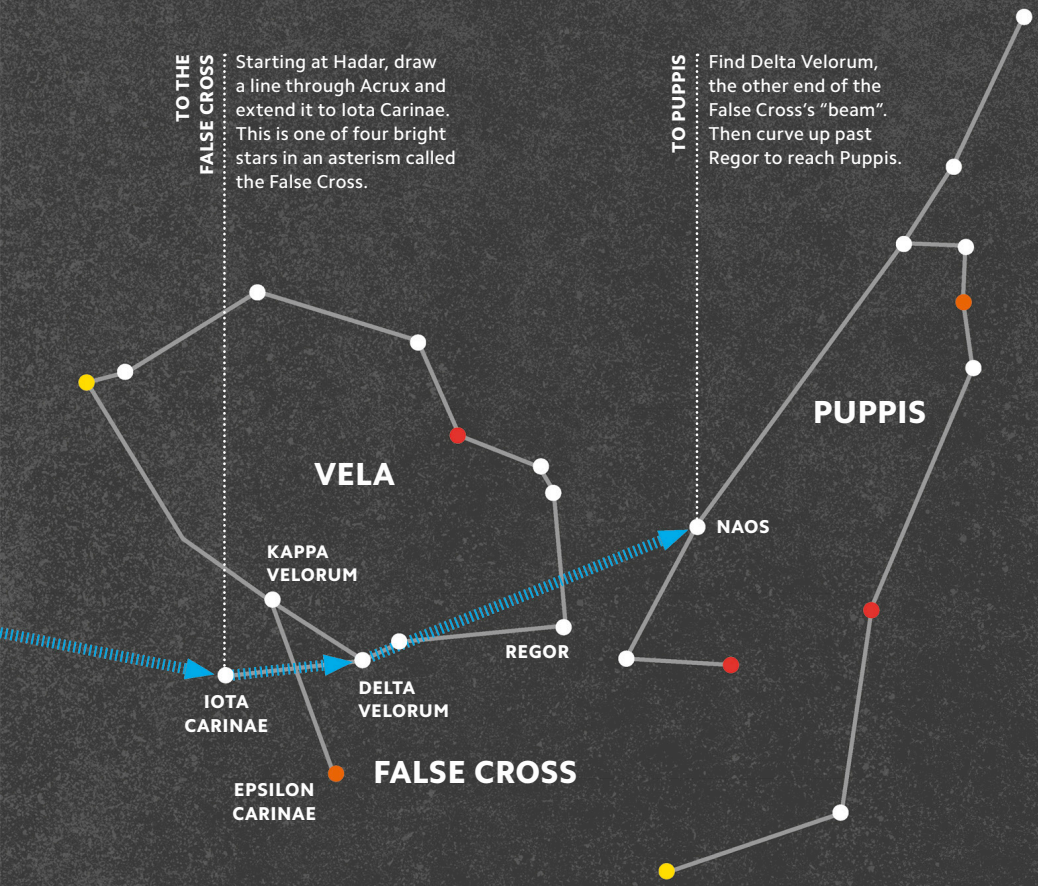
Armed centaur
Kaus Australis (ε) is found in the Archer's bow, drawn back by his hand, where Nunki (σ) is located. Just below, in the Archer's torso, lies Ascella (ζ). Rukbat (α) and Arkab (β^1), meanwhile, form one of his front legs.



KEY DATA

Fully visible: *44°N–90°S*
Brightest star: *Epsilon (ε)*
Highest in sky at 10pm: *July–August*


CENTAURUS
Draw a line from Delta Crucis through Mimosa and onwards. Soon you will see Hadar, and then Rigil Kentaurus, the two main stars of Centaurus.
TO CENTAURUS
AL BIRDHAUN
START AT CRUX
GAMMA CRUCIS
MIMOSA
HADAR
DELTA CRUCIS
RIGIL KENTAURUS
ACRUX
TRIANGULUM AUSTRALE
ALPHA TRIANGULI AUSTRALIS
TO TRIANGULUM AUSTRALE
Extend a line from Al Birdhaun through Rigil Kentaurus and onwards until you reach Alpha Trianguli Australis.
TO OCTANS AND THE SOUTH CELESTIAL POLE
Between Hadar and Rigil Kentaurus, draw a line at a right angle. From Gamma Crucis, draw a line through Acrux. These two lines meet at the southern pole.
SOUTH CELESTIAL POLE
OCTANS



STARHOPPING FROM CRUX

In the southern hemisphere, Crux, much like the Plough in the north, is easily visible throughout the year, even for those observing from areas burdened with heavy light pollution. The smallest constellation in our skies, Crux is an invaluable signpost towards several other constellations – and even the south celestial pole.

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